



TotalPlant Solution (TPS) System

Global User Station
GUS Solution Pack
User's Guide

Doc. No.:	GU2E200
Release:	200.1
Last Revision Date:	9/01



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Release 200.1 September 18, 2001**

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About This Document

Document Purpose

The *GUS Solution Pack User's Guide* provides information on using the GUS Solution Pack (GSP) toolkit for creating an operator's workspace environment and process displays for the Global User Station (GUS). The sections of this document fall into four categories. These categories are:

Overview of the GSP components and how they are used together to produce an integrated process operator interface.

Tutorial for a quick start in using GSP to create an operator's view.

Detailed instructions on how to build process displays and task panel displays.

Detailed definitions of the GSP components.

Before getting started using the GSP, read the following sections to understand the integrated view of the operator's workspace environment provided by using GSP.

"Introduction"

"GUS Solution Pack Fundamentals"

"SafeView Configurations"

To quickly implement a simple operator's workspace managing process displays and a task panel display created from GSP components, follow the instructions in the "Quick Start" tutorial.

When implementing a GSP process display, follow the instructions in the "Building Process Displays" section.

When implementing a GSP Task Panel display, follow the instructions in the "Building Task Panels" section.

Use the following sections as references to detailed definitions of the GSP components:

- "GUS-Provided Displays Reference"
- "Display Navigation Buttons Reference"
- "Display Templates Reference"
- "Function Buttons Reference"
- "Display Components Reference"
- "GSP Components Library"

To change the values of the GSP constants or add custom special constants, follow the instruction in the "GSP Configuration Utility" section.

When upgrading from an earlier version of GSP, follow the instructions in the "GSP Upgrade Utility" section to upgrade the GSP components in existing displays.

Revision Notes

The following list provides notes concerning all revisions of this document:

Doc ID	Rel ID	Date	Notes
GU2E100	R100	8/99	Initial Publication
GU2E100A	R100.2	3/00	2 nd Publication
GU2E200	R200.1	9/01	3 RD Publication

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title

Display Builder User's Guide

SafeView User's Guide

Display Authoring Tutorial

Faceplate, Alarm, and Message Scripting User's Guide

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Contents

Introduction

Overview

About the GUS Solution Pack

The GUS Solution Pack (GSP) is a toolkit for creating an operator's workspace environment and process displays for the Global User Station (GUS). GSP provides a library of display templates, display components, faceplate displays, and SafeView configurations.

The benefits of using GSP are as follows:

- Reduces the cost of building displays.
- Provides a best-practice operator environment.
- Provides a synchronized view across displays in an operator's workspace.
- Provides the status of an operator's scope and navigation to the display of interest
- Provides a workspace for safe, effective operations.

Reduces the Cost of Building Displays

The cost of authoring is significantly reduced through the use of the easy-to-use, prebuilt display components. GSP eliminates much of the learning curve associated with making full use of the power and functionality of an open, control-system human interface. Display authors do not need to design and script their reusable display components prior to building operational displays. Rather they may start building operational displays by inserting the appropriate display components from the GSP library and entering the parameter values for each inserted component (e.g., assigning a tagname). Having completed inserting GSP components and entering parameter values, a display author has a fully functional, operational display showing the behavior of the process.

If a display author identifies the need for an application-specific component, then a GSP display component may serve as a template. Since the display components are GUS displays, they are easily modified and/or copied. For example, the digital pump display in the GSP library could be used as a template for a motor display. The scripts for the digital pump are reusable in the motor display. Simply change the pump symbol to a motor symbol.

Provides a Best Practice Operator Environment

To provide a best-practice operator environment, GSP:

Provides performant call-up of displays and performant user-component interaction.

Uses consistent color coding and component behavior throughout the interface.

Enables the operator to see the entire process under the scope of his/her control.

Facilitates use of a multilevel display view for monitoring and control by providing on-screen navigation to access key displays.

Provides detailed information as needed.

Provides a Synchronized View Across Displays in an Operator's Workspace

All the GSP library components using both Microsoft technology and GUS display functionality are designed and implemented to provide a synchronized view across displays in an operator's workspace. The following information is synchronized across displays:

Point-of-Interest in the Faceplate Display – All GSP display components, referencing the point being shown in the Faceplate Display, will appear selected on all running displays in the operator's workspace.

Show/Hide Toggle – Some GSP display components such as the digital alarm indicators have a show/hide state that may be toggled by the operator. Pressing the GSP-supplied Show/Hide button on any display toggles the visibility of these GSP display components on all running displays in the operator's workspace. This show/hide toggle feature synchronized across running displays enables the operator to reduce clutter on his entire workspace and view detail information as needed.

Show Setpoint/Show Tagname Toggle – Some GSP display components such as large control boxes show either the setpoint or tagname for the associated point depending on the Show Setpoint/Show Tagname state. A button is provided that toggles between the display of tagname and setpoint for control boxes on any running display.

Provides the Status of an Operator's Scope and Navigation to the Display of Interest

GSP provides components to build a Task Panel display. The Task Panel is a user-built display, providing two-level display navigation, alarm annunciation, and other operator functions. Typically, there is one Task Panel per operator area, and it appears in each station of an operator's console.

GSP navigation buttons on the Task Panel display provide both rapid access to high- and low-level displays and composite alarm annunciation, based on PRIMMOD associations.

Thus, the Task Panel display enables the operator to view the status of his/her area of control and to quickly navigate to the display of interest.

Provides a Workspace for Safe, Effective Operations

GSP provides a set of workspaces for safe, effective operations.

The workspace:

Allows the operator to focus his attention on managing the process not display windows.

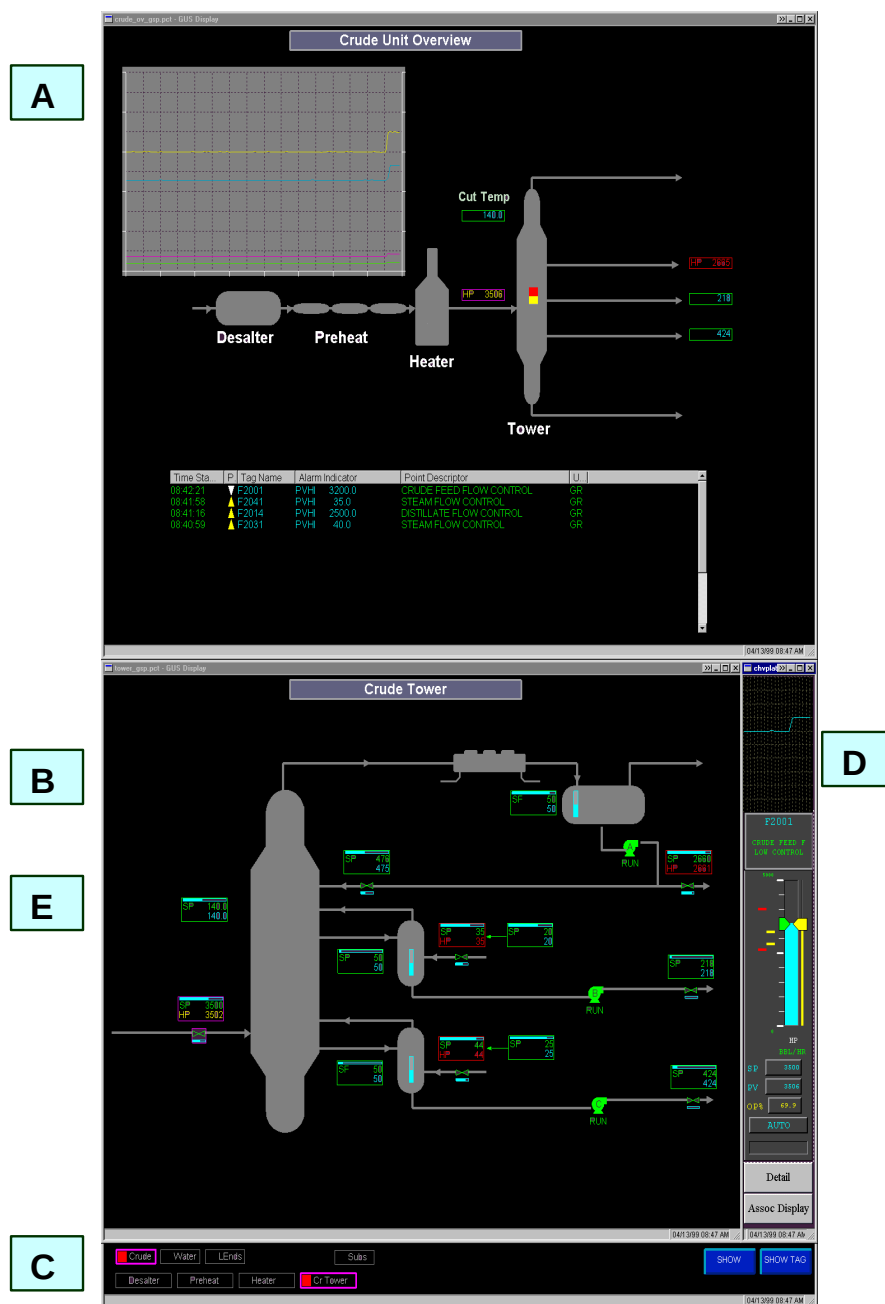
Supports multilevel display view for monitoring and control.

Shows the status of the operator's scope of control and enables quick access to displays via the Task Panel.

Appears as an integrated panel since GSP enables a synchronized view across displays.

For example, the two-screen workspace defines five windows. The windows are as follows (see the workspace figure on the following page):

- A. High-Level Display Window
- B. Low-Level Display Window
- C. Task Panel Display Window
- D. Faceplate Display Window
- E. Native Window (occupies the same window area as the low-level display)



Software Requirements

GUS Solution Pack Buildtime Files

The GUS Solution Pack Buildtime Files must be installed on the GUS prior to building any GSP displays. User must start with one of the GSP display templates (DisplayTemplate, TaskPanelTemplate, or LowLevelNavTemplate) when building a new GSP display.

GUS Solution Pack Common and Runtime Files

The GUS Solution Pack Common and Runtime Files must be installed on the GUS prior to running any GSP displays.

GUS Display Builder

GUS Display Builder must be installed before building any GSP displays. Both on-line and off-line versions of GUS Display Builder are acceptable.

GUS Display Runtime / Remote Client

GUS Display Runtime must be installed and the HOPC connection established before running any GSP displays.

GUS Multiple Displays

GUS Multiple Displays is required to allow process display, task panel and faceplate to co-exist on a GUS. Most of the GSP display components are designed to callup the faceplate and all of the GSP navigation buttons are designed to callup a process display. Therefore, at runtime, the Multiple Displays capability is required to benefit from the GSP package. However, it is not required to build GSP displays.

SafeView

SafeView is required for operator console to fully benefit from the GSP package. GSP provides SafeView configuration files to allow user to quickly configure a useable environment for the operator console. However, it is not required to build or test GSP displays.

GUS Solution Pack Fundamentals

Purpose of This Section

This section provides an overview of the GUS Solution Pack components, and how they are used together to produce an integrated process operator interface.

Overview of Components

The GUS Solution Pack consists of the following components:

Display Templates – These are GUS displays to be used as starting points for building process displays and task panels. All displays that use GUS Solution Pack components must be built from a copy of one of these templates. The templates are required for the components to function properly since they provide synchronization of indications (such as tag selection) across displays within a station. Detailed descriptions of the display templates are provided in the “Display Templates” section.

Display Components – These are embedded displays, such as control valves, control boxes, and alarm indicators, used as components for building process displays.

Buttons – These are embedded displays, such as display navigation buttons and function buttons, used as components for building task panels.

Faceplate Display – This is a prebuilt display that provides an operator interface for viewing and manipulating operating parameters (e.g., setpoint, output, mode, etc.) of a single point. It is based on the GUS faceplate ActiveX control.

SaveView Configurations – These are a set of prebuilt SafeView configurations to manage the windows on single- or dual-screen Global User Stations.

General Concepts

Color Conventions – The GUS Solution Pack is delivered with a fixed set of color conventions. All colors, including display background and all display component parts, are based on these conventions predefined in the system registry. A GUS display (GSPColor.pct) is provided that shows the color conventions. A tool to modify the color conventions will be provided in a future release.

Selectability – Some display components are selectable and are indicated as such in the following summaries. When selected (by mouse click or touch screen), they highlight to indicate selection and invoke the faceplate for the referenced point. The navigation buttons are selectable, but selecting one invokes the referenced display rather than a faceplate.

Point Alarm Annunciation – Some components annunciate alarms for a single associated point. They indicate the highest priority of alarm on the point and acknowledgment status. Blinking indicates that at least one alarm on the point is unacknowledged. Pressing the ACK key will acknowledge all alarms associated with these components on a selected display.

Composite Alarm Annunciation – Some components annunciate alarms for a group of points based on the user-defined Primary Module (PRIMMOD) associations. Composite alarm priority is shown, and blinking indicates composite acknowledgment status. Alarms associated with these components are not acknowledged from that display when the ACK key is pressed. However, acknowledgment status is updated when alarms are acknowledged from other displays.

Show/Hide Toggle – Some GSP display components can be made visible or invisible based on the Show/Hide state (using a display database Boolean). A Show/Hide Button on any display toggles the Show/Hide state for the station. This is useful to remove clutter from displays.

Show Setpoint/Tagname Toggle – The large control boxes show either the setpoint or tagname for the associated point depending on the Show Setpoint/Tagname state (using a display database Boolean). A Show Setpoint/Tagname button on any display toggles the Show Setpoint/Tagname state for the station.

Display Components

The following types of display components are provided for building process graphics. While this provides a summary of the types of display components, detailed descriptions of the components are provided in the “Display Components Reference” section.

Control Boxes

Control Boxes are selectable components that show operational information such as PV, Mode, and Alarm Status. The control boxes are typically used with Regulatory Control points running PID algorithm. Large and small control boxes are provided. Large Control Boxes toggle to show either setpoint or tagname, using the Show Setpoint/Tagname toggle button as described above. Small Control Boxes take up less room, but do not show tagname, setpoint, or output. They are intended to be used where display space is not sufficient for Large Control Boxes.

Control Valves

Control Valves are selectable components that show operating information such as the current valve position, red tag, and mode. Typically, they are used in conjunction with a Control Box connected to the same point.

Output Boxes

Output Boxes are components that show the output value of a point. There are three types of output boxes included in the GSP package: controller output boxes, analog output boxes and output value only boxes.

Selector Boxes

Selector Boxes are selectable components that help to describe a control strategy on the display. They are typically associated with a Regulatory Control Point.

Level Bars

Level Bars are selectable bars that show the level in a tank or vessel. They are typically associated with a Regulatory Control Point or Analog Input Point.

Cascade Symbols

Cascade Symbols are non-selectable components that show the state of a cascade connection between two Regulatory Control Points. They are lines, arrows, and continuation symbols, used in conjunction with two Control Boxes, and are associated with the secondary control points.

Measurement Boxes

Measurement Boxes are selectable components that show operational information such as process variable and point alarm annunciation. Typically, they are used with an Analog Input Point.

Digital State Boxes

Digital State Boxes are selectable components that show operational information such as current state and point alarm annunciation. Typically, they are used with a Digital Input, Digital Composite, or Device Control Point.

Digital Alarm Graphic Symbols

Digital Alarm Graphic Symbols are selectable components that annunciate point alarms from discrete alarm inputs, such as pressure and level switches. Typically, they are used with a Digital Input Point. However they may also be used with Digital Composites and Device Control Points. They indicate the type of measurement such as pressure or level. The user must decide if the digital status is a high, high-high, low, or low-low limit and use the appropriate symbol.

Digital Alarm Message Boxes

Digital Alarm Message Boxes are selectable components that represent devices, such as digital valves and pumps. Typically, they are used with a Digital Composite or Device Control Point and show the current state by behavior

Digital Valves

Digital Valves are selectable components that show the digital state of valves. Typically, they are associated with dynamic valves that have limit switches coming to the DCS to indicate status.

Digital Pumps and Motors

Digital Pumps and Motors are selectable components that show the digital state of pumps and motors. Typically, they are associated with dynamic pumps and motors that have limit switches coming to the DCS to indicate status.

Primmod Status Indicators and Aggregate Primmod Status Indicators

Primmod Status Indicators are non-selectable components that annunciate composite alarms for one or more Primary Module (PRIMMOD) associations. They blink if any alarm in the PRIMMOD association is unacknowledged.

Custom Parameter Boxes

Custom Parameter Boxes are used to display any point.parameter with numeric or text values. They do not reference other LCN parameters, therefore, would not cause any problem during display validation.

Buttons

The following types of buttons are provided for building task panels. While this provides a summary of the types of buttons, detailed descriptions of the buttons are provided in the “Display Navigation Buttons Reference” and “Function Buttons Reference” sections.

Display Navigation Buttons

Navigation Buttons are selectable components that annunciate alarms for one or more PRIMMOD associations. Each button is connected to one display, which is invoked when the button is selected.

Function Buttons

Function Buttons are provided to perform the following functions:

- Invoke a GUS Display
- Invoke an Application or Document
- Toggle the Show Setpoint/Tagname State
- Toggle the Show/Hide State
- Toggle the Show/Hide State of user scripted objects
- Switch among Multiple State of user scripted objects
- View Point to keep track of PV of a tag in task panel

Display Organization

Display Hierarchy

A display hierarchy is an efficient and logical framework for a system of displays. Techniques such as summaries, prioritized data, and composite information enable the operator to understand large areas of the plant through a manageable number of displays. In the upper levels of the hierarchy, full detail is not possible, but the summary should provide enough information to alert the operator to investigate the lower levels of the hierarchy. The hierarchical display structure should enable navigation in all directions.

The specifics of display organization depend on a number of factors such as the information and tasks of a particular application, and the scope of control and number of points for which an operator is responsible. A display hierarchy can provide the formal organization for the system while providing the designer the flexibility to meet the specific needs. The navigation mechanisms provided by the GUS Solution Pack include specific support for two major levels of displays. This is provided by the two-level hierarchy of navigation buttons in the Task Panel, however the user may implement multiple sublevels within these major levels by providing links in the displays themselves. In other words, multiple sublevels may be implemented within either of the two major levels, but navigation between these sublevels is done by display invocation buttons within the displays, rather than the navigation buttons in the Task Panel.

High-Level Displays

High-Level displays generally represent large portions of an operator's process area, and although they typically support control changes, they are often used for monitoring the process. These are typically overview and summary displays that provide the operator key information to quickly ascertain the operational status of the process area. They may contain:

Composite alarm annunciation.

All Emergency and/or High priority alarms for the specific area.

List of currently active alarms (Alarm Summary).

Key trends.

Critical control loops and process measurements, shown as values with point alarm annunciation.

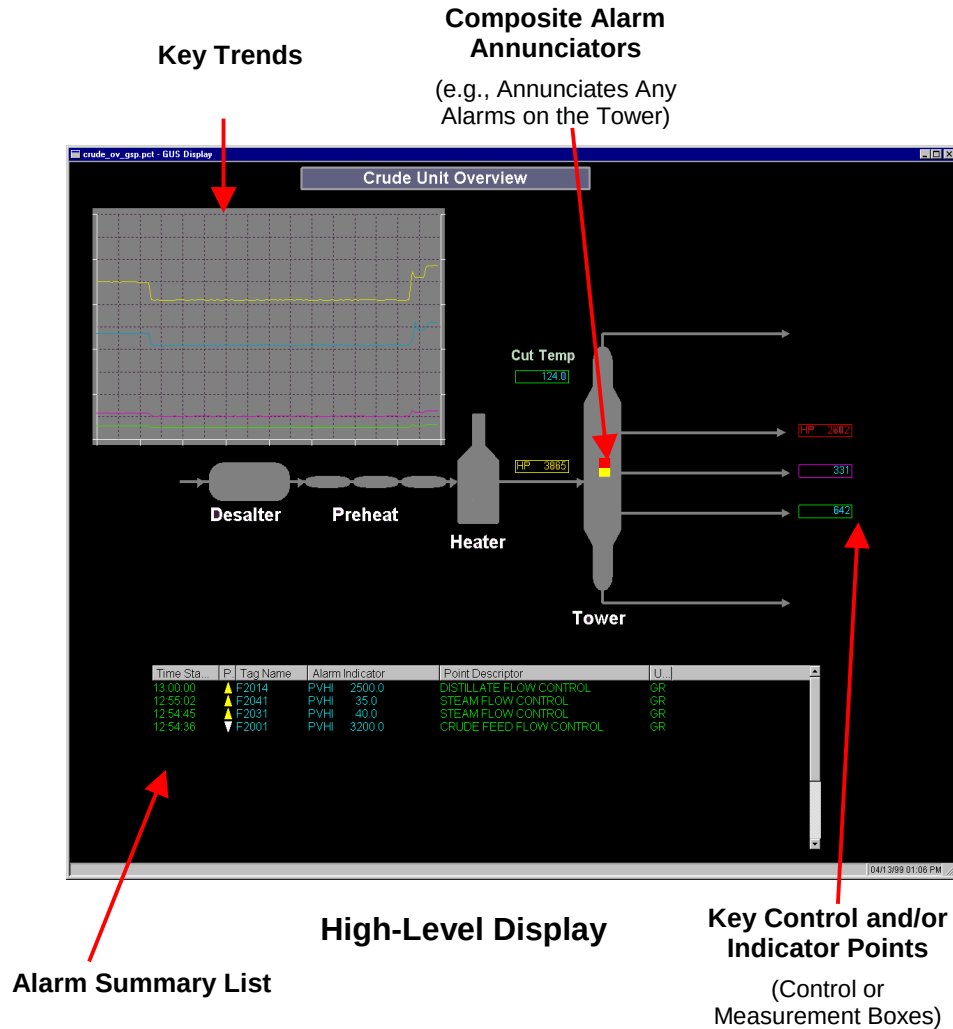
Key process performance indicators (e.g., product analysis, utility usage, efficiency factors).

Key safety indicators.

Safety interlock and advanced control status.

Buttons to invoke more detailed displays (e.g., Low-Level displays).

A simplified example of a Crude Unit overview display, shown below, illustrates the types of information presented on a High-Level display.



Low-Level Displays

Low-Level displays are generally more detailed representations of smaller portions of the process area. They provide sufficient detail for the operator to determine detailed process conditions, perform abnormal condition analysis, and make process changes. Typically they are detailed process graphics, showing the following:

All control loops (with process variables, setpoints, and valve positions) and process measurements shown as values with point alarm annunciation.

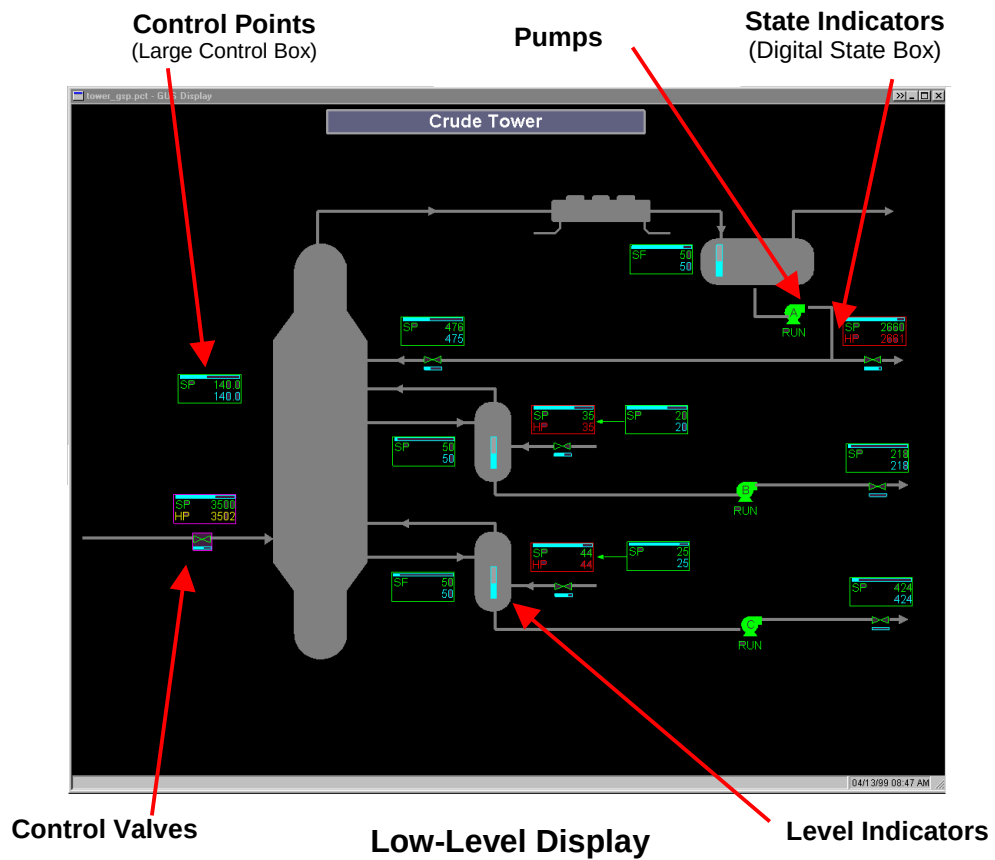
Appropriate trends.

Point alarm indications.

Links to other displays (e.g., upstream and downstream displays, other Low-Level displays).

The operator makes process changes from a faceplate that is invoked by selecting a point from these displays. The Low-Level displays should provide the operator sufficient detail and context for making process changes and seeing the results of those changes.

A simplified example of a Crude Tower display, shown below, illustrates the types of information presented on a sample Low-Level display.



Display Navigation and Interaction

SafeView Configuration

A set of SafeView configurations is provided to manage the operator displays. Although single screen configurations are provided, optimal operations may be achieved using dual-screen configurations such as the one shown below. The configuration consists of five windows. The upper screen has one window assigned to High-Level displays. The lower screen has four windows, assigned to the following categories of displays:

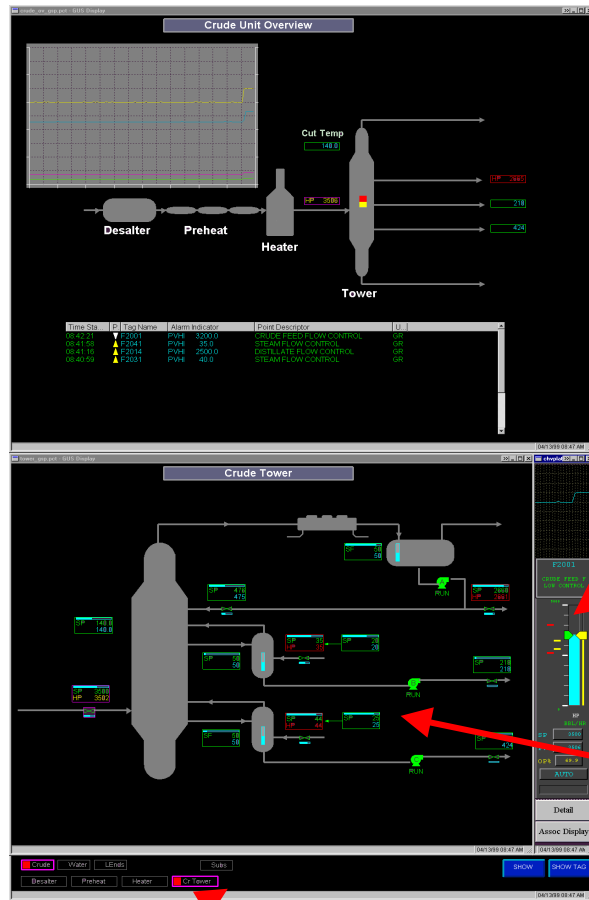
Low-Level Displays

Faceplates

Task Panel

Native Window

The Native Window becomes visible when it is invoked, and it occupies the same area as the Low-Level display, as shown below. Invocation of a Low-Level displays overlays the Native Window.



Upper Screen

High-Level Display

- Provides process overview.
- Provides high-level alarm annunciation.
- Provides links to more detailed displays.

Lower Screen

Faceplate

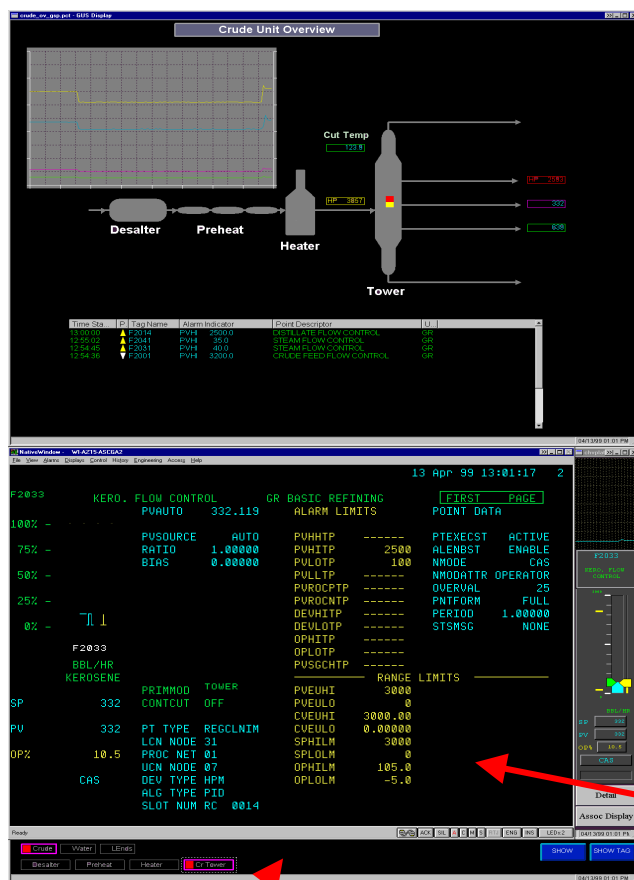
- Provide single point interface for making process changes.

Low-Level Display

- Provide detailed view of the process.
- Shows detailed point alarm annunciation.
- Operator typically invokes faceplates from this display to make process changes.

Task Panel

- Provides buttons for annunciating alarms and navigating to displays.
- Provides other function buttons for toggling show/hide and setpoint/tagname states.
- May provide user defined function buttons.



Upper Screen

High-Level Display

Lower Screen

Faceplate

Native Window

Task Panel

In the single screen configuration, only the lower screen configuration is provided, and the High-Level and Low-Level displays share the same window.

Task Panel

The Task Panel is a user-built display using GUS Solution Pack components to provide multilevel display navigation, alarm annunciation, and other operator functions. It is built starting with a copy of the Task Panel Display Template. Typically, there is one Task Panel per operator area, and it appears in each station of an operator's console. The Task Panel template (the GUS display that the user starts with in building the Task Panel) contains the following function buttons as defaults, but the user can add other function buttons as appropriate:

Show/Hide Button

Show Setpoint/Show Tagname Button



Task Panel

Navigation Buttons

- Provides buttons to invoke high- and low-level displays.
- Provides composite alarm annunciation, based on PRIMMOD associations.

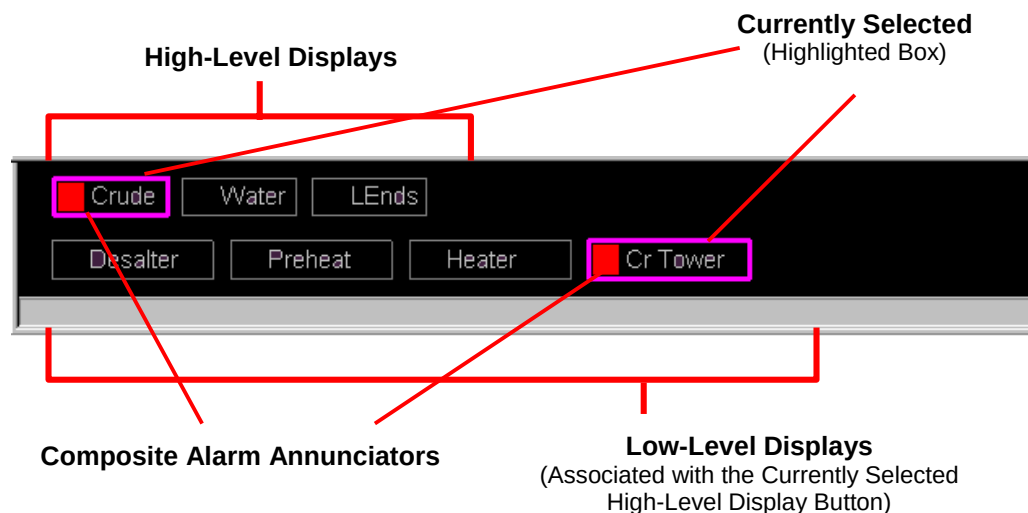
Function Buttons

- Provides Show/Hide and Show Setpoint/Show Tagname toggle functions (default).
- May contain user-defined function buttons (e.g., invoking applications and documents).

Display Navigation

Navigation buttons on the Task Panel allow the operator to navigate to High-Level and Low-Level displays. In the following simplified example, the operator's area has three High-Level displays: Crude Unit (Crude), Water Treatment Unit (Water), and Light Ends Unit (LEnds). The Crude Unit has the following Low-Level displays: Desalter, Preheat, Heater, and CrTower. The High-Level display navigation buttons appear on the

top row of the Task Panel. When one of these is selected, it highlights, invokes the associated High-Level display, and shows the associated Low-Level display navigation button on the bottom row of the Task Panel. Note that since the Task Panel is a user-built display, the format can be customized for a users specific application. In the example below, the Crude High-Level display is selected, showing the associated Low-Level displays, Desalter, Preheat, Heater, and CrTower. Selecting another High-Level display button invokes that display and shows the Low-Level navigation buttons, associated with it. Selecting a Low-Level display button highlights it and invokes the associated Low-Level display.



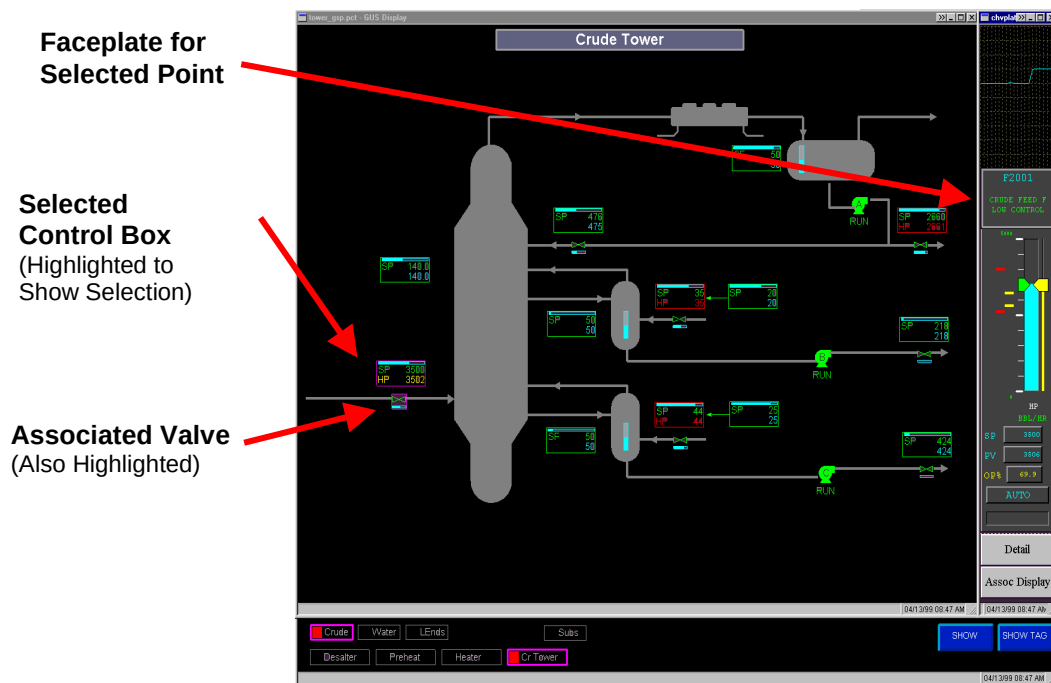
Alarm Annunciation

The navigation buttons also provide composite alarm annunciation based on PRIMMOD association, thus enabling quick navigation to where the operator can diagnose the problem and take corrective action. Typically, a Low-Level navigation button is associated with a PRIMMOD point, with which all alarmable points in the corresponding Low-Level displays are associated. Any alarms on that display will be annunciated by the composite alarm annunciator on the button. Each High-Level navigation button is associated with the set of PRIMMOD points of its Low-Level navigation buttons. Any alarm on a Low-Level display is also annunciated on the associated High-Level navigation button. The result is a two-level alarm annunciation scheme that corresponds to the two-level display navigation scheme. This allows the operator to see the alarms associated with both levels of displays, and easily navigate to the appropriate display. This can be extended to multilevel display hierarchies.

Making Process Changes

The primary interface for making process changes is the faceplate. Selecting a selectable component (e.g., control box) on a display invokes the faceplate for that point, as shown below. The selected component is highlighted and any components that reference that same point on any display on the same station are highlighted. This allows the operator to identify the current point of focus on both High-Level and Low-Level displays being shown.

The faceplate is specific to the selected point type, it shows the key operating parameters of the point, and allows the operator to make process changes. For example, the faceplate for a Regulatory Control Point (PID) shows setpoint, process variable, and output (numerically and graphically). Also shown are graphical marks for alarm limits, mode, alarm state, and any abnormal state of the point. A trend of the process variable is provided. The operator is allowed to change setpoint, mode, and output. With the faceplate active, the setpoint, output, mode, and raise/lower keys can also be used to make process changes.



Reserved Display Database Variables

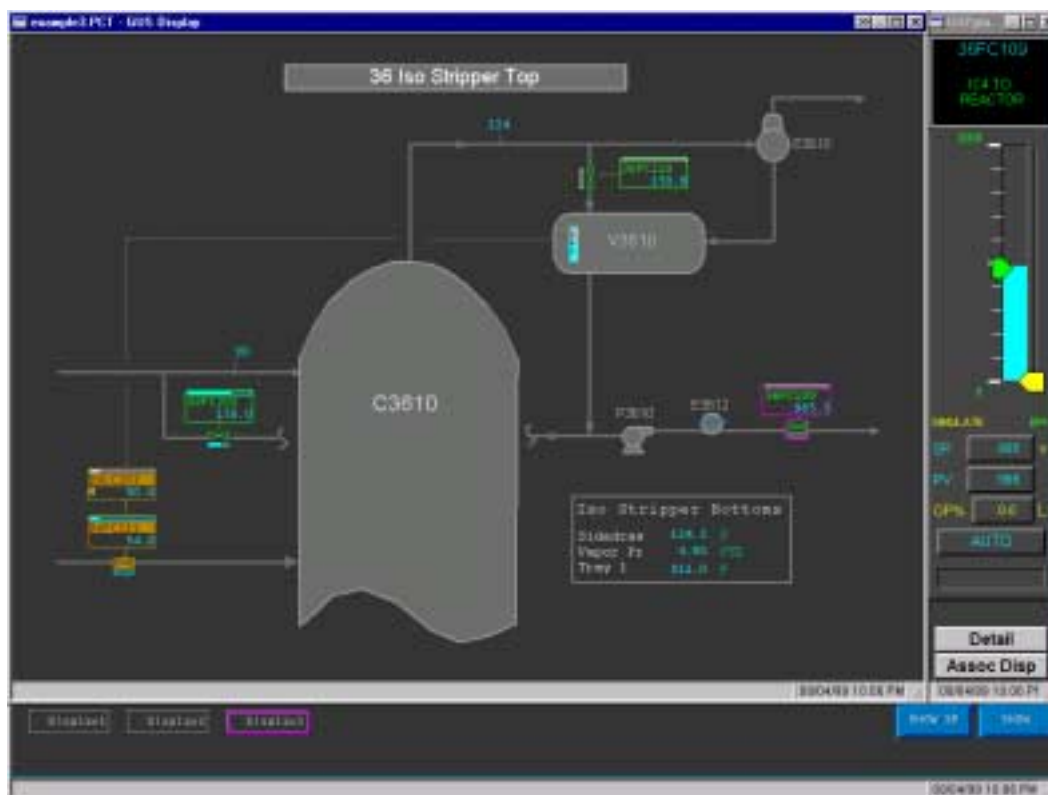
Reserved Display Database Variables GSP reserves the variables between 201-256 of all variable types of the GUS Local Display Database (e.g., ENT256, BOOL256). To avoid conflict with the GSP display components, users must not use the variables within this range.

Refer to “Reserved GUS Display Database Variables” in the Appendix for a complete list of display database variables currently used by GUS Solution Pack.

Quick Start

This section provides a step-by-step tutorial to help the user start building GUS displays using the “GUS Solution Pack” with minimum effort. At the end of the tutorial, you will have a working user interface managed under a SafeView configuration, a customized task panel with navigation buttons to call-up process displays. You will also be able to call-up a faceplate when selecting a point from the process display.

Below is a figure showing the process display, task panel, and faceplate on a single-screen GUS at run-time.



For more complete details, please refer to the specific sections in this document listed below:

Building Process Displays

Building Task Panels

SafeView Configurations

Before You Start

Before you start to build displays using the GUS Solution Pack, review the topics below. They contain information that will help you to plan ahead in order to avoid rework.

Display Resolution

Determine the Display Resolution of the GUS on which the displays will be used. For new projects, you may want to investigate the options before you decide. It is important to decide this before you build displays because it will determine which SafeView configuration to use, and therefore, affect the display size to build. Currently, GSP supports two resolutions: 1280 x 1024 and 1024 x 768. The main difference between the two SafeView configurations is the faceplate to be used under that resolution. Look at the faceplate provided below to decide which resolution to implement.

Faceplate

GSP provides a choice of faceplates to handle point manipulation.

GSPplate_fp1_127x871.pct	This is a GUS display with a Honeywell GUS Faceplate Control and a Honeywell GUS Trend Control. It is designed to work under 1280 x 1024 Display Resolution.
GSPplate_fp2_123x626.pct	This is a GUS display with a Honeywell GUS Faceplate Control. It is designed to work under 1024 x 768 Display Resolution.

The user must select one of the above displays to implement throughout an operator interface if the default GSP configuration is used. If you decide to develop a customized faceplate to be used under the GSP environment, it can be implemented through the GSP Configuration Utility. Refer to the GSP Configuration Utility section in this document for more detail.

To use the default faceplate display:

Copying the selected faceplate from the \Program Files\Honeywell\TPS\GSP\Display folder to a folder that is included in the Pathname Catalog on the station.

Renaming the file to GSPplate.pct. All point selection from GSP process displays will look for this filename.

Below are sample bitmaps of the faceplates:

GSPplate_fp1_127x871	GSPplate_fp2_123x626
	

Display Templates

Process Display Template

Create a display work folder on your system.

Open the **DisplayTemplate.pct** in the \Program Files\Honeywell\TPS\GSP\Template folder. Press “YES” to acknowledge that the template is a read-only file.

Check the size of the display on the “Size” tab of the “Display Properties” dialog window. Change the “Display Size” to match the Build-time Low-Level Display size determined according to the following table.

Delete the ResizeReminder object.

Use “Save As” to save the file in the display work folder.

Make your Display Template read-only via the file’s Properties page of NT Explorer to avoid it from being over-written by mistake.

Task Panel Template

Open the TaskPanelTemplate.pct in the \Program Files\Honeywell\TPS\GSP\ Template folder. Press “YES” to acknowledge that the template is a read-only file.

Check the size of the template on the “Size” tab of the “Display Properties” dialog window. Change the “Display Size” to match the Task Panel Display size determined according to the following table.

Delete the ResizeReminder object.

Use “Save As” to save the file in the display work folder.

Make your Task Panel Template read-only to avoid it from being over-written by mistake.

Display Size at Build-Time

GSP provides recommendations on Display Sizes for both single-screen and dual-screen GUS. Refer to “SafeView Configurations” section for more details. For quick start, use the default values below:

Display Size	Display Resolution	1280 x 1024	1024 x 768
	High-Level Display (top-screen)	1014 x 779	1012 x 718
	Low-Level Display (bottom-screen or single-screen)	908 x 697	883 x 626
Task Panel	Using faceplate GSPplate_fp1_127x871.pct	1014 x 62	N/A
	Using faceplate GSPplate_fp2_123x626.pct	N/A	1012 x 71

Build Process Display

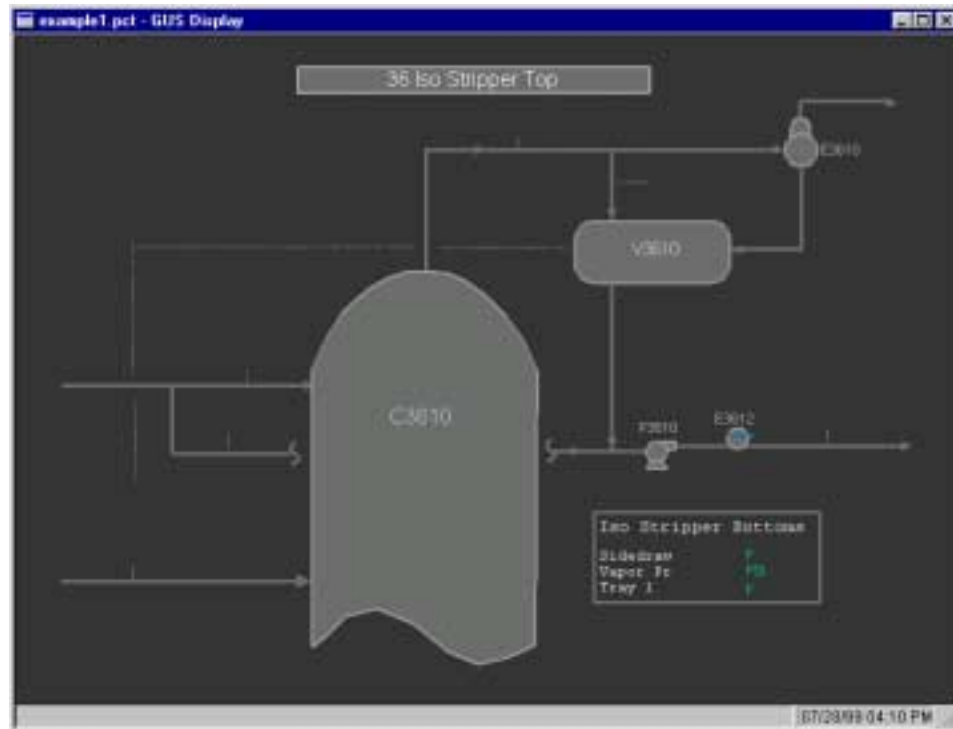
Open the Process Display Template in the display work folder. Press “YES” to acknowledge that the template is a read-only file if needed.

Change the text on the title object via the “Text” tab on the Object Properties dialog window of “Title” object.

Draw a few static objects (e.g., vessel, lines, etc.).

Validate and Save the display using a **different filename** (e.g., example1.pct). **Do not write-over the template.**

Run the display. Below is example1.pct at run-time. Note that the background color was modified for printing purpose, and may not match what you see on the screen.



Using the display you built above (e.g., example1.pct) as a starting point, continue with the following steps to add GSP display components:

Select View/Invisible Objects.

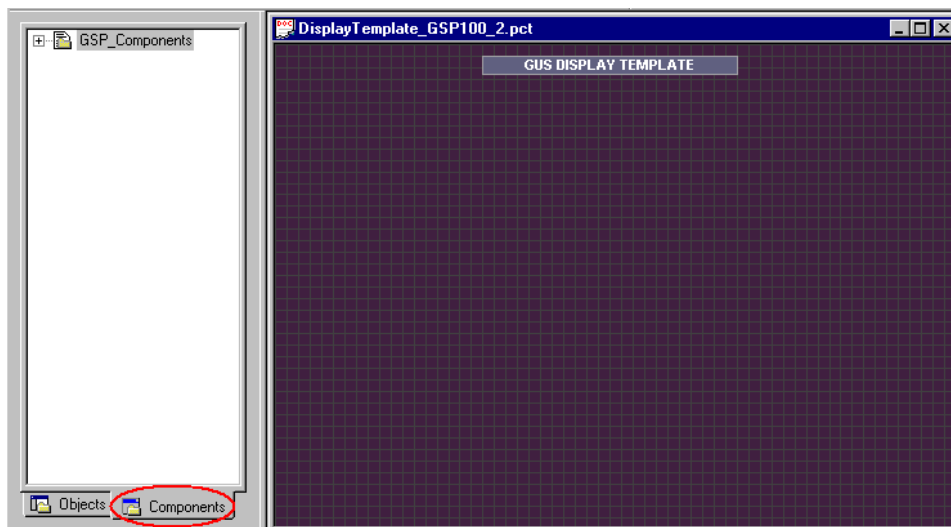
Two methods to add GSP display components exist:

1. Select display component from GSP Component Library.
2. Select display component using Insert/Display from GUS Display Builder menu bar.

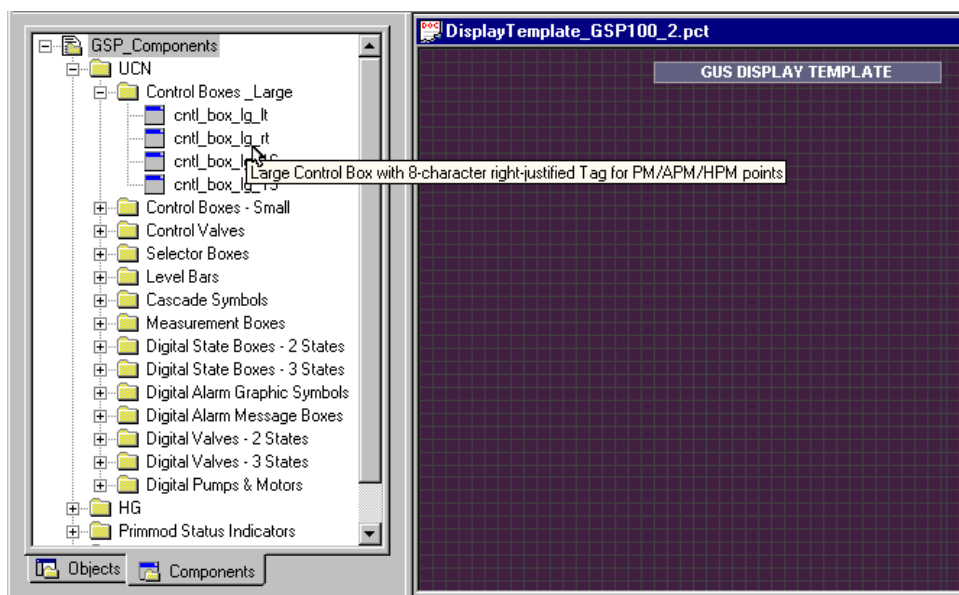
Select Display Component From GSP Component Library

Select GUS Explorer on the View menu (or select the GUS Explorer button on the Standard toolbar). Either the Object Browser or the Component Explorer window is displayed.

If the Object Browser is displayed, select the Components tab. A view of the GSP Component Library is displayed.



The Components Library is displayed in the standard Windows hierarchical tree format. Expand the tree to gain access to the individual display components you want to insert to the display. A one-line description (a tooltip) is displayed when the cursor is left on a display component.

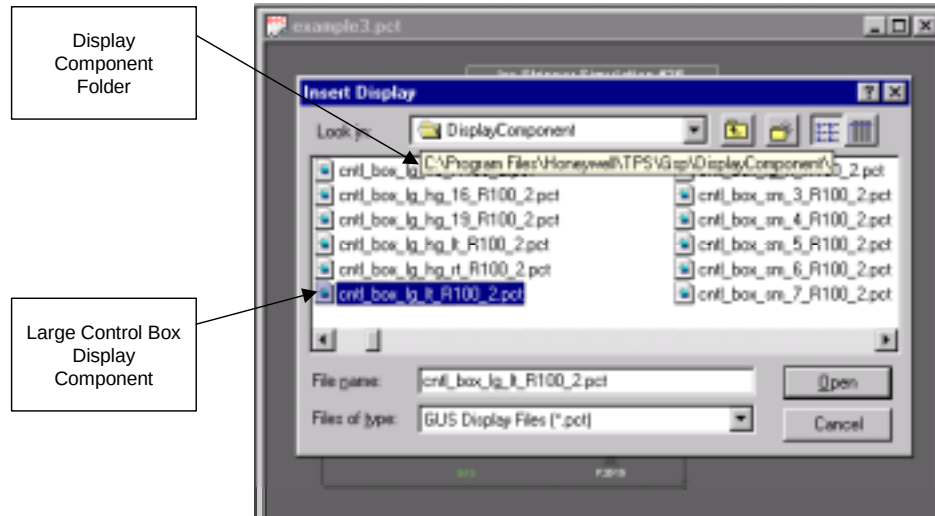


Double-click the component you want to insert. The Display Object Properties page is displayed, and you are prompted to enter the parameters required by the display component.

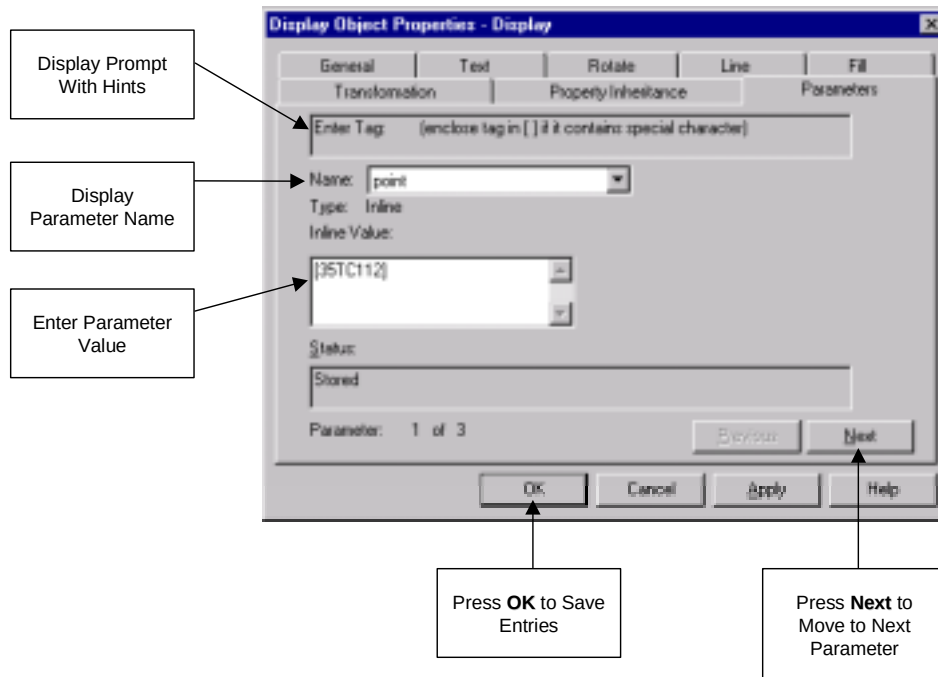
Select Display Component Using Insert/Display on GUS Display Builder Menu

From the GUS Display Builder menu bar, select Insert/Display.

The following figures and steps explain the insertion of a large control box (e.g., cntl_box_lg_lt.pct) using this method.



Enter a valid Tag for display parameter “point.” It must be an existing point on the TPN system, with the correct point type that the particular display component can handle. For quick start, skip the remaining display parameters.



Insert a few GSP display components (e.g., cntl_box_lg_lt, cntl_vlv_hor, casc_line, cntl_vlv_ver, casc_continue, and and casc_arrow) using either the GSP Component Library or Insert/Display from the GUS Display Builder menu bar. (You may want to try both methods.)

Validate and Save the display using a different filename (e.g., example2.pct). This is to provide multiple displays at the end of the tutorial in order to demonstrate the navigation feature in Task Panel later. Normally you don't need to use a different filename.

If you encounter validation errors, check to make sure that the tags you entered are existing points on the TPN system, and they are connected to the point types that the display components can handle. Refer to the specific component in the "Display Component Reference" for point type information.



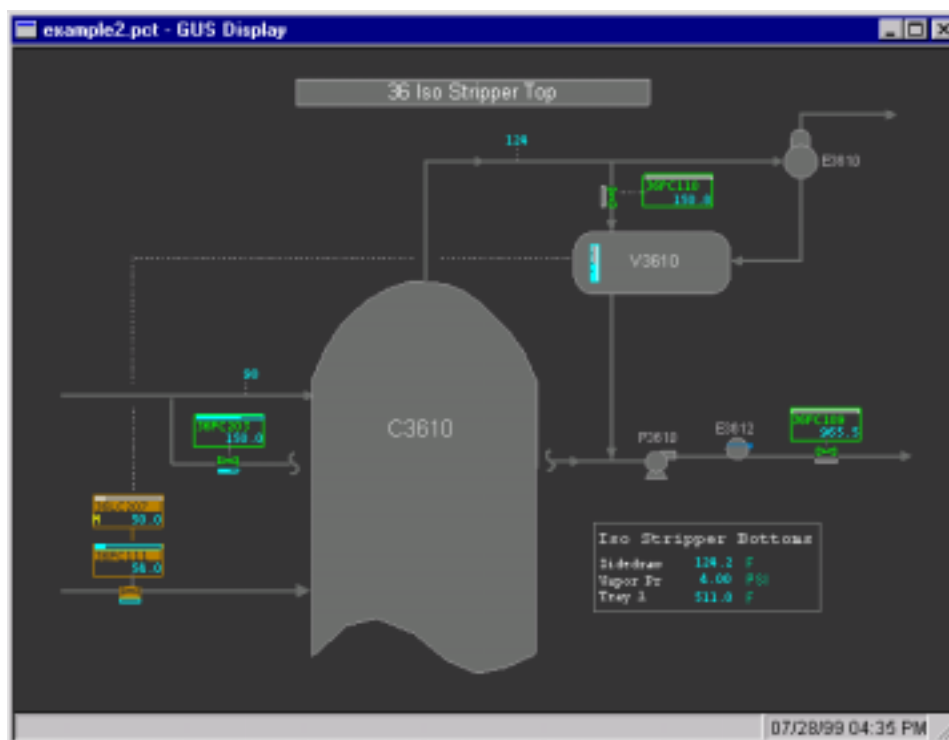
Attention

The validation process gives a warning message when a null primmod collector is encountered in a script. Ignore these validation warnings. GSP components handle null primmod collectors at runtime; thus, no errors concerning null primmod collectors will occur during the running of a GSP display.

Expect this type of warning if you have not assigned a primmod reference to all display parameters concerned with primmods in the following classes of GSP display components:

- Aggregate Primmod Status Indicators
 - Primmod Status Indicators
 - High-Level Navigation Buttons with Small Primmod Status Box
 - High-Level Navigation Buttons with Aggregate Primmod Status Box
 - Low-Level Navigation Buttons with Small Primmod Status Box
 - Low-Level Navigation Buttons with Aggregate Primmod Status Box
-

Run the display. Following is example2.pct at run-time.



Using the display you built above (e.g., example2.pct) as a starting point, continue with the steps below.

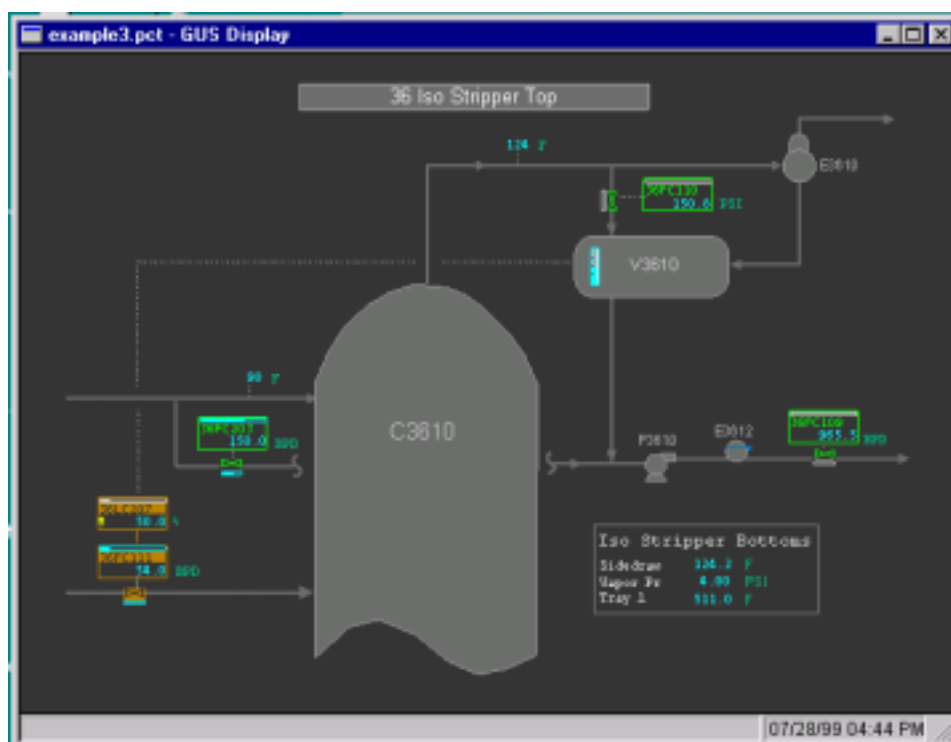
Add a few finishing touches (e.g., engineering units).

Validate and Save the display using a different filename (e.g., example3.pct).

Run the display. Below is example3.pct at run-time.

Quick Start - Build Process Display

You should be able to monitor the point status and values on live TPN data. Refer to “GSP Color Constants” in the Appendix for default color settings.



Build Task Panel With Navigation

The Task Panel is a powerful, user-configured window used in the GUS Solution Pack that can contain an alarm overview, navigation-to-displays, and system tasks. The alarm overview and navigation-to-displays work together to provide a graphics-based alarm management system that enables the operator to acknowledge alarms and take action from the same display. User-configured PRIMMOD associations provide aggregate alarm status of displays through GSP objects; by selecting the target, the operator navigates to the display to possibly see patterns in the alarm data and take action. The Task Panel is a convenient place to house objects for handling "system functions" such as objects that change the content of the displays (show/hide text), or objects that open other applications.

Below are the steps to illustrate how to build a customized Task Panel:

Open the Task Panel Template in the display work folder. Press "YES" to acknowledge that the template is a read-only file if needed.

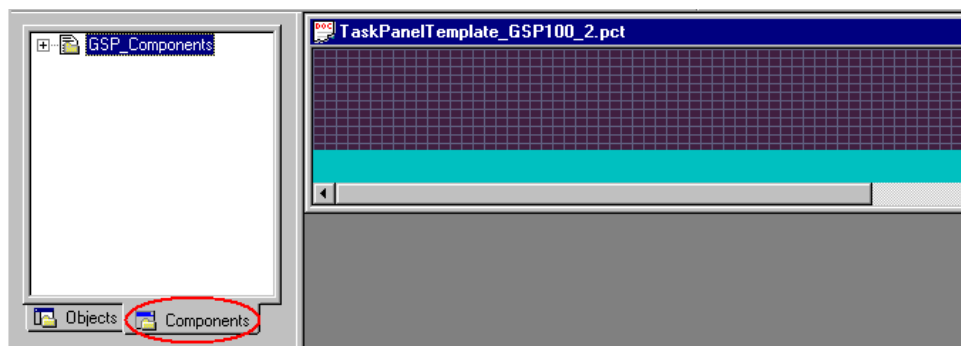
Select View/Invisible Objects.

Two methods for adding GSP display components exist:

1. Select display component from the GSP Component Library.
2. Select display component using Insert/Display from the GUS Display Builder menu bar.

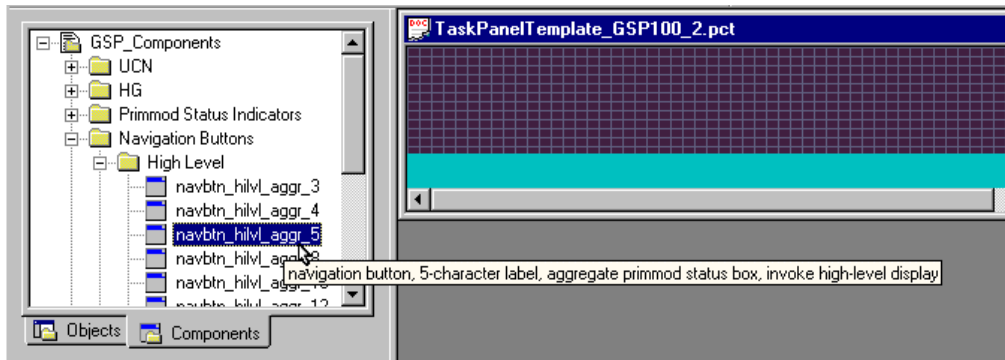
Select GUS Explorer on the View menu (or select the GUS Explorer button on the Standard toolbar). Either the Object Browser or the Component Explorer window is displayed.

If the Object Browser is displayed, select the Components tab. A view of the GSP Component Library is displayed.



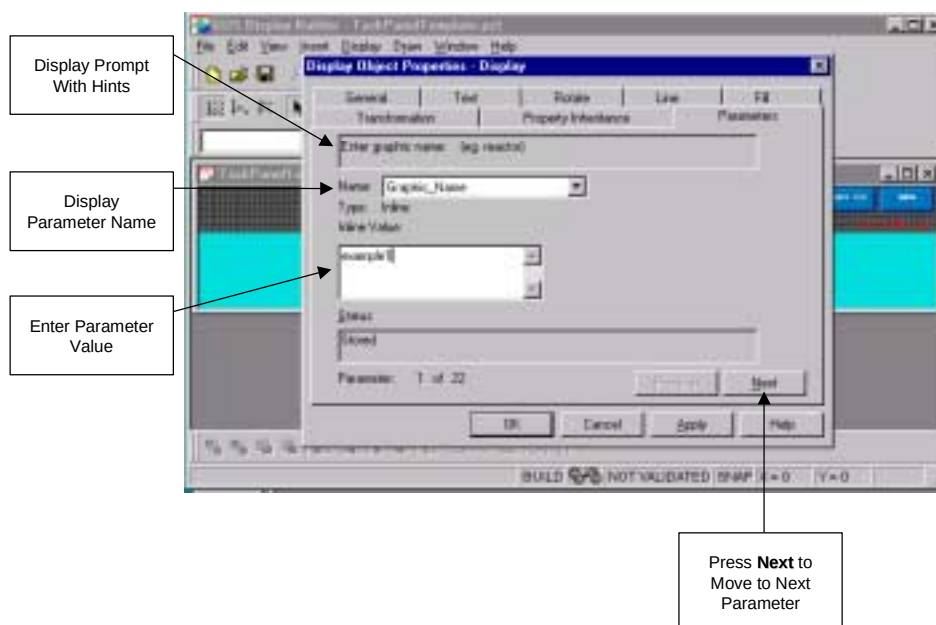
Quick Start - Build Task Panel With Navigation

The Components Library is displayed in the standard Windows hierarchical tree format. Expand the tree to gain access to the individual display components you want to insert to the display. A one-line description (a tooltip) is displayed when the cursor is left on a display component.



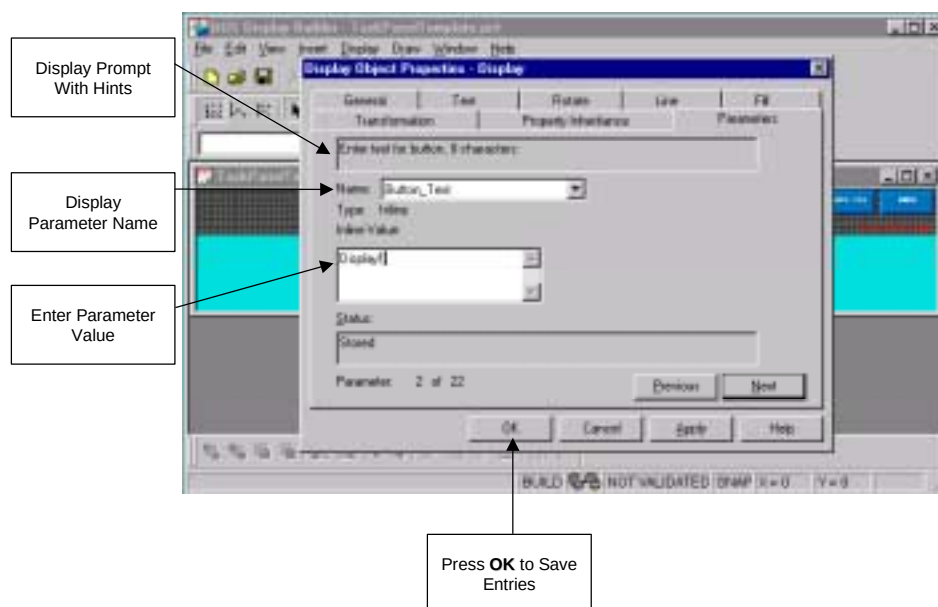
Double-click the component you want to insert. The Display Object Properties page is displayed, and you are prompted to enter the parameter required by the display component.

- Enter the Graphic_Name display parameter. Use the simple filename and allow the Pathname Catalog to locate the file. Do not enter the full pathname of the graphic filename. This provides a lot more flexibility to where the displays need to reside on your system. If the display is a GUS display, you may enter the extension (.pct). Make sure that you use the same display naming convention (e.g., “crudetower” or “crudetower.pct”) when entering the graphic name in an object’s display parameter. During runtime, if the display can’t be found, an error message will be displayed in task panel instead of searching through the TPN database.

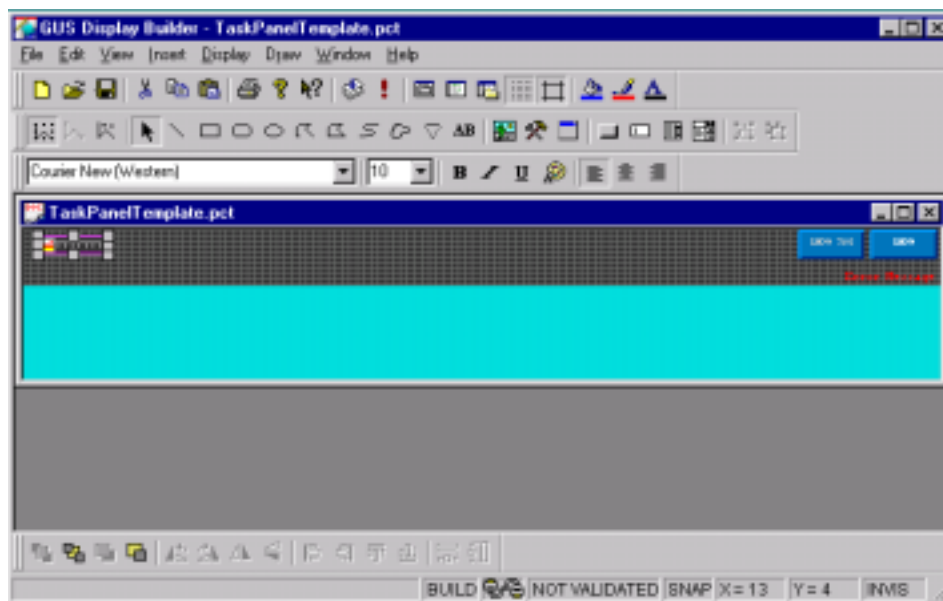


- Select “Next”, enter the Button_Text display parameter, then click OK. For quick start, skip the remaining display parameters regarding PRIMMOD flags.

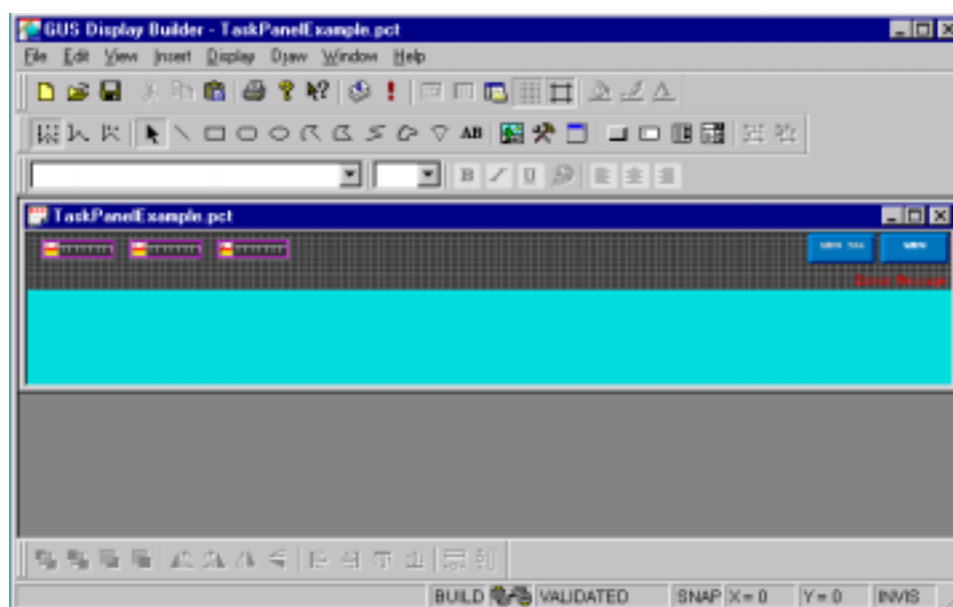
Quick Start - Build Task Panel With Navigation



- Locate the position to insert the button.



Add two more navigation buttons and connect them to example2 and example3.



Validate and Save the display using a **different filename** (e.g., TP_TEST.pct). **Do not write-over the template.**



Attention

The validation process gives a warning message when a null primmod collector is encountered in a script. Ignore these validation warnings. GSP components handle null primmod collectors at runtime; thus, no errors concerning null primmod collectors will occur during the running of a GSP display.

Expect this type of warning if you have not assigned a primmod reference to all display parameters concerned with primmods in the following classes of GSP display components:

- Aggregate Primmod Status Indicators
 - Primmod Status Indicators
 - High-Level Navigation Buttons with Small Primmod Status Box
 - High-Level Navigation Buttons with Aggregate Primmod Status Box
 - Low-Level Navigation Buttons with Small Primmod Status Box
 - Low-Level Navigation Buttons with Aggregate Primmod Status Box
-

Quick Start - Configure SafeView

Copy the process displays and task panel display from the display work folder to a folder that is included in the Pathname Catalog on the station.

Run the display.



Click on one of the navigation buttons, it will highlight the selected button and call-up the referenced display.

Configure SafeView

Identify the GSP SafeView configuration template to use according to the Display Resolution and faceplate that you chose before you started building. These templates are located in \Program Files\Honeywell\TPS\GSP\Template folder.

The available options are:

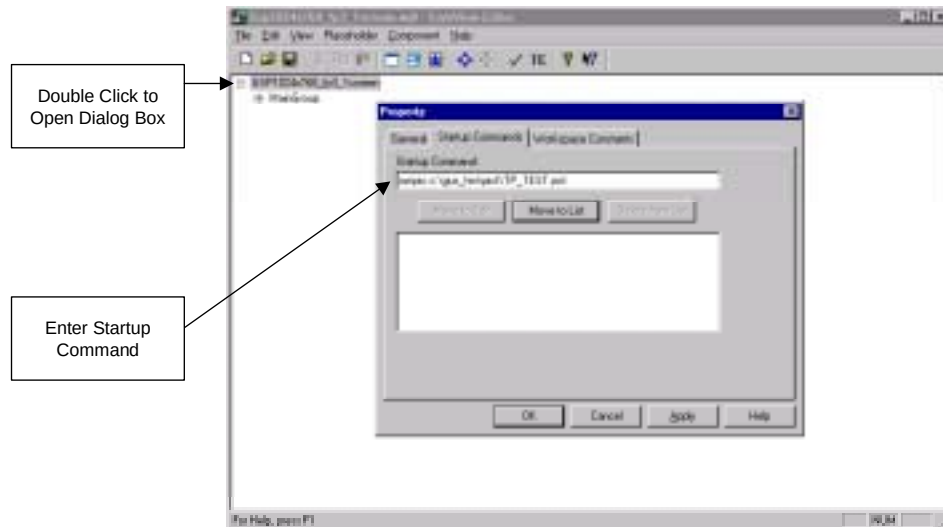
Display Resolution	Faceplate	SafeView Template
1280 x 1024	GSPplate_fp1_127x871	GSP1280x1024_fp1_1screen.wdl
1280 x 1024	GSPplate_fp1_127x871	GSP1280x1024_fp1_left_right.wdl
1280 x 1024	GSPplate_fp1_127x871	GSP1280x1024_fp1_top_bottom.wdl
1024 x 768	GSPplate_fp2_123x626	GSP1024x768_fp2_1screen.wdl
1024 x 768	GSPplate_fp2_123x626	GSP1024x768_fp2_left_right.wdl
1024 x 768	GSPplate_fp2_123x626	GSP1024x768_fp2_top_bottom.wdl

Open the selected SafeView template using the SafeView Workspace Editor.

Double click on the Workspace Title.

Select the “Startup Commands” tab.

Add a command to run the Task Panel Display that was created earlier.



Save the configuration using a **different filename** (e.g., SafeView_test.wdl). **Do not write-over the template.**

You are now ready to test the display environment:

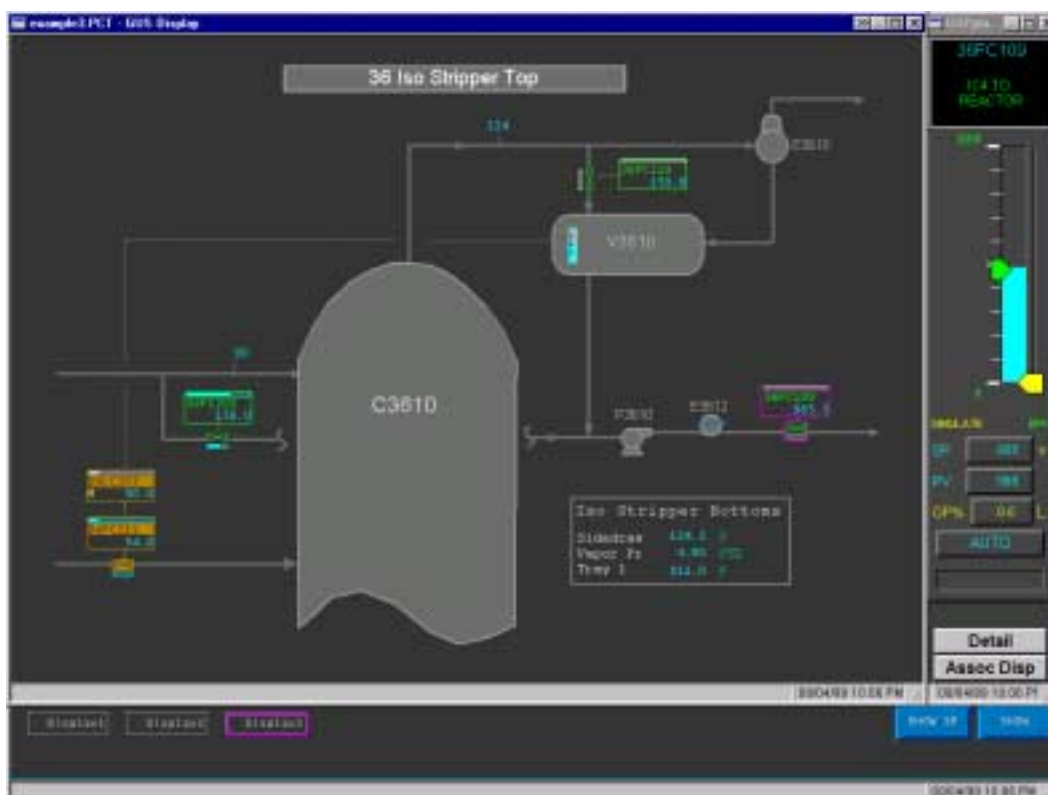
Load the above SafeView configuration workspace using SafeView.

Click on any navigation buttons in the task panel to call-up the referenced process display.

Click on any selectable display component in the process display to call-up the faceplate.

Quick Start - Configure SafeView

A figure showing the process display, task panel, and faceplate on a single-screen GUS at run-time follows. You should be able to monitor the point status and values on live TPN data and manipulate the point via faceplate. Refer to “GSP Color Constants” in the Appendix for default color settings.



SafeView Configurations

Overview

Six SafeView configurations are provided that integrate the various displays into an integrated process operator's interface. Configurations supporting 1280 x 1024 and 1024 x 768 display resolutions are provided for single-screen, left-right-dual-screen and top-bottom-dual-screen configurations. The main difference between the SafeView configurations is the faceplate used under that resolution. The faceplate for 1280 x 1024 resolution shows a trend of the point, while the faceplate for 1024 x 768 resolution does not. Refer to "Faceplate Display" section in the "GSP-Provided Displays Reference" for the faceplate provided to decide which resolution to implement.

For best result, GSP recommend setting the Windows Taskbar property on GUS to "Not Always on top" (un-check the checkbox). This allows GSP Task Panel to move on top of the Windows Taskbar whenever user click on it.



Attention

Warning: The GSP components were built and tested with the default font, font size and display sizes. If you decide to use font or display size other than the default, make sure all information displayed by the display components are clearly visible when running under your safeview configuration. For example, the top part of a character "T", the bottom part of a character "Q".

The following describes the default SafeView configurations provided with the product. The user may modify these configurations as applicable by modifying the appropriate wdl file(s).

Dual-Screen Configuration

Top-Bottom-Dual-Screen

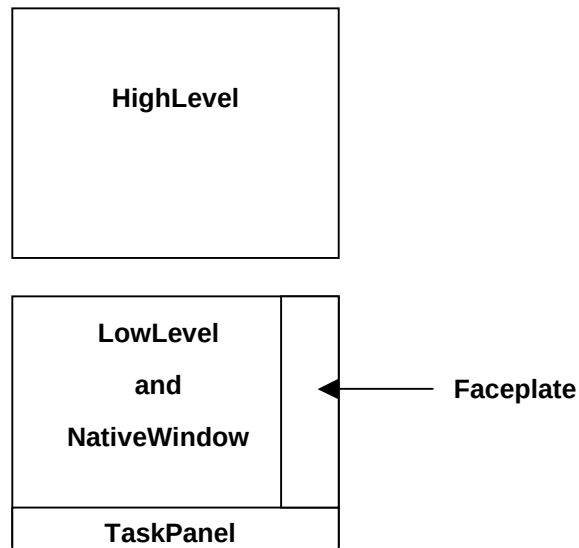
The workspace provided by GSP assumes that the monitor on the top is defined to be the primary monitor under the Windows Display Settings (available in the Control Panel). To ensure Top-Bottom-Dual-Screen SafeView Configuration works properly, the Display Settings have to be set as follows:

1. The layout of the top and bottom monitors in the Display Setting must match their physical layout.
2. The resolutions of the top and bottom monitors must match the corresponding SafeView workspace configuration files.

The top-bottom-dual-screen SafeView configuration is shown below, indicating each window specification name, as defined in the SafeView configuration.

The primary monitor always displays "HighLevel" displays and secondary monitor displays Native Window, Faceplate and TaskPanel.

If the user chooses to configure the bottom monitor as the primary monitor, the Top-Bottom-Dual-Screen SafeView workspace configuration file should be modified to reflect the different origin position.



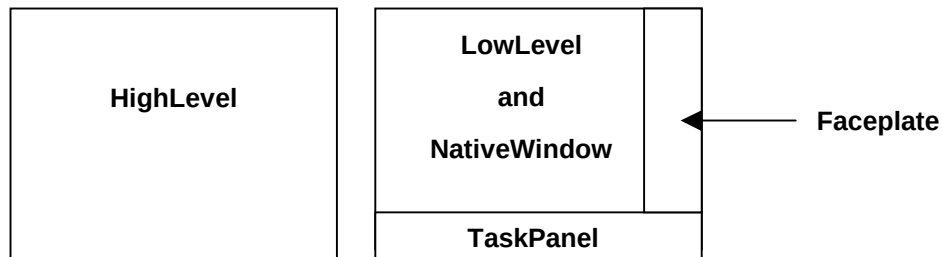
Left-Right-Dual-Screen

The workspace provided by GSP assumes that the monitor on the left is defined to be the primary monitor under the Windows Display Settings (available in the Control Panel). To ensure Left-Right-Dual-Screen SafeView Configuration works properly, the Display Settings have to be set as follows:

1. The layout of the left and right monitors in the Display Setting must match their physical layout.
2. The resolutions of the left and right monitors must match the corresponding SafeView workspace configuration files.

The left-right-dual-screen SafeView configuration is shown below, indicating each window specification name, as defined in the SafeView configuration.

The primary monitor always displays "HighLevel" displays and secondary monitor displays Native Window, Faceplate and TaskPanel.



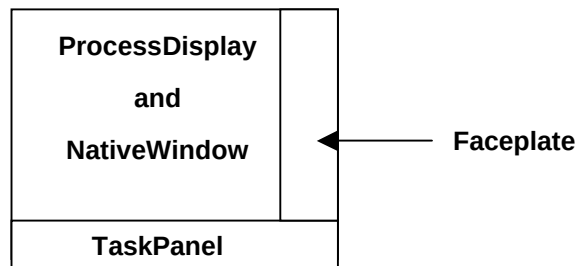
The display category assigned to each window has the same name as the window specification name. The faceplate displays and task panel display templates have their category property set to "Faceplate" and "TaskPanel," respectively. The user must set the category property of the process displays to "HighLevel" or "LowLevel" when building them. The standard display applications (Group and Alarm Summary) are also assigned to the LowLevel window.

Native Window is a separate window specification that has the same size and coordinates as the LowLevel window. It is assigned to the Native Window application, and is used to show all Native Window displays. When a process display is invoked in the overlaying LowLevel window, if the Native Window is up and visible, it is possible to have the Native Window minimized by a GSP navigation button. Invoking a Native Window display restores the window to its initial size and overlays the LowLevel window. Minimization of the Native Window is accomplished by components in the GUS Solution Pack and is not an inherent function of GUS. Not closing the Native Window

when it is not in use improves call-up time of Native Window displays, because the Native Window application does not need to be started each time.

Single-Screen Configuration

The single-screen configuration is shown below, showing the window specification names, as defined in the SafeView configuration.



HighLevel and LowLevel category displays are both assigned to the Process Display window. The user must specify the category of process displays as “HighLevel” or “LowLevel” as in the dual-screen configuration. The standard display applications (Group and Alarm Summary) are also assigned to the Process Display window.

Native Window is a separate window specification, as in the dual-screen configuration. Faceplate and Task Panel are the same as in the dual-screen configuration.

Window and Display Sizing

The windows are sized such that the sizes of High-Level and Low-Level displays have the same height-to-width ratio. This allows either Low-Level displays to be shown in the HighLevel window and vice versa. The operator can manually select the window in which to invoke a display, overriding the category specification in the SafeView configuration.

In the SafeView configuration tables below, window size is specified for each window. The “Display Run Size” defines the area inside the window. For the default Windows NT configuration, the window frame, title bar, and status bar consume 50 pixels of height and 12 pixels of width. Therefore, the “Display Run Size” is 50 pixels less in height and 12 pixels less in width than the window size. Note that this is only true with the default Windows NT setting. If these are changed, SafeView configurations and display sizes must be recalculated accordingly.

Screen Resolution Considerations

The display components were originally built to be viewed at 1024 x 768 resolution. To preserve their actual size when using 1280 x 1024 screen resolution, the display size property must be 80% of the “Display Run Size” (as shown in the tables below) with the “zoom to fit (run-time)” set to true. When the display is run, it will fill the window and appear full size. In the configuration tables below, the “Display Build Size” field indicates the size to use when building the display. The “Display Run Size” field indicates the size of the display when it is run.

Customize SafeView Workspace

For applications other than the GSP displays, the user should add appropriate windows in the SafeView workspace and match the applications to the desired window.

SafeView Configuration Details

The following tables define the SafeView configurations. Since most of the configuration properties are common to single- and dual-screen configurations, common tables are shown for them with differences indicated where appropriate. For single-screen configurations, the Process Display window has the same properties as the LowLevel window in the dual-screen configuration except where noted in the tables. For each window specification, a coordinate table is provided for each resolution.

Match expressions are provided that handle the window assignment of high-level displays, low-level displays, the faceplate, task panel, and the Native Window, but the user must add match expressions to handle third party applications, including the GUS Group display application, as well as the GUS Alarm Summary Display, if desired.

GSP supplies SafeView workspace definition files that support the following configurations. The filenames are:

gsp1024x768_fp2_1screen.wdl – single-screen configuration for 1024 x 768 screen resolution

gsp1024x768_fp2_left_right.wdl – left-right-dual-screen configuration for 1024 x 768 screen resolution

gsp1024x768_fp2_top_bottom.wdl – top-bottom-dual-screen configuration for 1024 x 768 screen resolution

gsp1280x1024_fp1_1screen.wdl – single-screen configuration for 1280 x 1024 screen resolution

gsp1280x1024_fp1_left_right.wdl – left-right-dual-screen configuration for 1280 x 1024 screen resolution

gsp1280x1024_fp1_top_bottom.wdl – top-bottom-dual-screen configuration for 1280 x 1024 screen resolution

1024 x 768 Resolution Window Specifications

HighLevel Window Coordinates - 1024 x 768 Resolution

Property	Size or Coordinate	Comments
Initial Position	0, 0	
Maximized Position	N/A	Not sizable
Initial Size	1024 x 768	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	1012 x 718	
Display Build Size	1012 x 718	

HighLevel Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	Y	
Global Focus:		
single-screen	N	
dual-screen	Y	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

LowLevel (dual-screen) & ProcessDisplay (single-screen) Window Coordinates – 1024x768 Resolution

Property	Size or Coordinate	Comments
Initial Position: single-screen left-right top-bottom	0, 0 1024, 0 0, 768	
Maximized Position	N/A	Not sizable
Initial Size	895 x 676	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	883 x 626	
Display Build Size	883 x 626	

LowLevel (dual-screen) & ProcessDisplay (single-screen) Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	Y	
Global Focus:		
single-screen	N	
dual-screen	Y	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

Native Window Coordinates – 1024 x 768 Resolution

Property	Size or Coordinate	Comments
Initial Position: single-screen left-right top-bottom	0, 0 1024, 0 0, 768	
Maximized Position	N/A	Not sizable
Initial Size	895 x 676	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	N/A	
Display Build Size	N/A	

Native Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	N	Allows minimization
Set Maximized Position	N	Not sizable
Placeholder	N	
Closable	N	
Global Focus	N	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

Task Panel Window Coordinates – 1024 x 768 Resolution

Property	Size or Coordinate	Comments
Initial Position: single-screen left-right top-bottom	0, 651 1024, 651 0, 1419	
Maximized Position	N/A	Not sizable
Initial Size	1024 x 121	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	1012 x 71	
Display Build Size	1012 x 71	

Task Panel Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	N	
Global Focus	N	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

Faceplate Window Coordinates – 1024 x 768 Resolution

Property	Size or Coordinate	Comments
Initial Position: single-screen left-right top-bottom	889, 0 1913, 0 889, 768	
Maximized Position	N/A	Not sizable
Initial Size	135 x 676	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	123 x 626	
Display Build Size	123 x 626	

Faceplate Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	Y	
Global Focus	N	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

1280 x 1024 Resolution Window Specifications

HighLevel Window Coordinates - 1280 x 1024 Resolution

Property	Size or Coordinate	Comments
Initial Position	0, 0	
Maximized Position	N/A	Not sizable
Initial Size	1280 x1024	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	1268 x 974	
Display Build Size	1014 x 779	Display is built 80% of run-time size

HighLevel Window Properties - Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	Y	
Global Focus:		
single-screen	N	
dual-screen	Y	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

LowLevel (dual-screen) & ProcessDisplay (single-screen) Window Coordinates – 1280x1024 Resolution

Property	Size or Coordinate	Comments
Initial Position:		
single-screen	0, 0	
left-right	1280,0	
top-bottom	0, 1024	
Maximized Position	N/A	Not sizable
Initial Size	1147 x 921	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	1135 x 871	
Display Build Size	908 x 697	Display is built 80% of run-time size

LowLevel (dual-screen) & ProcessDisplay (single-screen) Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	Y	
Global Focus:		
single-screen	N	
dual-screen	Y	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

Native Window Coordinates – 1280 x 1024 Resolution

Property	Size or Coordinate	Comments
Initial Position: single-screen left-right top-bottom	0, 0 1280, 0 0, 1024	
Maximized Position	N/A	Not sizable
Initial Size	1147 x 921	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	N/A	
Display Build Size	N/A	

Native Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	N	Allows minimization
Set Maximized Position	N	Not sizable
Placeholder	N	
Closable	N	
Global Focus	N	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

Task Panel Window Coordinates – 1280 x 1024 Resolution

Property	Size or Coordinate	Comments
Initial Position: single-screen left-right top-bottom	0, 897 1280, 897 0, 1921	
Maximized Position	N/A	Not sizable
Initial Size	1280 x 128	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	1268 x 78	
Display Build Size	1014 x 62	Display is built 80% of run-time size

Task Panel Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	N	
Global Focus	N	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

Faceplate Window Coordinates – 1280 x 1024 Resolution

Property	Size or Coordinate	Comments
Initial Position: single-screen left-right top-bottom	1141, 0 2421, 0 1141, 1024	
Maximized Position	N/A	Not sizable
Initial Size	139 x 921	
Maximum Size	N/A	Not sizable
Minimum Size	N/A	Not sizable
Display Run Size	127 x 871	
Display Build Size	127 x 871	

Faceplate Window Properties – Same for Both Resolutions

Property	Y/N	Comments
Set Init Position	Y	
Set Init Size	Y	
Draggable	N	
Restrict Min Size	N	Not sizable
Restrict Max Size	N	Not sizable
Override Group Region	N	
No Minimize Button	Y	
Set Maximized Position	N	Not sizable
Placeholder	Y	
Closable	Y	
Global Focus	N	
Multiple Windows	N	
Sizable	N	
Always On Top	N	

Building Process Displays

This section explains how to use the GUS Solution Pack to build process displays. It assumes that the GSP users are familiar with basic features of Windows NT and the Honeywell GUS Display Builder.

Before You Start

Determine Display Size

Before you start to build displays using the GUS Solution Pack, make sure you have read the “SafeView Configurations” section. It contains information that will help you to plan ahead in order to avoid rework.

Below are the GSP recommended Display Sizes for single-screen and dual-screen GUS under two different Display Resolution settings. The display components were built to be viewed at 1024 x 768 resolution. To preserve their actual size when using 1280 x 1024 screen resolution, the display size property must be 80% of the “Display Run Size” with the “zoom to fit (run-time)” set to true. The size of the Low-Level Displays and the High-Level Displays should have the same height-to-width ratio. The remaining space is allocated for Task Panel and Faceplate windows.

Display Size at Build-Time	Display Resolution	1280 x 1024	1024 x 768
	High-Level Display (top-screen)	1014 x 779	1012 x 718
	Low-Level Display (bottom-screen or single-screen)	908 x 697	883 x 626

Setup Display Building Environment

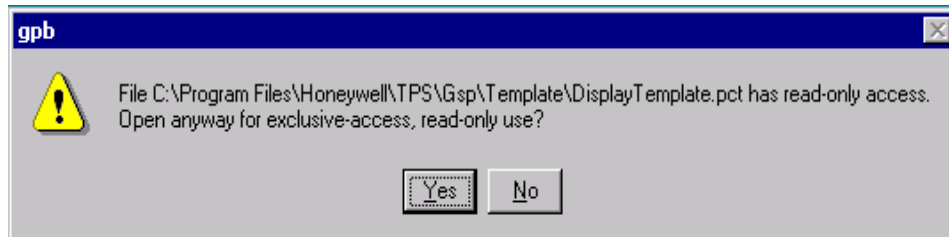
Before you start to build process displays, you should customized the master Display Template and make a set of customized display templates (e.g., High-Level and Low-Level displays) to use throughout the project. The main reason to customize the template is to set the category and the display size to match the SafeView configuration.

Building Process Displays - Before You Start

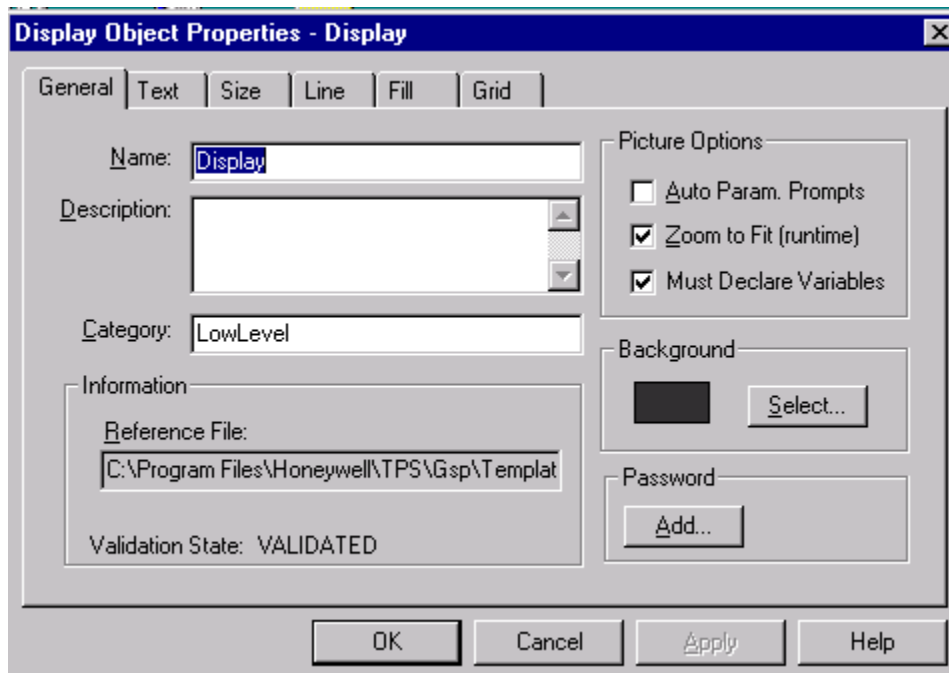
Follow the steps below to customize the template.

Create a display work folder on your system.

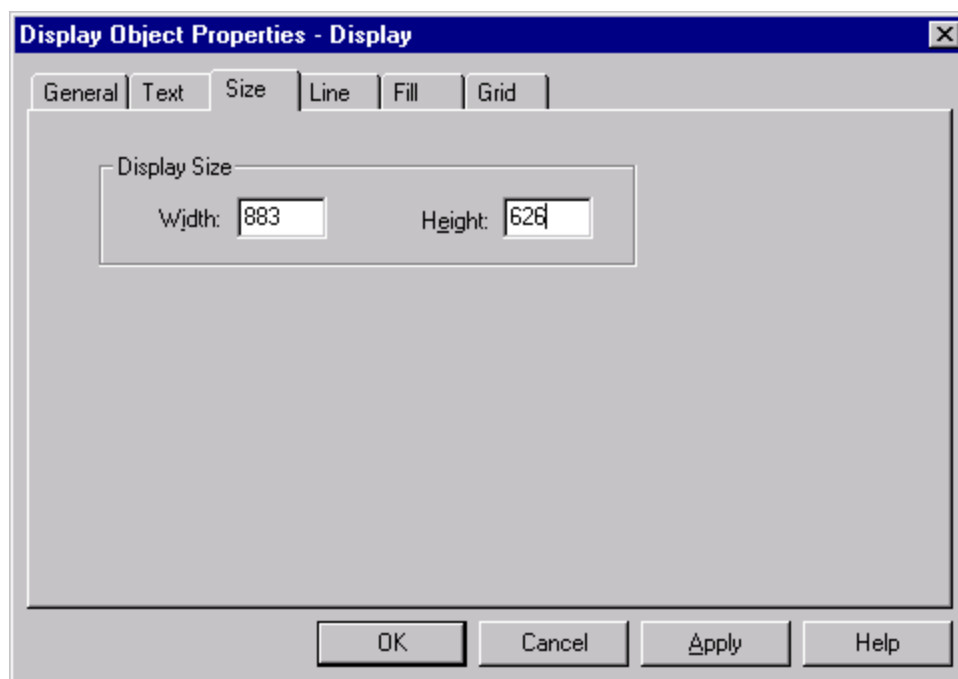
Open the **DisplayTemplate.pct** located in the \Program Files\Honeywell\TPS\GSP\Template folder. Click on “Yes” to acknowledge that the template is a read-only file.



Check the Category on the “General” tab of the “Display Properties” dialog window. The default is “LowLevel.” If you plan to build both High-Level and Low-Level displays, you should create two display templates – using category “HighLevel” and “LowLevel.” Note that the category is used in SafeView configuration to match the default window for each display. Refer to “Display Templates Reference” section for more detail.

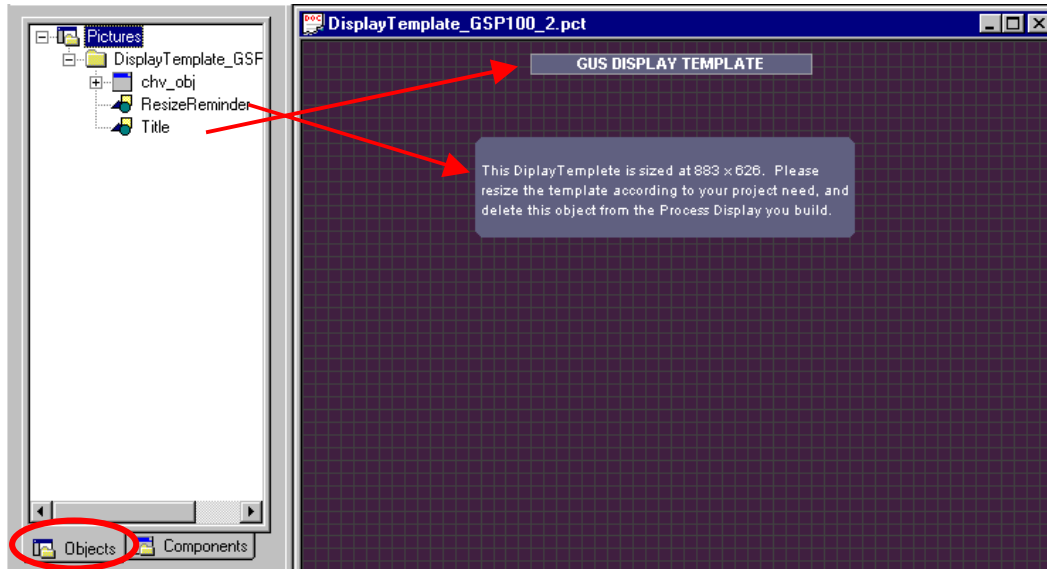


Check the size of the display on the “Size” tab of the “Display Properties” dialog window. Change the “Display Size” to match the configuration determined earlier.



Building Process Displays - Before You Start

Delete the ResizeReminder object.



Use “Save As” to save the file in the display work folder.

Make your Display Template read-only via the file’s Properties page of NT Explorer to avoid it from being over-written by mistake.

Reserved GUS Display Database

GSP reserves the variables between 201-256 of all variable types of the GUS Local Display Database (e.g., ENT256, BOOL256). To avoid conflict with the GSP display components, users must not use the variables within this range.

Refer to “Reserved GUS Display Database Variables” in the Appendix for a complete list of display database variables currently used by GUS Solution Pack.

Build Process Display

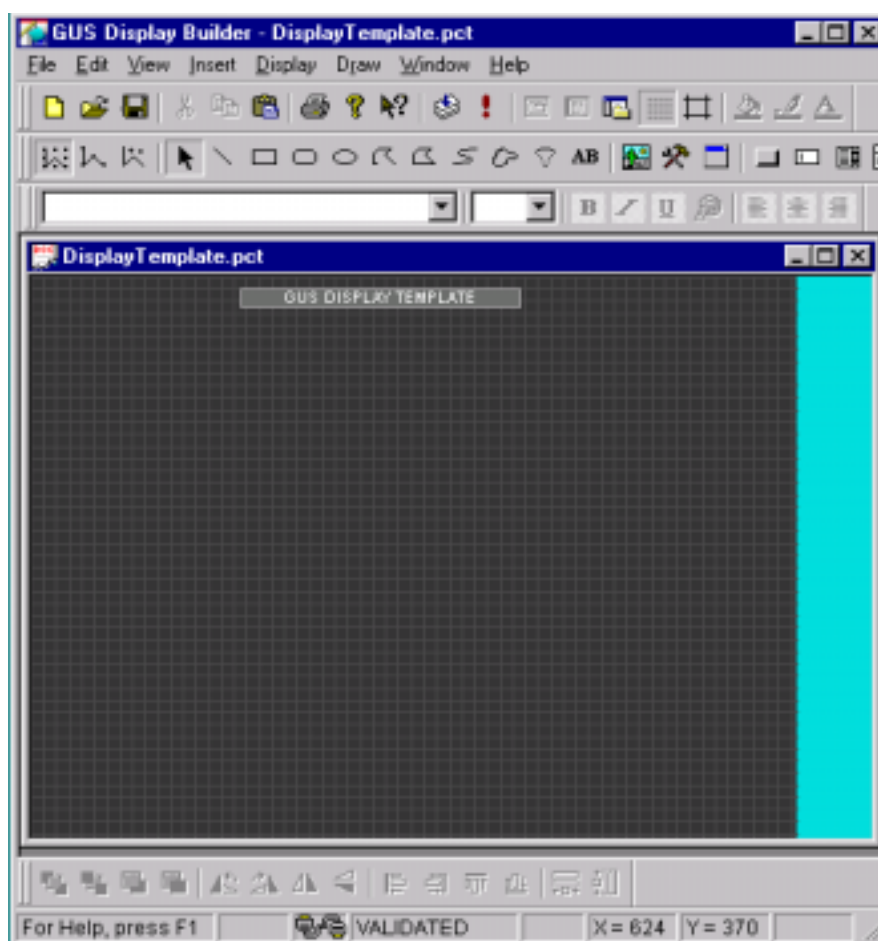
The process display may include any standard GUS display objects, controls, and scripts.

Caution:

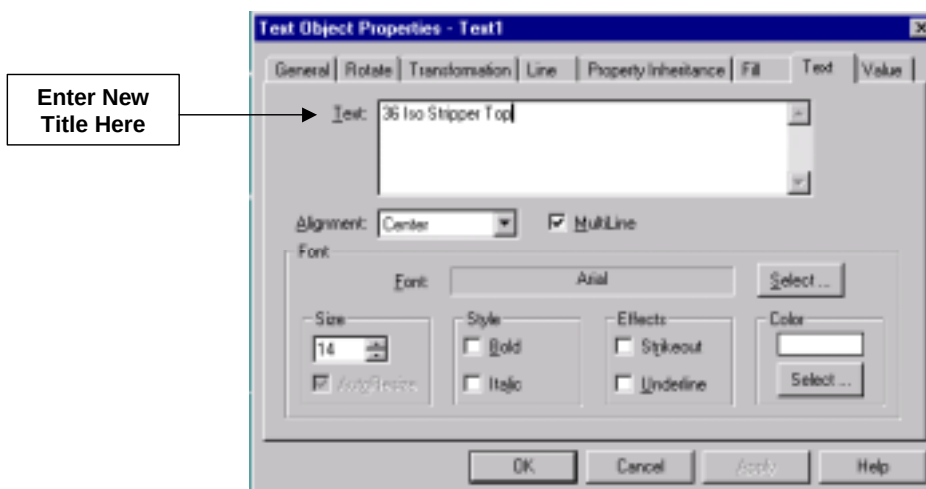
- Do not modify or delete the standard portion of the script or the embedded displays that come with the templates.
- Make sure that you do not modify the intended use of the reserved database variables. Refer to the Appendix for Reserved GUS Display Database Variables.

Draw Static Objects

Open one of the customized process display templates (e.g., High-Level, Low-Level) that you created earlier. It is critical to always start with a Display Template which contains the script and objects required for GSP to function properly.



Change the text of the title object or delete the title object from the display.
The figure below illustrates how to change the title to “36 Iso Stripper Top.”



Draw the static objects as needed (e.g., vessel, lines, etc.). Use low intensity color for all the static information so that they don't distract the operator from scanning the live data on the display.

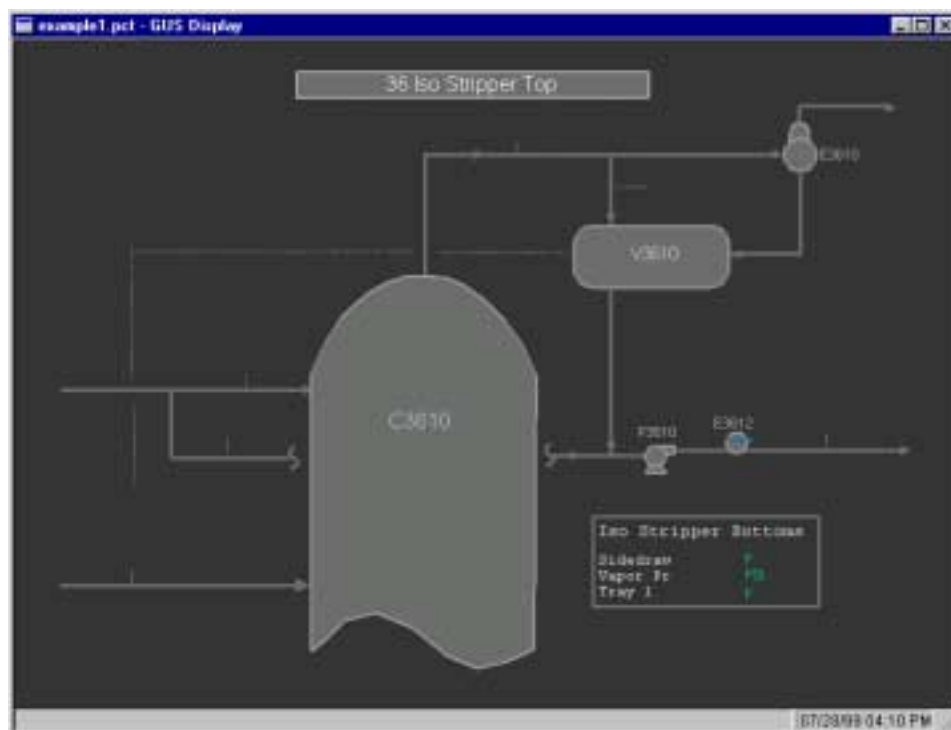
Validate and Save the display using a **different filename** (e.g., example1.pct).

Do not write-over the template.

Keep the filename within 8 characters if you intend to call-up the display from Native Window schematics or Keyboard buttons.

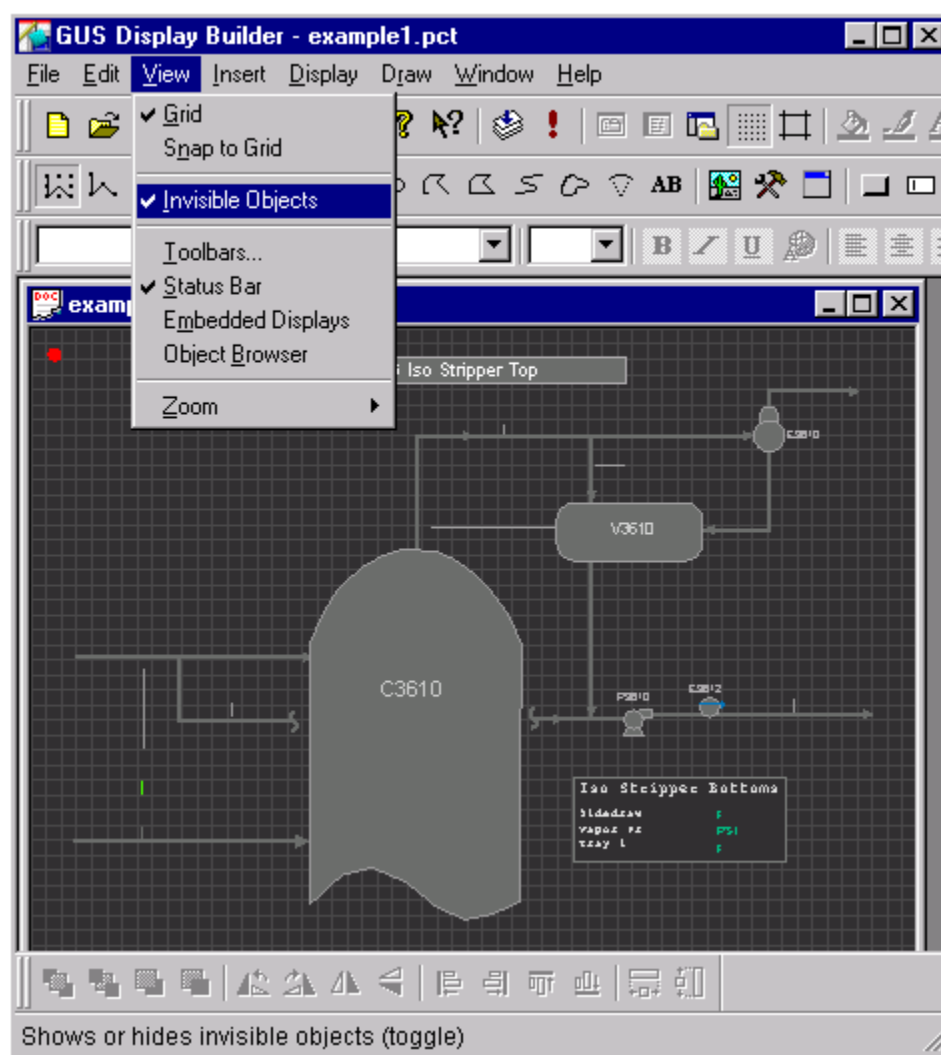
Building Process Displays - Build Process Display

Run the display. Below is example1.pct at run-time. (Note that the background color was modified for printing purpose and may not match what you see on the screen.)



Select View/Invisible Objects

Most of the GSP display components are initially invisible by design. The primary purpose is to eliminate the flickering of the dummy objects (before live data become available). Select “View Invisible Objects” during display build-time so that you can see all the GSP display components being inserted.



Insert GSP Display Components

Select an appropriate GSP display component to display information for a point on the TPN system. Refer to “Display Component Reference” section for details of each GSP display component and the point types it can handle.

Warning:

It is important that you use the appropriate component to represent each point, otherwise, the display will either fail to validate, or worse, display inaccurate information to the operator. Refer to the individual component in the “Display Components Reference” section for the correct point type for the particular component.

For example, if a 2-state digital state box was used to display information of a 3-state digital composite point, or a highway digital valve component was used to display information of a PM digital valve, the display may show wrong information. The similar situation holds true for Native Window schematics.

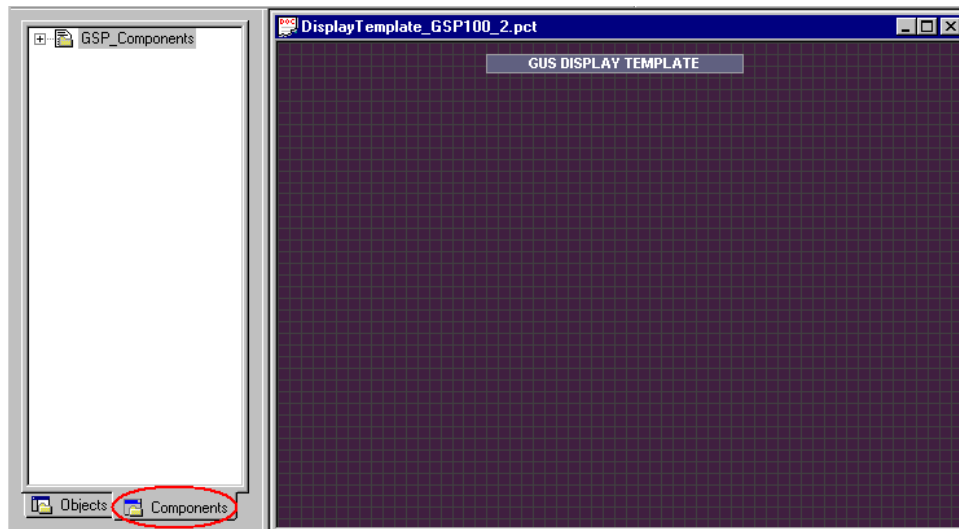
Two methods to add GSP display components exist:

- Select display component from GSP Component Library.
- Select display component using Insert/Display from GUS Display Builder menu bar.

Select Display Component From GSP Component Library

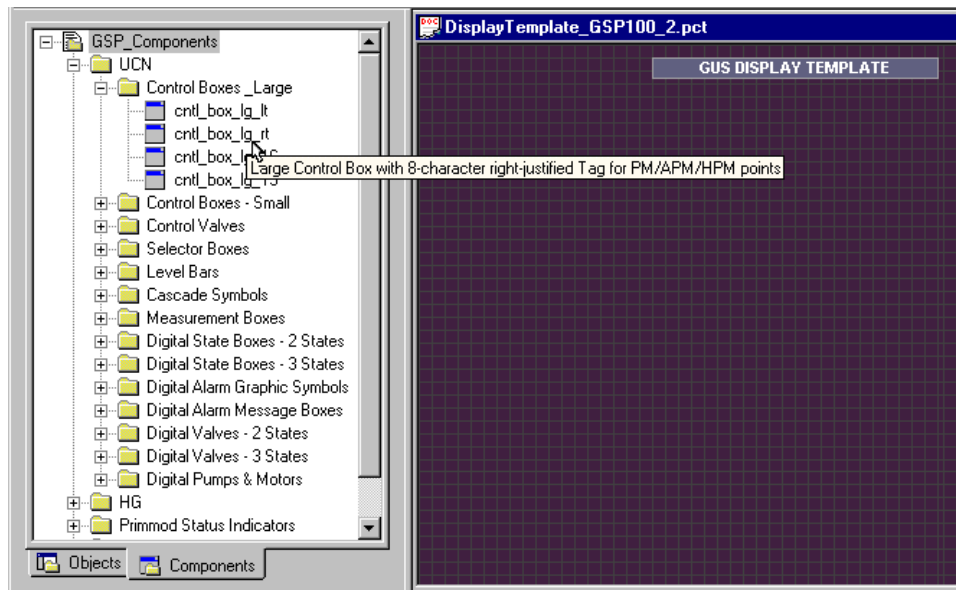
Select GUS Explorer on the View menu (or select the GUS Explorer button on the Standard toolbar). Either the Object Browser or the Component Explorer window is displayed.

If the Object Browser is displayed, select the Components tab. A view of the GSP Component Library is displayed.



Building Process Displays - Build Process Display

The Components Library is displayed in the standard Windows hierarchical tree format. Expand the tree to locate the individual display components you want to insert to the display. A one-line description (a tooltip) is displayed when the cursor is positioned on a display component.

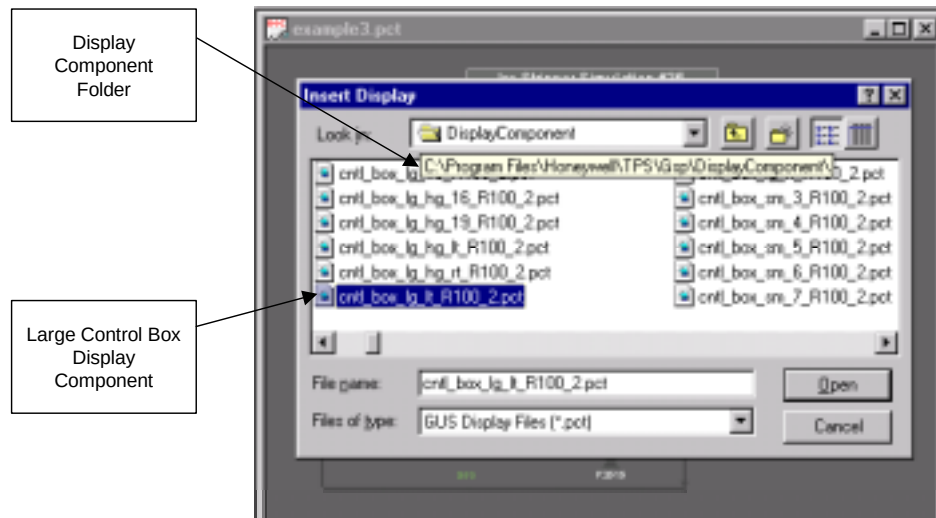


Double-click the component you want to insert. The Display Object Properties page is displayed, and you are prompted to enter the parameters required by the display component.

Select Display Component Using Insert/Display From GUS Display Builder Menu Bar

From the main GUS Display Builder menu bar, select Insert/Display.

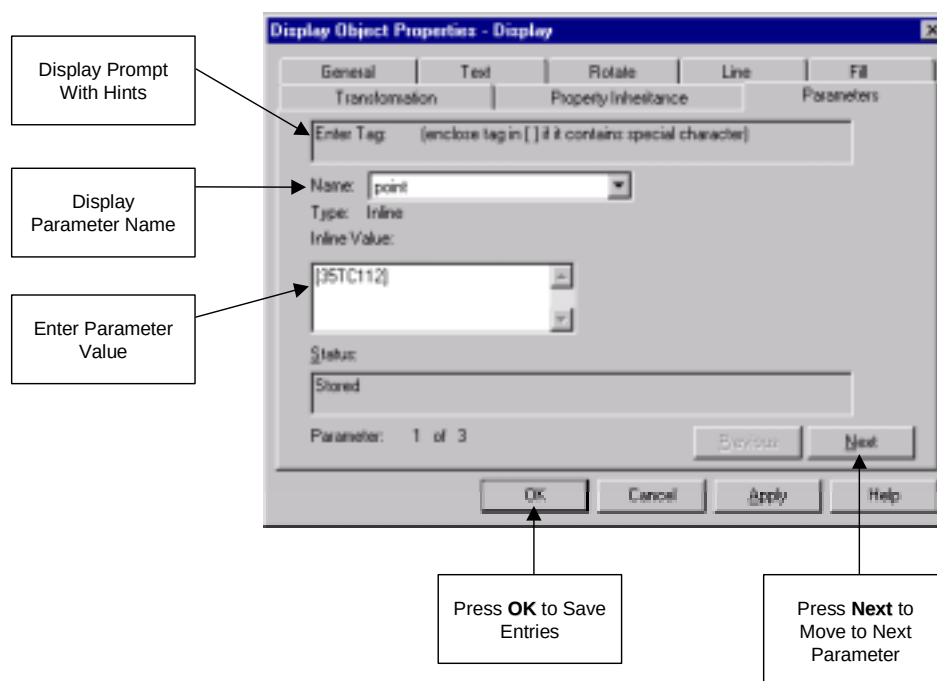
The following figures and discussion explain the insertion of a large control box (e.g., cntl_box_lg_lt.pct) using this method.



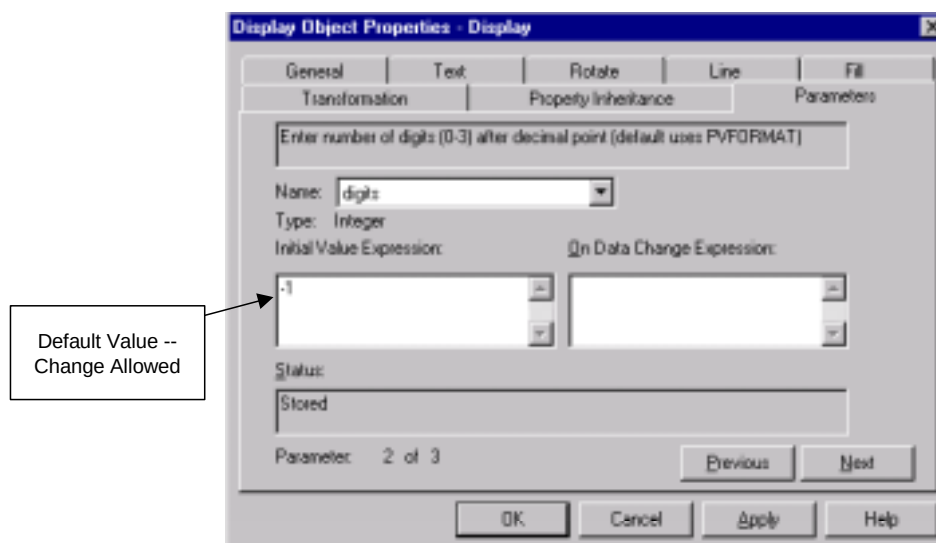
Enter a valid Tag for display parameter “point.” It must be an existing point on the TPN system with the correct point type that the particular display component can handle.

Hints:

- Most of the GSP display components require the user to enter a tag. You need to enclose the tagname in [] if it contains any special characters, such as \$, !, \, or starts with a number. It is **not** case-sensitive.
- Refer to the Prompt Message that provides information regarding the particular display parameter.



Select "Next" to enter the rest of the display parameters. Most of the other display parameters are not critical, and sometimes have default values.

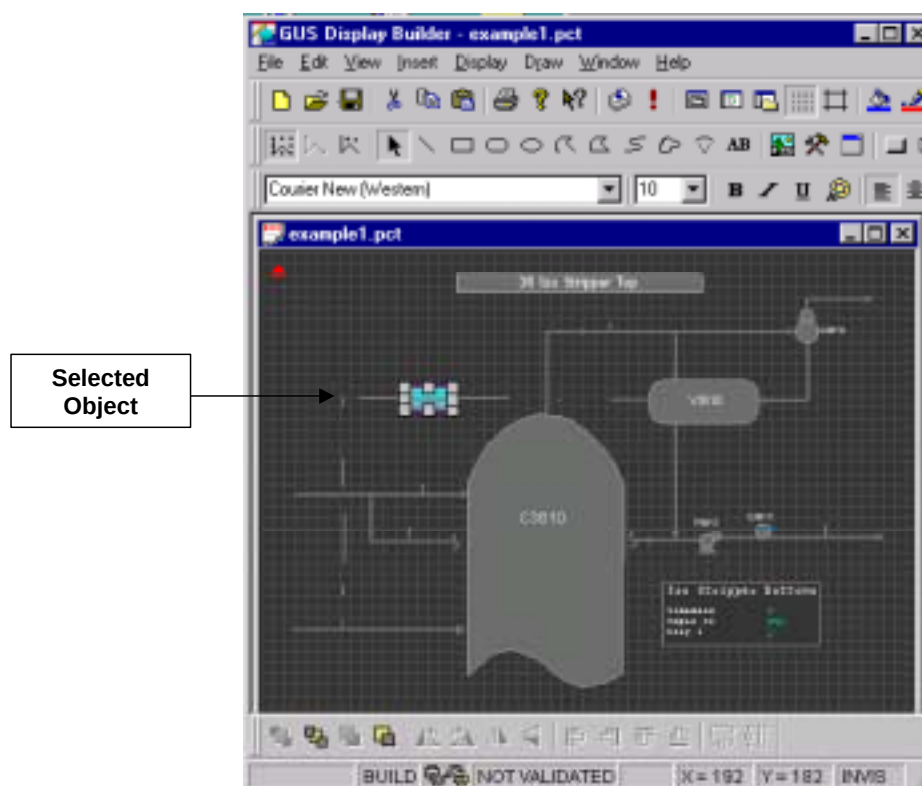


Select “OK” when all the display parameters values are set.

Position the display component at the desired location.

Hint: To fine-tune the alignment of an object, you can use the cursor keys to move the object while it is selected.

Note that most of the GSP display components are not resizable by design. This helps to prevent the components from being resized inadvertently during build-time. The exceptions to this include cascade symbols, level bars, and some of the function buttons.



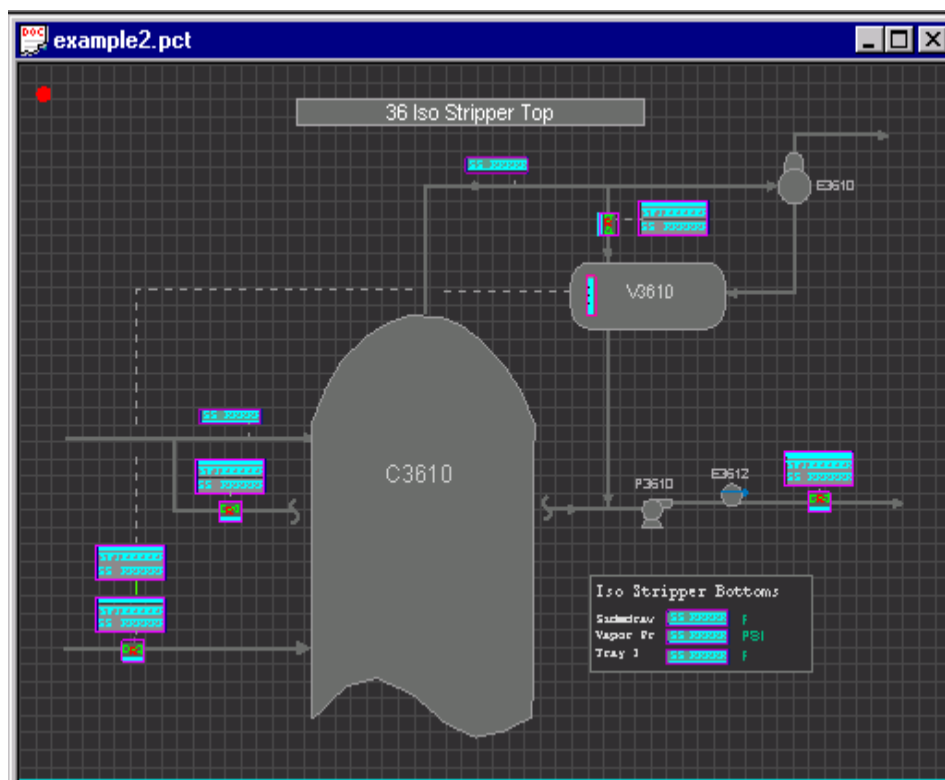
Insert a few GSP display components (e.g., cntl_box_lg_lt, cntl_vlv_hor, cntl_vlv_ver, casc_line, casc_continue and casc_arrow) using either the GSP Component Library or Insert/Display from the GUS Display Builder menu bar.

You may want to try both methods.

Note: When adding the same display component to connect to a different tag, you can use “Copy/Paste” in the Display Builder, then edit the display parameters (e.g., tagname) on the “Parameters” tab of the “Object Properties” dialog window.

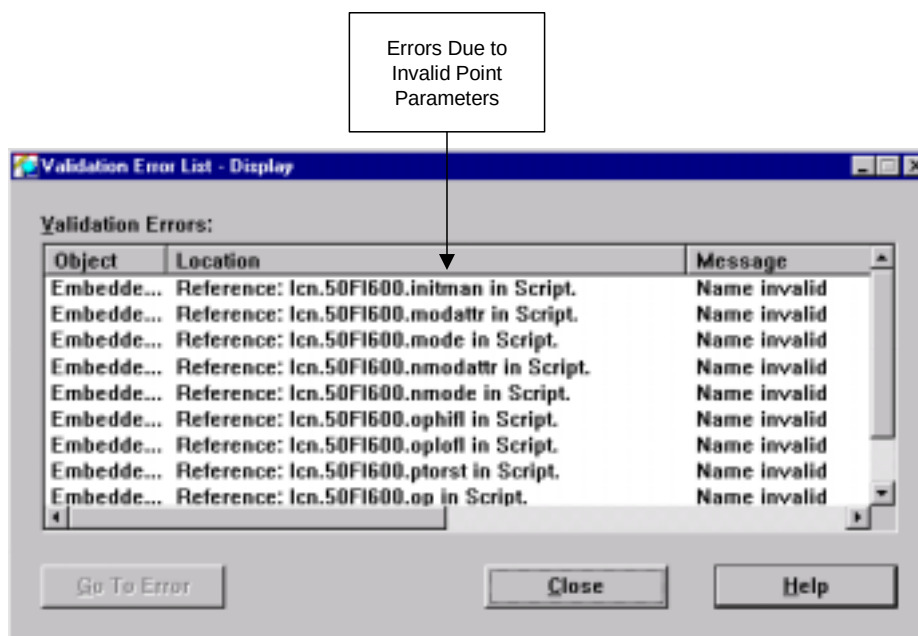
The screenshot shows a dialog box titled "Display Object Properties - EmbeddedPicture1". It has several tabs: "General", "Text", "Rotate", "Line", "Fill", "Transformation", "Property Inheritance", and "Parameters". The "Parameters" tab is currently selected. Inside the dialog, there is a text field labeled "Enter Tag:" with a hint "(enclose tag in [] if it contains special character)". Below this is a "Name:" dropdown menu showing "point". The "Type:" is set to "Inline". The "Inline Value:" is a text field containing "[50fc070]". Below that is a "Status:" dropdown menu showing "Validated". At the bottom left, it says "Parameter: 1 of 3". At the bottom right, there are "Previous" and "Next" buttons. At the very bottom of the dialog are "OK", "Cancel", "Apply", and "Help" buttons.

The figure below is an illustration of the display at build-time



Validate and Save the display.

Below is an example of validation errors that occurred when a display component (control box in this case) is connected to a point with a wrong point type (RegPV in this case). Refer to the "Display Component Reference" for point type information.



Attention

The validation process gives a warning message when a null primmod collector is encountered in a script. Ignore these validation warnings. GSP components handle null primmod collectors at runtime; thus, no errors concerning null primmod collectors will occur during the running of a GSP display.

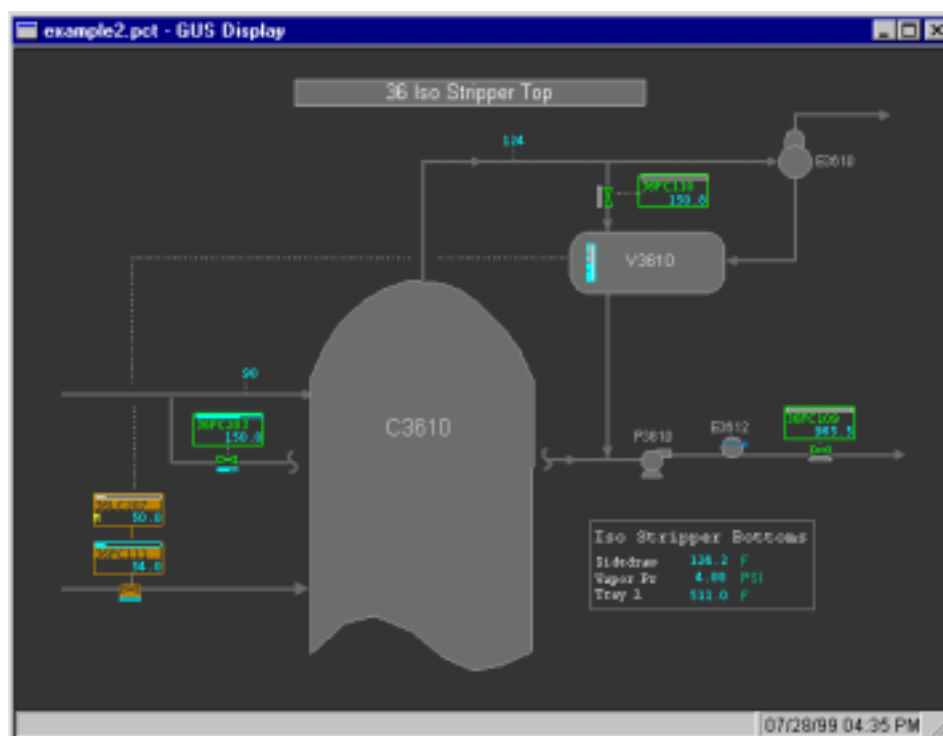
Expect this type of warning if you have not assigned a primmod reference to all display parameters concerned with primmods in the following classes of GSP display components:

- Aggregate Primmod Status Indicators
- Primmod Status Indicators
- High-Level Navigation Buttons with Small Primmod Status Box
- High-Level Navigation Buttons with Aggregate Primmod Status Box
- Low-Level Navigation Buttons with Small Primmod Status Box
- Low-Level Navigation Buttons with Aggregate Primmod Status Box

Building Process Displays - Build Process Display

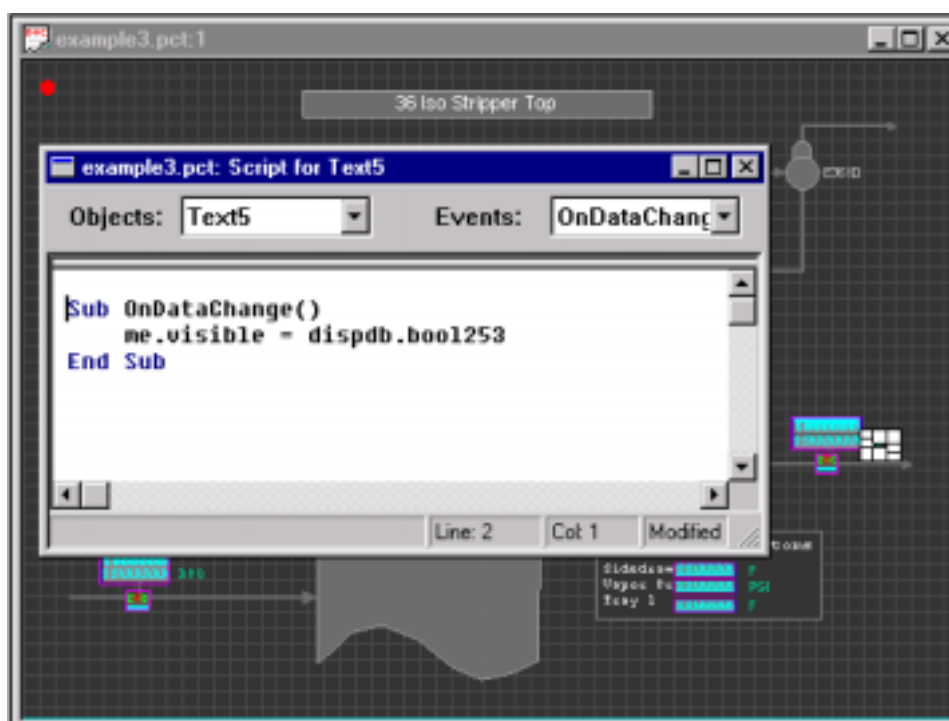
Run the display. Confirm that all of the components appear with the correct tag values.

Below is a figure showing the process display at run-time. You should be able to monitor the point status and values on live TPN data and invoke the faceplate by selecting a GSP component. Refer to “GSP Color Constants” in the Appendix for default color settings.

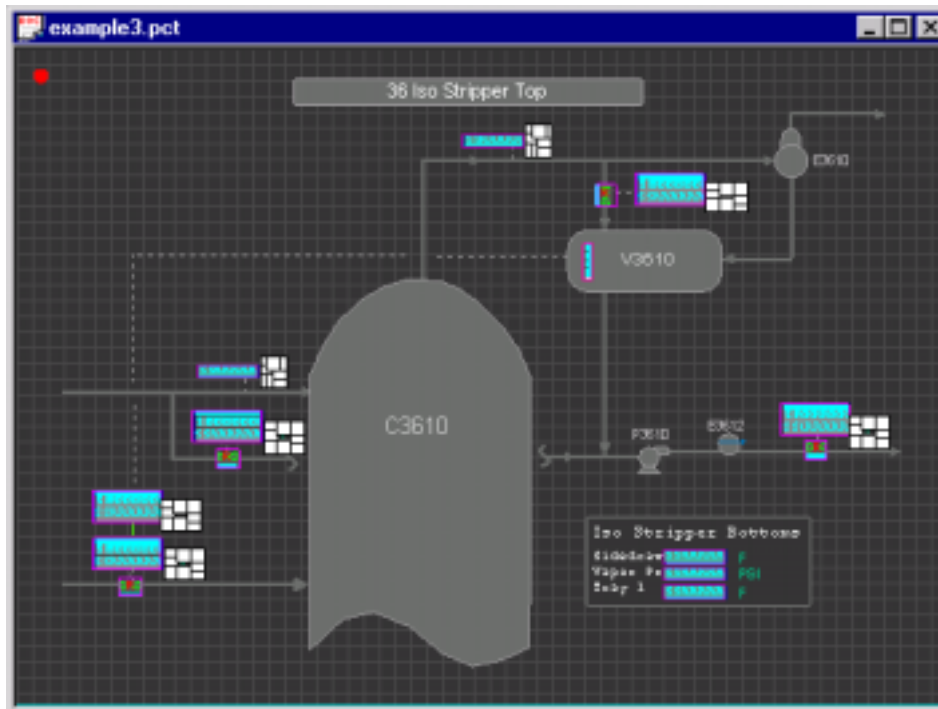


Add Finishing Touches

Once the static objects and the display components are in place, you may want to add a few finishing touches to enhance the usability of the display, such as adding the engineering units next to the control boxes. This can be accomplished by simply adding a text object. However, you can further enhance the display by making the object “appear and disappear” when the operator selects a show_hide_text button (a GSP Function Button). This can be accomplished by adding the script to the text object as shown below.



You can use the same technique for other text information you wish to add. You can also select multiple objects (e.g., all the engineering units) using the Shift-Click, group the objects, then add the above script to the group instead of adding it to individual object.



The advantage of this is to reduce the clutter on the display when accessory information is not needed, but make them visible when desired with the single touch of a button.

Refer to “Reserved GUS Display Database Variables” in the Appendix for a complete list of display database variables currently used by GUS Solution Pack.

Set Collection Rate

View the Collection List from GUS Display Builder. Modify the Collection Rate for the variables if desired. Refer to the “Fine Tuning Performance” section in the “Display Authoring Tutorial” for more information.

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Journal of Management Studies, 36(7), 809–826.

1. *Journal of the American Medical Association*, 1997; 278: 1039-1044.

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Building Task Panels

This section explains how to use the GUS Solution Pack to build task panels. It assumes that the GSP users are familiar with basic features of Windows NT and the Honeywell GUS Display Builder.

The Task Panel is a powerful, user-configured window used in the GUS Solution Pack that can contain an alarm overview, navigation-to-displays, and system tasks. The alarm overview and navigation-to-displays work together to provide a graphics-based alarm management system that enables the operator to acknowledge alarms and take action from the same display. User-configured PRIMMOD associations provide aggregate alarm status of displays through GSP objects; by selecting the target, the operator navigates to the display to possibly see patterns in the alarm data and take action. The Task Panel is a convenient place to house objects for handling "system functions" such as objects that change the content of the displays (show/hide text) or objects that open other applications.

Before You Start

Determine Task Panel Size

Before you start to build task panels using the GUS Solution Pack, you should have read the "SafeView Configurations" section. It contains information that will help you to plan ahead in order to avoid rework.

Below are the GSP recommended Task Panel Sizes for single-screen and dual-screen GUS under two different Display Resolution settings. The display components were built to be viewed at 1024 x 768 resolution. To preserve their actual size when using 1280 x 1024 screen resolution, the display size property must be 80% of the "Display Run Size" with the "zoom to fit (run-time)" set to true. Refer to "GSP-Provided Displays Reference" section for more detail on Faceplates.

Task Panel Size	Display Resolution	1280 x 1024	1024 x 768
	Using faceplate GSPplate_fp1_127x871.pct	1014 x 62	N/A
	Using faceplate GSPplate_fp2_123x626.pct	N/A	1012 x 71

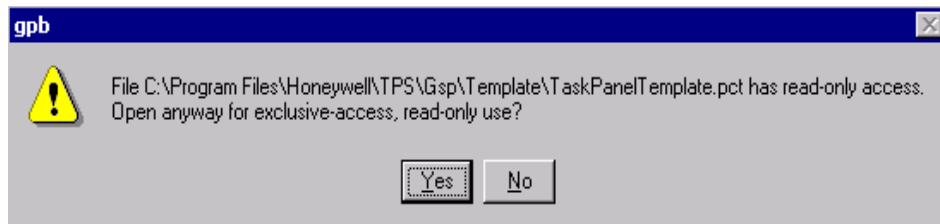
Setup Task Panel Building Environment

Before you start to build task panels, you should customized the master Task Panel Template and make a customized task panel template to use throughout the project. The main reason to customize the template is to set the category and the display size to match the SafeView configuration.

Follow the steps below to customize the template.

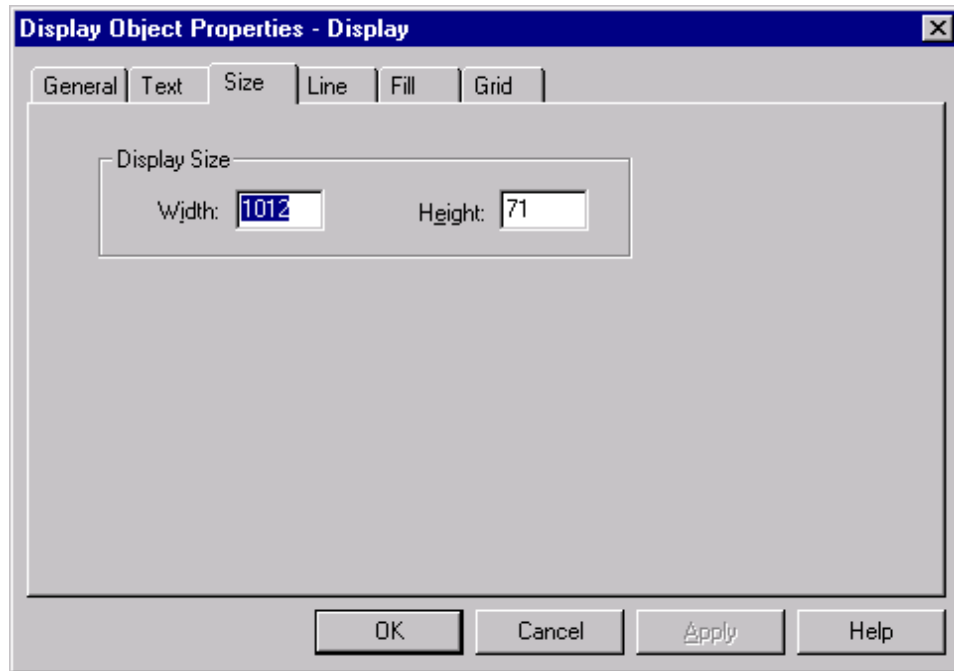
Create a display work folder on your system if you don't have one already.

Open the master TaskPanelTemplate.pct located in the \Program Files\Honeywell\TPS\GSP\Template folder. Click "Yes" to acknowledge that the template is a read-only file.

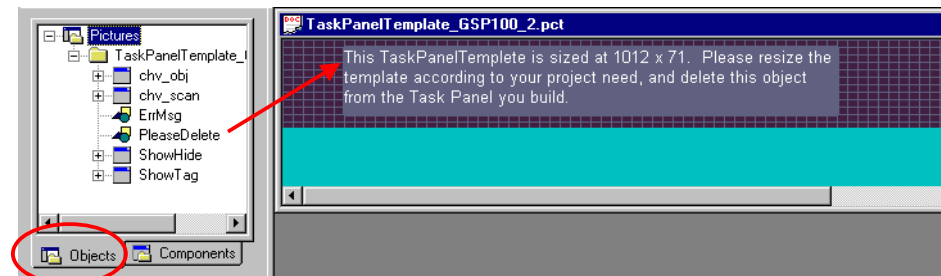


Check the Category on the "General" tab of the "Display Properties" dialog window. The default is "TaskPanel." Note that the category is used in SafeView configuration to match the default window for the task panel display. Refer to "Display Templates Reference" section for more detail.

Check the size of the display on the “Size” tab of the “Display Properties” dialog window. Change the “Display Size” to match the configuration determined earlier.



Delete the ResizeReminder object.

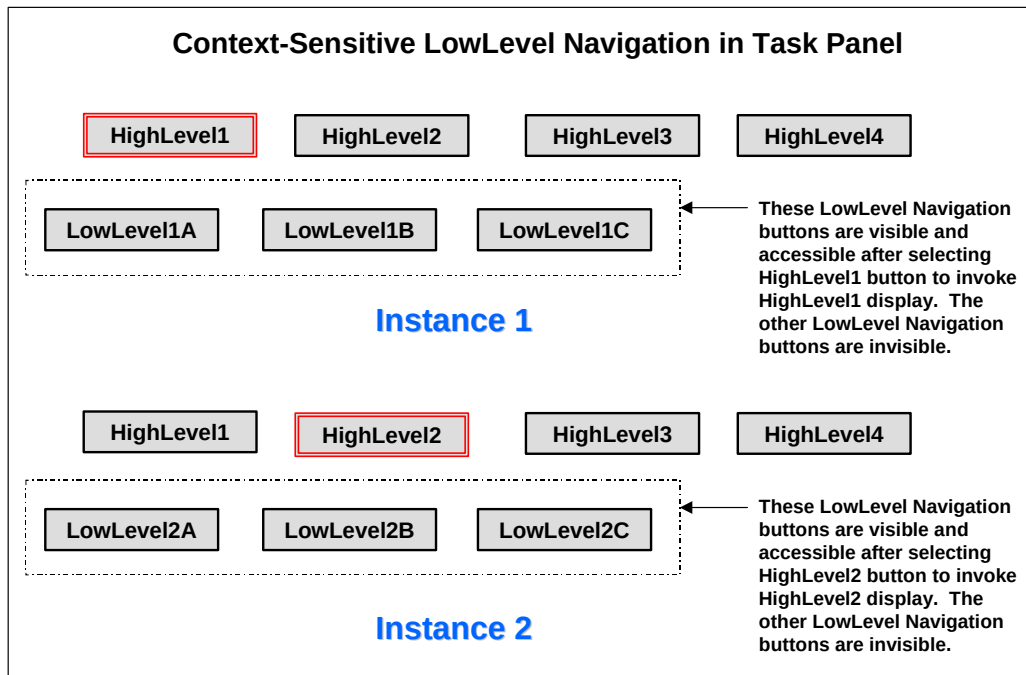


Use “Save As” to save the file in the display work folder.

Make your Display Template read-only via the file’s Properties page of NT Explorer to avoid it from being over-written by mistake.

Setup Low-Level Navigation Display Building Environment

GSP provides users the capability to build two-level navigation within the task panel display via the use of the Low-Level Navigation displays and Low-Level navigation buttons. This allows users to incorporate context sensitive Low-Level navigation buttons that are only visible and accessible when the referenced High-Level display is running. The figure below illustrates the two-level navigation scheme in task panel.



If you decide to take advantage of the two-level navigation capability, follow the steps below to set up the building environment:

Open the LowLevelNavTemplate.pct located in the \Program Files\Honeywell\TPS\GSP\Template folder. Click "Yes" to acknowledge that the template is a read-only file. Check the size of the display on the "Size" tab of the "Display Properties" dialog window. The display size should be no wider than the TaskPanel, and about half the height of the TaskPanel.

Use "Save As" to save the file in the display work folder.

Make your Display Template read-only via the file's Properties page of NT Explorer to avoid it from being over-written by mistake.

Reserved GUS Display Database

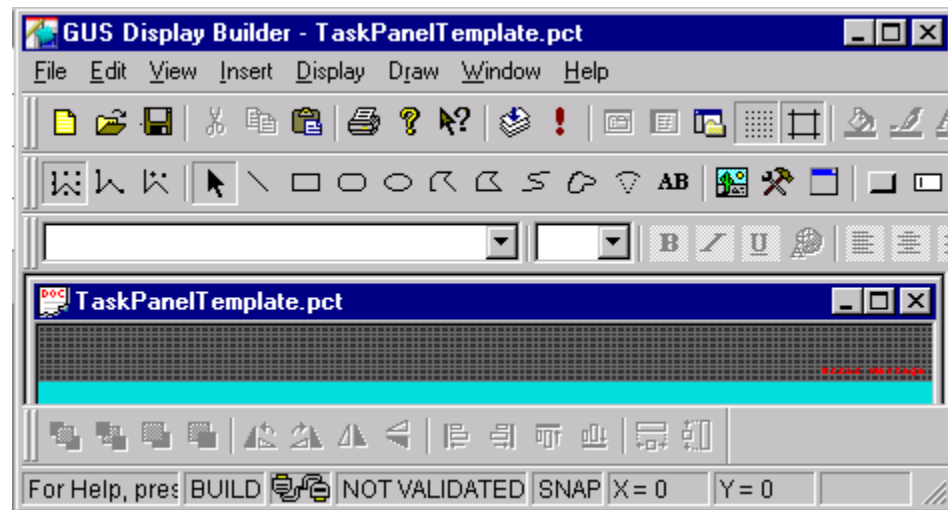
GSP reserves the variables between 201-256 of all variable types of the GUS Local Display Database (e.g., ENT256, BOOL256). To avoid conflict with the GSP display components, users must not use the variables within this range.

Refer to “Reserved GUS Display Database Variables” in the Appendix for a complete list of display database variables currently used by GUS Solution Pack.

Build Task Panel With Navigation

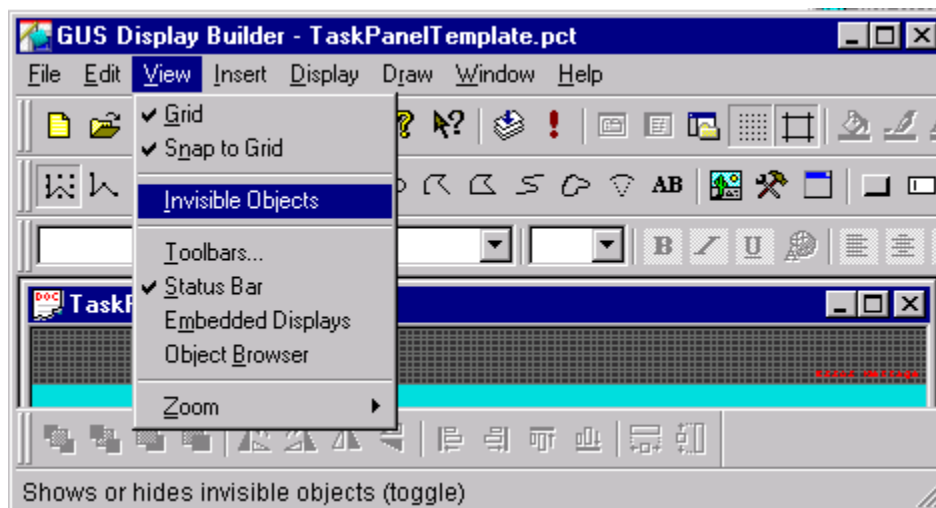
Open Task Panel Template

Open the customized task panel template that you created earlier. **It is critical to always start with a Task Panel Template which contains the script and objects required for GSP to function properly.**



Select View/Invisible Objects

Most of the GSP display components are initially invisible by design. The primary purpose is to eliminate the flickering of the dummy objects (before live data become available). Select “View Invisible Objects” during display build-time so that you can see all the GSP display components being inserted.

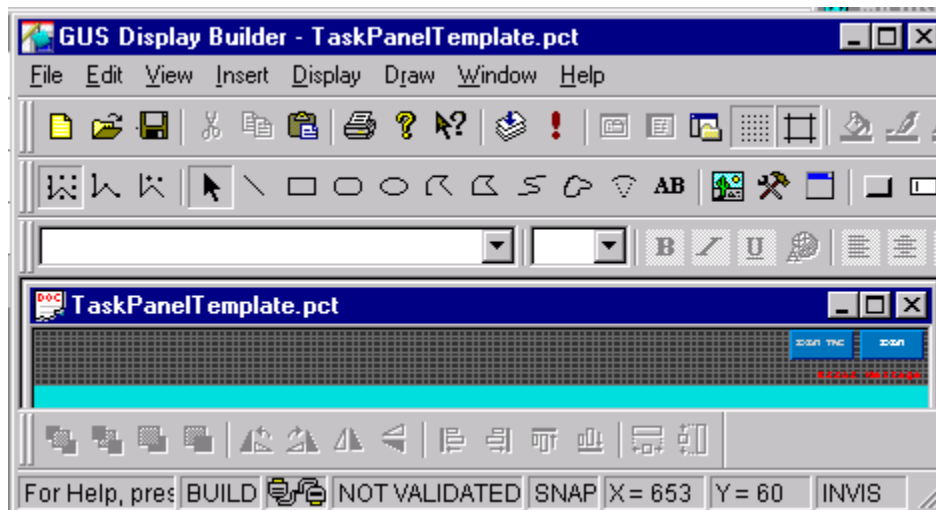


There are two function buttons included in the Task Panel Template:

Tag/Setpoint Toggle Button (used by Large Control Boxes).

Show/Hide Toggle Button (used by Measurement Boxes with Tag, Digital Alarm Graphic Symbols, Digital Alarm Message Boxes and GSP Version).

You may delete the above buttons if you don't use the above-mentioned display components (listed in parentheses).



Insert GSP High-Level Navigation Buttons

There are two types of navigation buttons for High-Level Display Navigation:

Buttons with Small Primmod Status Box

Buttons with Aggregate Primmod Status Box

The user should select one type to implement throughout an operator interface. Both types include a primmod status indicator in addition to their navigation function. Refer to “Display Navigation Button Reference” section for more details.

The following discussion and figures explain the insertion of a display navigation button.

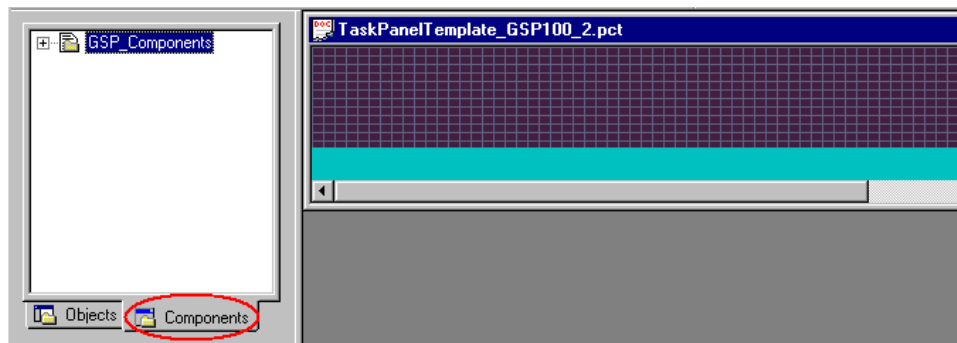
Two methods to add GSP display components exist:

1. Select display component from GSP Component Library.
2. Select display component using Insert/Display from GUS Display Builder menu bar.

Select Display Component From GSP Component Library

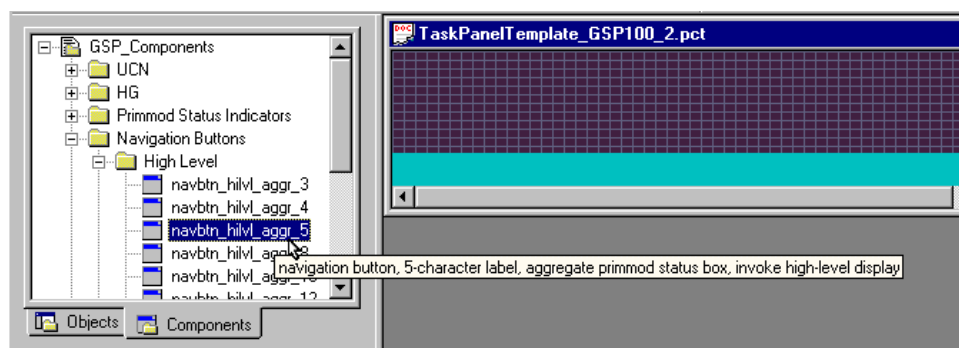
Select GUS Explorer on the View menu (or select the GUS Explorer button on the Standard toolbar). Either the Object Browser or the Component Explorer window is displayed.

If the Object Browser is displayed, select the Components tab. A view of the GSP Component Library is displayed.



The Components Library is displayed in the standard Windows hierarchical tree format. Expand the tree to gain access to the individual display components you want to insert to the display. A one-line description (a tooltip) is displayed when the cursor is left on a display component.

Building Task Panels - Build Task Panel With Navigation

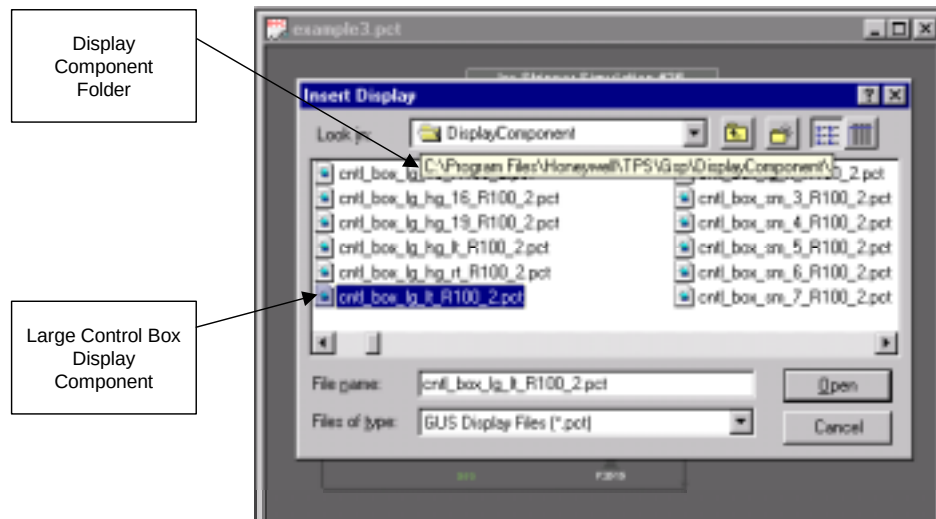


Double-click the desired high-level navigation button component you want to insert. The Display Object Properties page is displayed, and you are prompted to enter the parameters required by the display component.

Select Display Component Using Insert/Display From GUS Display Builder Menu Bar

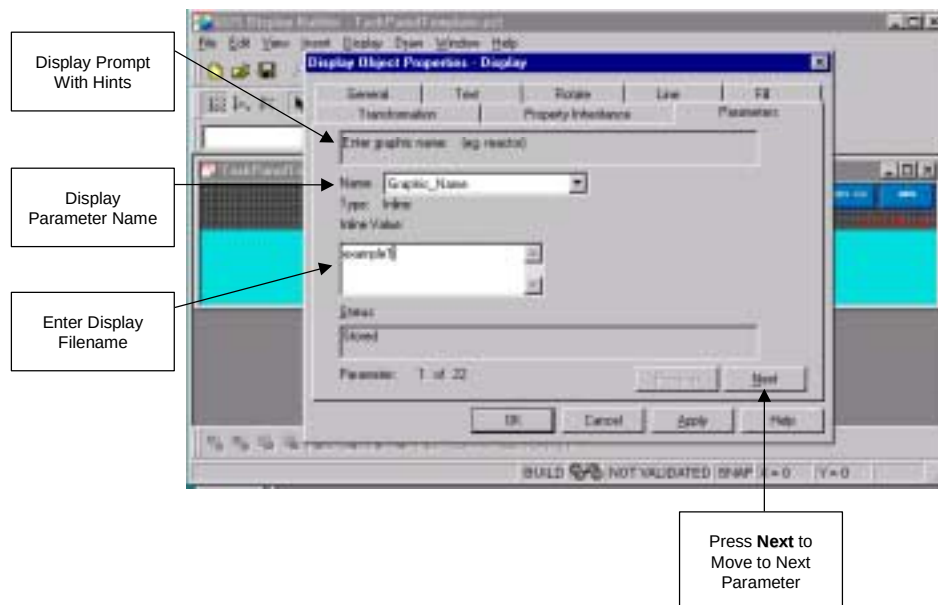
From the GUS Display Builder menu bar, select Insert/Display.

The following figures and paragraphs explain the insertion of a large control box (e.g., cntl_box_lg_lt.pct) using this method.

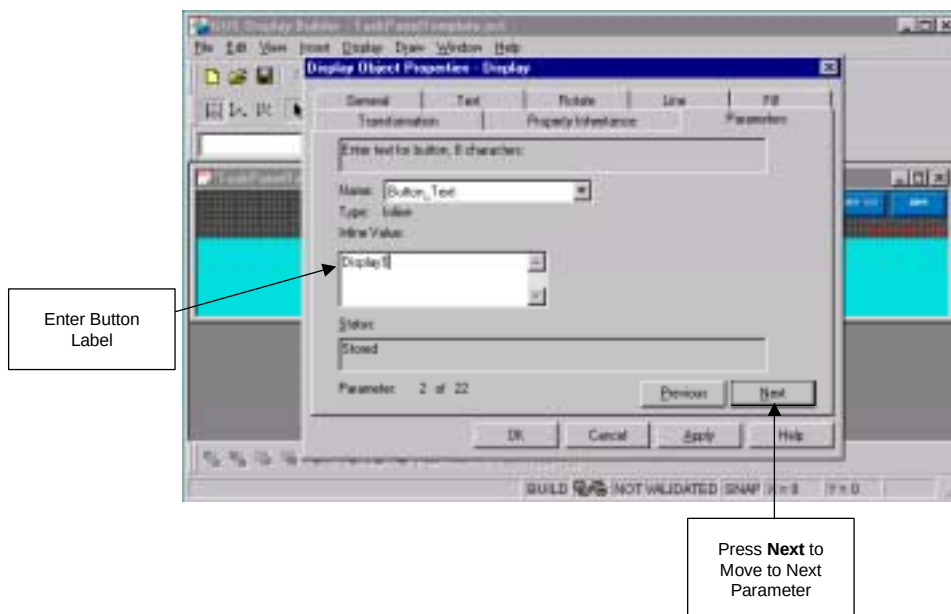


Building Task Panels - Build Task Panel With Navigation

Enter the Graphic_Name display parameter. Use the simple filename and allow the Pathname Catalog to locate the file. Do not enter the full pathname of the graphic filename. This approach provides a lot more flexibility to where the displays need to reside on your system. If the display is a GUS display, you may enter the extension (.pct). Make sure that you use the same display naming convention (e.g., “crudetower” or “crudetower.pct”) when entering the graphic name in an object’s display parameter. During runtime, if the display can’t be found, an error message will be displayed in task panel instead of searching through the TPN database.



Select “Next”, enter the Button_Text display parameter.

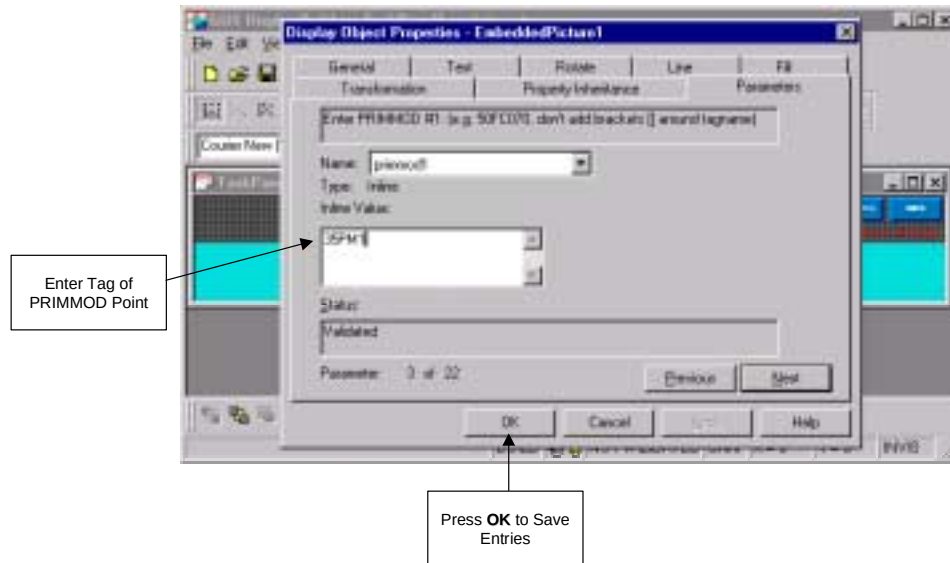


The next 20 display parameters are to be used for PRIMMODs. Enter all the PRIMMODs you want this button to reference to. Normally, it would be all the PRIMMODs that are configured for the tags that are on this particular display.

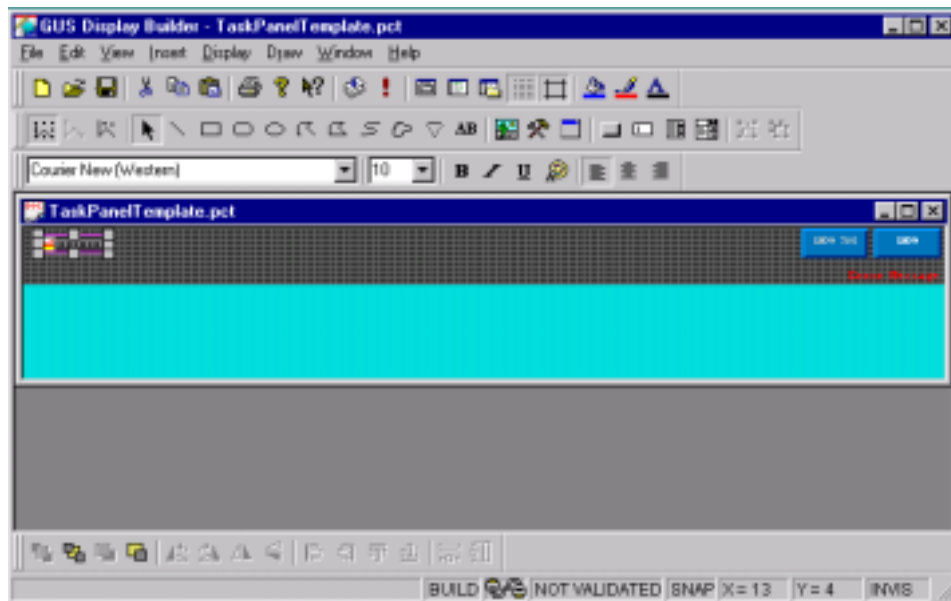
Hints:

- Enter one PRIMMOD for each display parameter. Do not skip parameters when entering the PRIMMOD tags.
- Users must make sure the correct PRIMMOD tag is entered. No error message is displayed during validation or run-time when encountering a non-existent PRIMMOD tag.
- Select “OK” when all the display parameter values are set.

Building Task Panels - Build Task Panel With Navigation



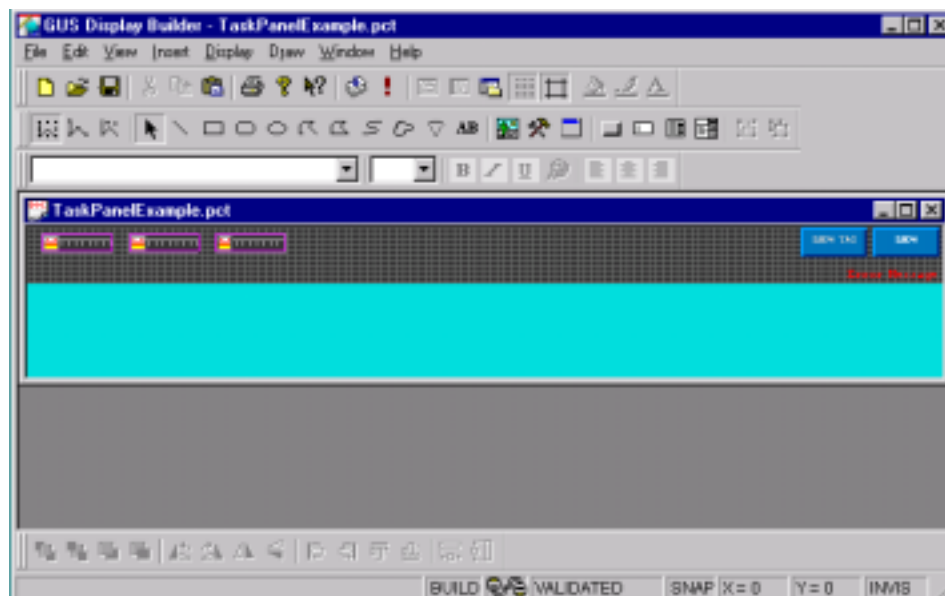
Position the Navigation Button at the desired location. Hint: To fine-tune the alignment of an object, you can use the cursor keys to move the object around while it is selected. Note that most of the GSP display components are not resizable by design. This helps to avoid the components being resized inadvertently during build-time.



Use the same steps to add other navigation buttons. When adding the same navigation buttons (where display labels have the same number of characters) to connect to a different display, you can use “Copy/Paste” in GUS Display Builder, then use “Edit/Enter Parameters” to edit the display parameters (e.g., display name, label, and PRIMMOD tags).

Note:

- You can insert other Function Buttons in the Task Panel.
- You can insert as many buttons as there is space in the display. However, if you plan to build a two-level navigation scheme, be sure to allocate space for the Low-Level navigation displays to be inserted in Task Panel later.



Validate and Save the display using a different filename (e.g., TP_TEST.pct). **Do not write-over the template.**



Attention

The validation process gives a warning message when a null primmod collector is encountered in a script. Ignore these validation warnings. GSP components handle null primmod collectors at runtime; thus, no errors concerning null primmod collectors will occur during the running of a GSP display.

Expect this type of warning if you have not assigned a primmod reference to all display parameters concerned with primmods in the following classes of GSP display components:

- Aggregate Primmod Status Indicators
- Primmod Status Indicators
- High-Level Navigation Buttons with Small Primmod Status Box
- High-Level Navigation Buttons with Aggregate Primmod Status Box
- Low-Level Navigation Buttons with Small Primmod Status Box
- Low-Level Navigation Buttons with Aggregate Primmod Status Box

Run the display.



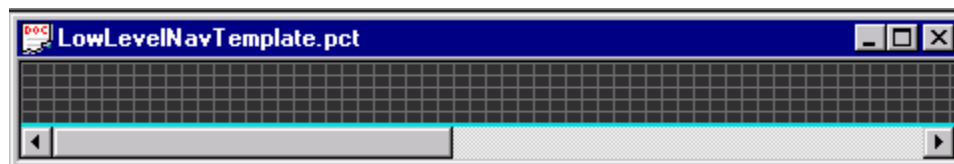
Click on one of the navigation buttons, it will highlight the selected button and call-up the referenced display.



Build Low-Level Navigation Display

If you decide to take advantage of the two-level navigation capability, follow the steps below to build the Low-Level navigation displays which are to be inserted into the Task Panel display you built earlier.

Open the customized LowLevelNavTemplate that you created earlier. It is critical to always **start with a Low-Level Navigation Template** which contains the script required for GSP to function properly.



Select View/Invisible Objects

Select View/Invisible Objects, so that you can see all the GSP display components being inserted.

Determine Navigation Button Type

There are two types of navigation buttons for **Low-Level** display navigation (same as for High-Level Navigation, but with different filenames):

Buttons with Small Primmod Status Box

Buttons with Aggregate Primmod Status Box

You should use the same type as what was used for High-Level Display Navigation in the Task Panel.

Choose the Desired Navigation Display Component

There are two decisions to be made when choosing the desired Low-Level Navigation buttons to use:

To show alarm status with the small primmod status box or aggregate primmod status box.

To minimize or not to minimize Native Window when the Low-Level navigation button is selected.

Warning:

Be sure to allow Native Window to minimize in the SafeView Configuration if you use the navigation buttons that minimize the Native Window.

The user should select one type of Low-Level navigation button to implement throughout an operator interface. Refer to “Display Navigation Button Reference” section for more details.

Insert GSP Low-Level Navigation Buttons

Insert the Low-Level Navigation buttons into the LowLevelNavTemplate.pct the same way you insert the High-Level Navigation buttons into the TaskPanelTemplate.pct. Enter the display filename (e.g., “crudetower” or crudetower.pct), button label, and PRIMMOD tagnames in the display parameter dialog boxes.

Save the display using a different filename. **Do not write-over the template.**

Note:

There is no need to validate the Low-Level Navigation displays.

You can insert as many Low-Level Navigation buttons as there is space in the display.

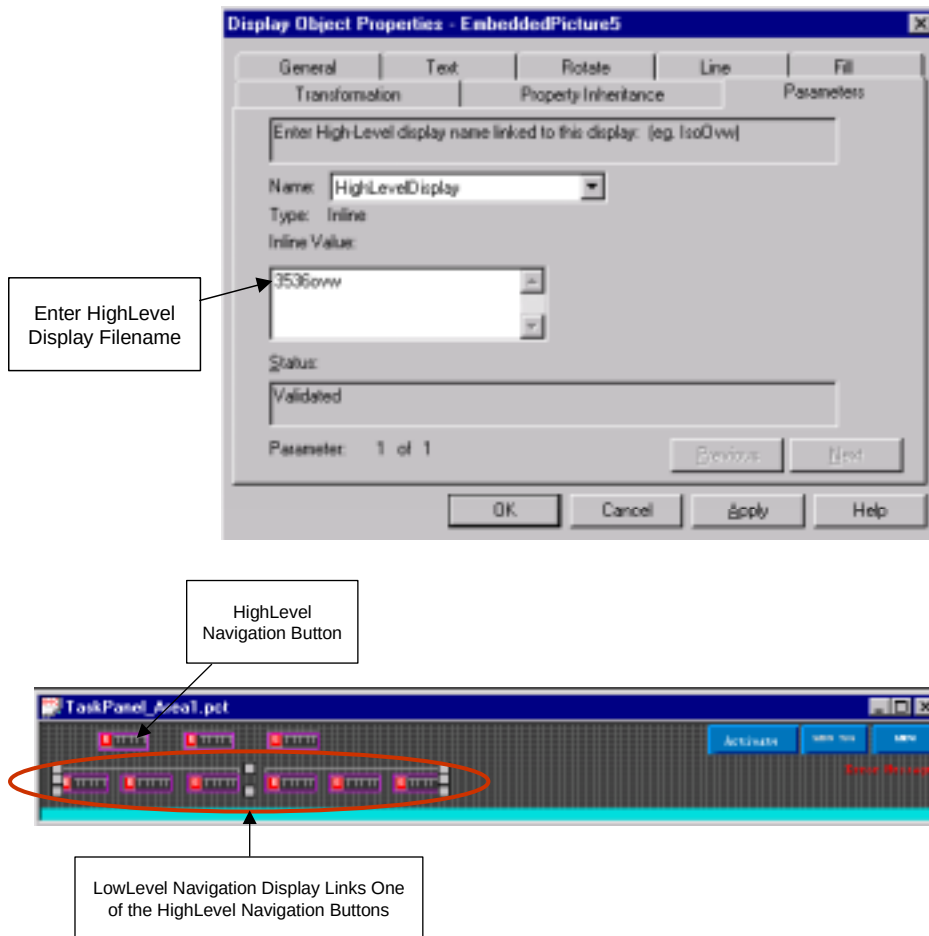
If desired, you can build one Low-Level Navigation display for each High-Level Navigation button included in the Task Panel display you built earlier.

Insert Low-Level Navigation Displays in Task Panel

Open the Task Panel that was built earlier.

Insert each Low-Level Navigation Display in the Task Panel.

Enter the filename of High-Level display to be referenced. Use the simple filename and allow the Pathname Catalog to locate the file. Do not enter the full pathname of the graphic filename. This provides a lot more flexibility to where the displays need to reside on your system. If the display is a GUS display, you may enter the extension (.pct). Make sure that you use the same display naming convention (e.g., “crudetower” or “crudetower.pct”) when entering the graphic name in an object’s display parameter. During runtime, if the display can’t be found, an error message will be displayed in task panel instead of searching through the TPN database.



There is no problem overlapping the Low-Level navigation displays. Only the one connected to the selected High-Level navigation button will be visible at any one time.

Validate and Save the Task Panel Display.



Attention

The validation process gives a warning message when a null primmod collector is encountered in a script. Ignore these validation warnings. GSP components handle null primmod collectors at runtime; thus, no errors concerning null primmod collectors will occur during the running of a GSP display.

Expect this type of warning if you have not assigned a primmod reference to all display parameters concerned with primmods in the following classes of GSP display components:

- Aggregate Primmod Status Indicators
- Primmod Status Indicators
- High-Level Navigation Buttons with Small Primmod Status Box
- High-Level Navigation Buttons with Aggregate Primmod Status Box
- Low-Level Navigation Buttons with Small Primmod Status Box
- Low-Level Navigation Buttons with Aggregate Primmod Status Box

Run Task Panel Display

First copy all of the Process Displays that will be invoked by navigation buttons into a folder included in the pathname catalog.

When clicking on a High-Level Navigation Button, the following will occur:

The button outline box will turn to “Selection” color.

The Process Display referenced by the button will be invoked.

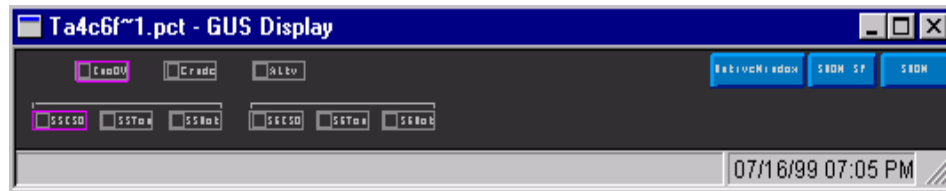
The Low-Level Navigation Display will become visible, and the buttons on the Low-Level Navigation Display will become selectable if there is a Low-Level Navigation Display connected to that High-Level Navigation Button.

When clicking on a Low-Level Navigation Button, the following will occur:

The button outline box will turn to “Selection” color.

The outline box of the High-Level Navigation Button connected to this Low-Level Navigation Display will be in “Selection” color. A thick-line box indicates that the High-Level Display is still running, whereas, a thin-line box indicates that the High-Level Display is no longer active.

The Process Display referenced by the button will be invoked.



Once you confirm that the Task Panel display is working properly, it is ready to be copied to a folder included in the station's Pathname Catalog. This allows the display to be invoked through the keyboard button, and fully integrated with the GSP SafeView workspace.

GSP-Provided Displays Reference

Faceplate Display

Description

The GSP Faceplate Display is a prebuilt display that provides an operator interface for viewing and manipulating the key operating parameters (e.g., setpoint, output, mode, etc.) of a single point.

The GSP Faceplate Display:

Participates in the GSP interwindow communication scheme. It enables all running GSP displays to operate on the same faceplate. Each GSP component in the running displays indicates if the faceplate is handling its point-of-interest.

Contains the GUS Faceplate ActiveX control. Thus, the Faceplate Display provides the operator interface for all point types supported by the GUS Faceplate ActiveX control in the installed GUS release.

Supports the point manipulation keys (e.g., SP, OP, MAN, ramp keys).

The GSP toolkit contains two faceplate displays. One is designed for the 1280 x 1024 resolution, and the other is designed for the 1024 x 768 resolution. Because of the higher resolution, the faceplate display designed for the 1280 x 1024 resolution is able to include a 20-minute trend of the PV value when it is available.

Filenames and Description

GSPplate_fp1_127x871.pct – Faceplate display designed for use in GUS nodes running 1280 x 1024 resolution. GSPplate_fp1_127x871.pct contains:

GUS Faceplate ActiveX Control

Command buttons to invoke the detail and associated displays for the point-of-interest in the faceplate.

20-minute real-time trend of the PV value when available.

GSPplate_fp2_123x626.pct - Faceplate display designed for use in GUS nodes running 1024 x 768 resolution. GSPplate_fp2_123x626.pct contains:

GUS Faceplate ActiveX Control

Command buttons to invoke the detail and associated displays for the point-of-interest in the faceplate.

GSP-Provided Displays Reference - Faceplate Display

The following table shows a picture of each faceplate.

GSPplate_fp1_127x871	GSPplate_fp2_123x626
	

Properties of the Faceplate Display

The display properties defined for the Faceplate Display are:

Category: “Faceplate” (category supported by the GSP SafeView configurations).

Picture Option: Zoom to Fit at run-time is true.

Size:

GSPplate_fp1_127x871.pct –

Display Size: 127 x 871

GSPplate_fp2_123x626.pct –

Display Size: 123 x 626

Properties of the Trend in the Faceplate Display

The properties defined for the trend in the Faceplate Display are:

Time Base: 20 Minutes

Data Source: Automatic (Set in script.)

Trend Color: Cyan (Set in script.)

YRangeHigh: The pveuhi value of the point being trended (Set in script.)

YRangeLow: The pveulo value of the point being trended (Set in script.)

Background Color: Black

Inherit Background: False

Hairline Cursor: False (The hairline cursor is turned off.)

Trend Style: Plot Area Only

Show Grid: True

Show Axes: False

Running the Faceplate in the GSP Environment

Important: Copy the GSP Faceplate that supports the screen resolution of your GUS mode to a folder included in the pathname catalog and then rename the faceplate to GSPplate.pct.

The interaction of a GSP display component and a faceplate display assumes that the name of the faceplate display is “GSPplate.pct” with GSP default. User can now use the GSP Configuration Utility to use customized faceplate display. However, the display must be a GUS display and reside in the folder included in the pathname catalog. Refer to the GSP Configuration Utility section in the document for more detail.

Run-Time Behavior

Only one GSP Faceplate Display may be running per GUS node. All running GSP displays operate on the same faceplate display.

When the faceplate window has focus, it interacts with the IKB keys. The user may interact with the faceplate using the point manipulation keys, SP, OUT, MAN, AUTO, NORM, and ramp keys.

The Faceplate Display's error handling is to display a message box containing the error message. The message box must be acknowledged before the user can continue working in the faceplate.

The Faceplate Display may be closed. All GSP targets invoke the Faceplate Display if it is not currently running.

GSP Color Display

This is a prebuilt display that provides users an easy way to check the health of the GSP installation. It is located in the \Program Files\Honeywell\TPS\GSP\Display folder. Users can run the display using GUS Display Builder anytime after the GSP is installed on a GUS.

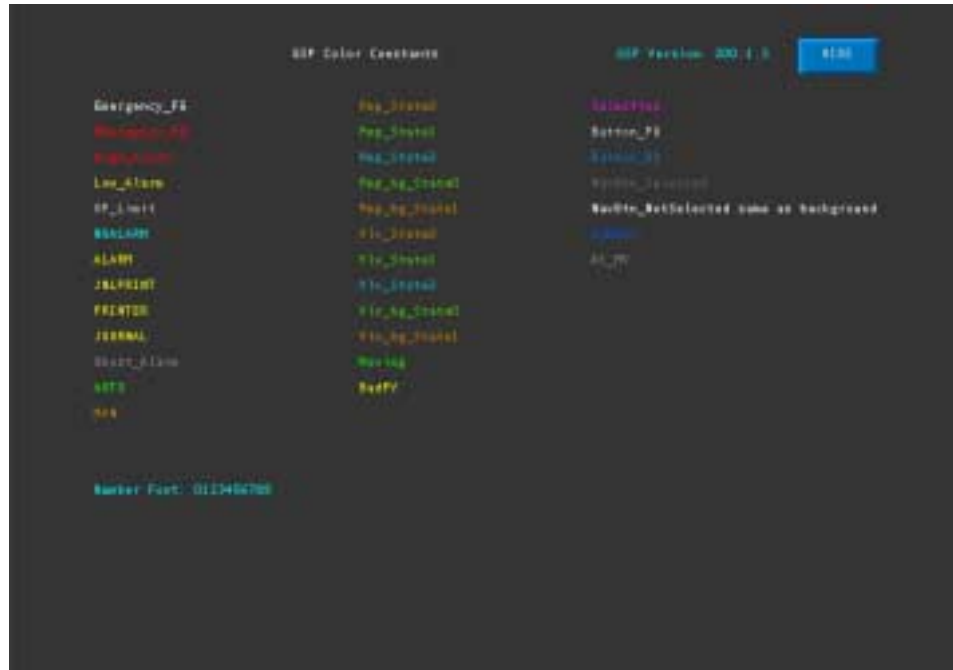
Below is a picture of the GSPColor.pct at run-time. It displays

Current color configuration installed on the GUS

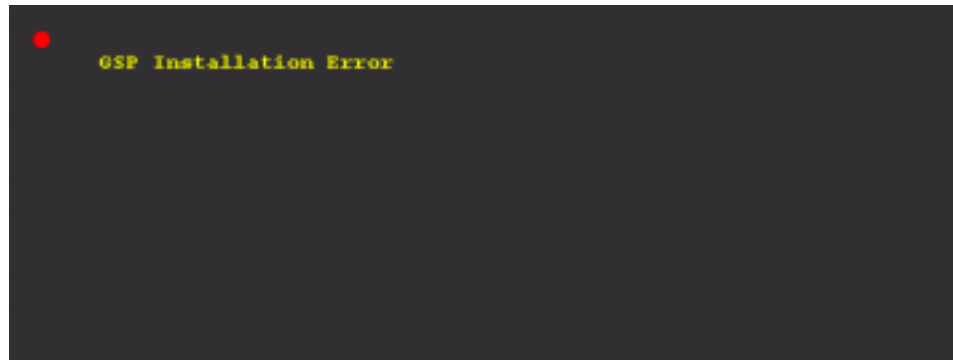
Name and color of the GSP Color Constants in the font used by GSP to display text information.

A line of numbers on the bottom screen in the font used by GSP to display numeric information.

A show/hide button to toggle the GSP version number.



Below is a picture of the GSPColor.pct at run-time when GSP encounters error. The blinking red circle showing the GSP status is the same object that is in the DisplayTemplate.pct. It is only visible when the display encounters problem communicating with GSP library.



Display Templates Reference

DisplayTemplate.pct

This GUS Display Template file must be used as the starting template to build all GSP process displays. The built-in display script and chv_obj object are required for the interwindow communication to function properly.

Parameters

There are no display parameters that users need to enter.

Objects

Display

Script

The script on the DisplayTemplate.pct consists of the onDataChange and onDisplayShutdown subroutines. **Users must not modify the standard portion of the script.** To handle special needs, users may append to the standard portion of the script.

Property

The properties referenced by GSP are described below:

Category: LowLevel (Users may change this to meet the SafeView configuration requirement.)

Picture Option: Zoom to Fit at run-time.

Background Color: Does not matter. At run-time, the display background color is set to the Display_BG color defined according to the “GSP Color Constants” Table in the Appendix.

Size: Users may change this. The default size is based on the recommended settings for ProcessDisplay (single-screen) and LowLevel (dual-screen) displays with 1024 x 768 display resolution.

Note: It is critical to decide the size of the process displays that meets the SafeView configuration requirement. Refer to the “SafeView Configurations” section for the recommended settings.

Title

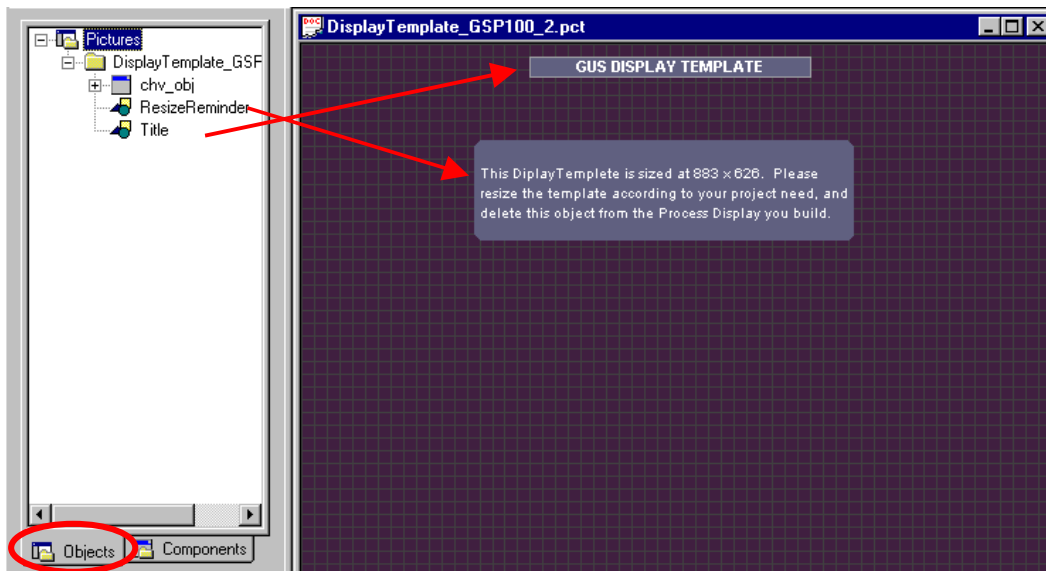
This is a text object on the DisplayTemplate.pct. Users may change or delete this object without affecting the functionality of the template.

ResizeReminder

This is a text object to remind the user to resize the Display Template to meet the project need. Refer to the SafeView Configurations section of this document on how to decide the window and display sizing. User should delete this object from the template before using it to build process displays.

chv_obj

This is an embedded object that handles the interwindow communication. **Do not delete or modify** this object. A blinking red circle on the upper-left hand corner of the display (GSP_status in chv_obj) will be visible if GSP is unable to establish reference to the GSP DLL. Most likely, this is due to failure during GSP installation.



TaskPanelTemplate.pct

This GUS Task Panel Template file must be used as the starting template to build all GSP Task Panel displays. The built-in display script, chv_obj object and chv_scan object are required for the interwindow communication to function properly.

Parameters

There are no display parameters that users need to enter.

Objects

Display

Script

The script on the TaskPanelTemplate.pct consists of an onDataChange and onDisplayShutdown subroutines. **Users must not modify the standard portion of the script.** To handle special needs, users may append to the standard portion of the script.

Property

The properties referenced by GSP are described below:

Category: TaskPanel (Users may change this to meet the SafeView configuration requirement.)

Picture Option: Zoom to Fit at run-time.

Background Color: Does not matter. At run-time, the display background color is set to the Display_BG color defined according to the “GSP Color Constants” Table in the Appendix.

Size: Users may change this. The default size is based on the recommended settings for Task Panel with 1024 x 768 display resolution.

Note: It is critical to decide the size of the task panel displays that meets the SafeView configuration requirement. Refer to the “SafeView Configurations” section for the recommended settings.

ShowHide

This is an embedded show_hide function button. We recommend users to keep the button in the Task Panel displays so it is always accessible. However, users may delete the object from the template if the “show/hide” feature is not desired.

ShowTag

This is an embedded tag_sp_btn function button. We recommend users to keep the button in the Task Panel displays so it is always accessible. However, users may delete the object from the template if the “show tag/setpoint” feature is not desired.

ErrMsg

This is a text object that displays error message when an error is encountered when invoking a display from the task panel. **Do not delete** this object from the template.

Below are possible errors:

Error Number	Error Message	Comment
1101	Graphic_Name is not a valid display file	Invalid .pct or .ds
1102	Graphic_Name display not found	.pct display not found in Search Path
1103	Unable to activate Native Window	Error trying to invoke a .ds display

ResizeReminder

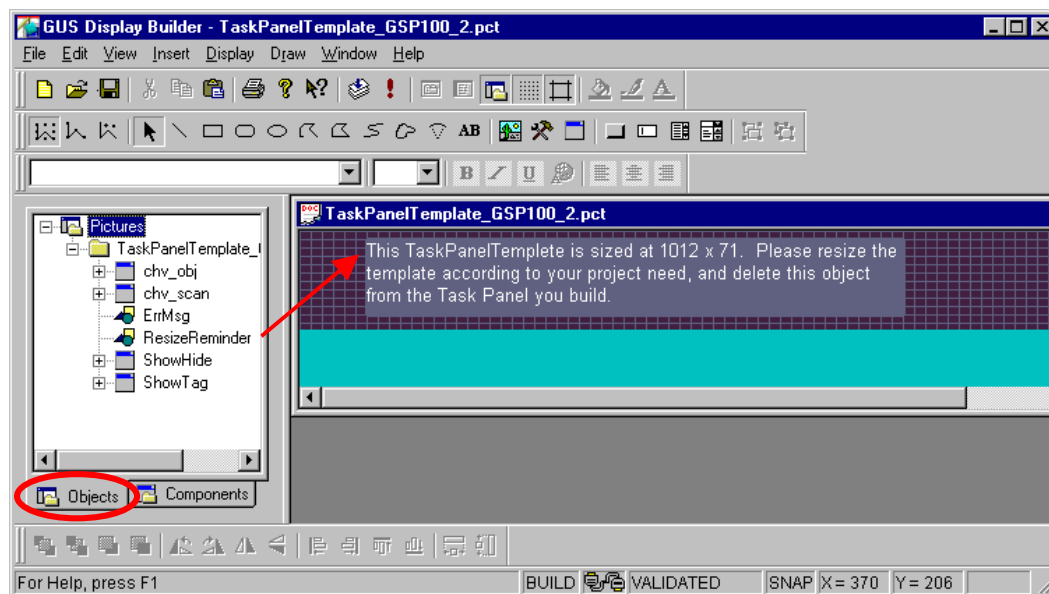
This is a text object to remind the user to resize the TaskPanelTemplate to meet the project need. Refer to the SafeView Configurations section of this document on how to decide the window and display sizing. User should delete this object from the template before using it to build task panel displays.

chv_obj

This is an embedded object that handles the interwindow communication. **Do not delete** this object from the template.

chv_scan

This is an embedded object that scans for currently active displays on the station. **Do not delete** this object from the template.



LowLevelNavTemplate.pct

The Low-Level Navigation Display Template must be used as the starting template to build all the Low Level navigation displays.

The Low-Level navigation displays provide a level of context sensitive navigation buttons within the task panel.

The Low-Level navigation displays are embedded displays that must be inserted into the TaskPanel display. Each Low-Level Navigation display is connected to a High-Level navigation button that the user inserted into the TaskPanel display. Multiple Low-Level navigation displays can be inserted into a TaskPanel display -- only the one connected to the currently selected High-Level navigation button will be visible.

Parameters

Parameter	Prompt	Required Entry
HighLevelDisplay	Enter High-Level display name connected to this display: (e.g., IsoOvw). Note: Only enter the display filename. Do not enter the pathname.	Yes

Objects

Display

Script

The script on the LowLevelNavTemplate.pct consists of an onDataChange and OnDisplayShutdown subroutines. **Users must not modify the standard portion of the script.** To handle special needs, users may append to the standard portion of the script.

Property

The properties referenced by GSP are described below:

Category: Does not matter.

Background Color: Does not matter. At run-time, the display background color is set to the Display_BG color defined according to the "GSP Color Constants" Table in the Appendix.

Size: Users may change this. It is critical to decide the size of the display that will fit into the available space in Task Panel display.

Display Components Reference

General Information

The following information applies to all the display components.

GSP Color Constants

GUS Solution Pack implements a set of Color Constants that are referenced throughout this document. A table is provided in the Appendix that describes all the color constants available.

Target Actions

The left-mouse-up on most of the display components will invoke a faceplate that displays the information of the point connected to the particular component. GSP refers to this as “the point is selected” throughout this document. Refer to the description of each component in the following sections for details.

The right-mouse-up on the display components is reserved by GSP.

File Version Identification

All GSP display components on the distribution media are provided with a suffix to indicate the version number. The version number is omitted throughout this document when the display components are mentioned.

Sizing at Build-Time and Run-Time

Unless explicitly mentioned, the GSP display components are non-resizable at build-time by design. This is to avoid users from inadvertently resizing the component and ending up with various sizes of the same component on the graphics.

All GSP display components will be resized appropriately according to the display size at run-time if zoom-to-fit property is set.

Outline Box

Many of the GSP display components include an Outline Box.

The color of the Outline Box indicates the alarm state when the point referenced by the display component is in alarm state.

A blinking Outline Box indicates an unacknowledged alarm for the point that is connected to the particular component. The blinking will stop when the alarm is acknowledged.

The color of the Outline Box turns to the “Selection” color (a GSP color constant) when the user left-mouse-clicks on the component. All components connected to the selected point will show the “Selection” color as long as the GUS Display windows are on the same GUS.

The Outline Box is always visible for control boxes, whereas it is invisible for measurement boxes unless the point is in alarm state or is selected. This helps the operator to differentiate the control boxes from the measurement boxes. Refer to the description of each component in the following sections for details.

How to Read the Behavior Tables

The behavior of each component is detailed through the tables included in the following sections. It describes the behavior of the objects in the component according to the combination of the parameter values. The parameter values are listed in the upper portion of the table and the behaviors of the objects are listed in the lower (shaded) portion of the table. The order of the priorities is listed from left to right. In some of the tables, the rightmost column shows the behavior when an error is encountered.

For example, the table below shows the behavior of the cascade symbols, which is set according to the following LCN parameters: MODE, INITMAN, RCASOPT, and CASREQ.

If the value of RCASOPT does not equal “NONE” and the value of CASREQ equals “REQUEST”, then the cascade symbols will be displayed in MAN color. Refer to the “GSP Color Constants” table in the Appendix.

The empty cell means the value of that parameter does not affect the behavior of the object. In this case, the values of MODE and INITMAN do not affect the behavior of the cascade symbols.

If the parameter values do not match the condition above, then check the next column to the right. In this case, if the value of MODE equals “CAS” and the value of INITMAN equals FALSE, then the cascade symbols will be displayed in AUTO color. The values of RCASOPT and CASREQ do not matter.

Similarly, if the parameter values do not match any of the conditions above, then check the next column. In this case, if the value of MODE equals “BCAS” and the value of INITMAN equals FALSE, then the cascade symbols will be displayed in AUTO color.

The next column indicates that if the values of MODE, INITMAN, RCASOPT, and CASREQ do not match any of the above conditions, then the cascade symbols will be displayed in MAN color.

Lastly, if any error was encountered, then the cascade symbols will be displayed in MAN color.

		Value				On Error
Parameter	mode		CAS	BCAS		
	initman		FALSE	FALSE		
	rcasopt	≠NONE				
	casreq	REQUEST				
Behavior	CascColor	MAN	AUTO	AUTO	MAN	MAN

Value Format

The value field displays the value (PV, SP, etc.) in the format according to the number of characters of the field and the number of digits configured whenever possible. In case the width of the field does not allow the integer portion of the value to be displayed in the desired format, the component will display ***** to indicate field overflow. Below are some of the examples:

Field Width	Digits	Value	Displayed	Comment
5	2	12.3	12.30	
5	2	123.4	123.4	digit adjusted
5	2	-12345	*****	overflow

Error Indication

The following strings will be displayed in the value field if any errors are detected.

Error Encountered	Displayed
COMMUNICATION_ERROR	?
CONFIGURATION_ERROR	@
VALUE_ERROR	-
FETCH/STORE_ERROR	!
Any other errors	?

Large Control Boxes

Control Boxes are display components used to provide operator with an easy access to all regulatory control loops displayed on a graphic. GSP includes two sets of control boxes: a) Large and b) Small Control Boxes. Users can decide which size to use. The Large Control Boxes provides the Setpoint, Tagname and Mode information, whereas the Small Control Boxes take less space on the screen and induce less load on the TPN system.

There are four versions of the Large Control Boxes that differ only in the number of characters used to display tagname. Use the 8-character version if possible. Use the 16-character version if you need to display tagname longer than 8 characters. Use the 19-character version if the tagname includes a network gateway identifier.

Point Types

Large Control Boxes are designed to be used for Regulatory Control Data Points with PNTFORM=Full, PVSRCOPT=All, and with any of the following algorithms:

PID

PID with External Reset Feedback (PidErfb)

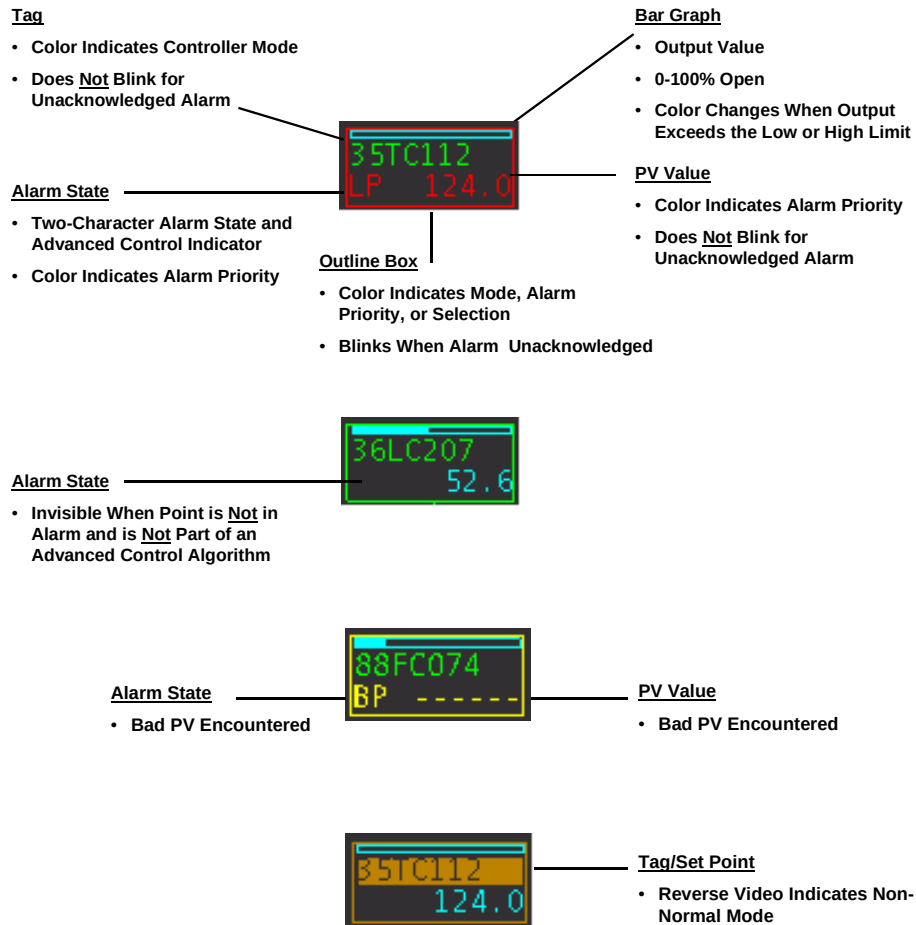
PID with Feed Forward (PidFf)

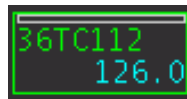
Position Proportional (PosProp)

Ratio Control (RatioCtl, PidRatio)

Basic Description

Large Control Box





Bar Graph

- Output Value
- 0-100% Open
- Color Changes When Output Exceeds the Low or High Limit



Set Point

- Color Indicates Controller Mode

Display switches between Tag and SetPoint through the use of the Tag/SP Switch (tag_sp_btn.pct) display component.

The color of Tag/SP indicates Mode. Non-normal mode is displayed in reverse video.

Process Variable Alarm State (2 characters).

Includes a target to invoke faceplate.

Filename

For PM/APM/HPM/AM:

cntl_box_lg_lt.pct – Includes 8-character tagname left aligned.

cntl_box_lg_rt.pct – Includes 8-character tagname right aligned.

cntl_box_lg_16.pct – Includes 16-character tagname centered.

cntl_box_lg_19.pct – Includes 19-character tagname centered.

For HG/PLCG:

cntl_box_lg_hg_lt.pct – Includes 8-character tagname left aligned.

cntl_box_lg_hg_rt.pct – Includes 8-character tagname right aligned.

cntl_box_lg_hg_16.pct – Includes 16-character tagname centered.

cntl_box_lg_hg_19.pct – Includes 19-character tagname centered.

LCN Parameters Referenced

Reference_name	Comment
ackstat collector of point	
point.alenbst	
point.highal	
point.highalpr	
point.initman	
point.modattr	
point.mode	
point.nmodattr	
point.nmode	
point.op	
point.ophifl	
point.oplofl	
point.ptorst	PM/APM/HPM/AM
point.pv	
point.pvformat	
point.pvsource	PVSRCOPT = All
point.pvsts	
point.sp	

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
digits	Enter Number of Digits After Decimal Point: (0~3, default uses PVFORMAT)	-1	
ac_label	Enter 2-Character Advance Control Indicator: (e.g., AC, CV, MV, etc.)		

The 2-character Advance Control Indicator can be used to remind the operators that the point is part of an advance control algorithm. If characters are entered, they will be displayed in the Alarm State field when the point is not in alarm state.

Behavior

Bar Graph

The Bar Graph shows the OP value of the point. The color of the Bar Graph is set according to the real-time value of OPHIFL and OPLOFL as shown in the table below:

		Value		
parameter	ophifl	TRUE		
	oplofl		TRUE	
Behavior	Bar Graph Color	OP_Limit	OP_Limit	NOALARM

Outline Box

The line color of the Outline Box is set according to the real-time value of HIGHAL, HIGHALPR, ACKSTAT, MODE, PTORST and INITMAN as shown in the table below:

		Value													
		Selected	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Parameter	highal			NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL
	highalpr														
	ackstat			UNAKALRM											
	mode				AUTO	CAS	BCAS								
	ptorst				SEL or NOTCON	SEL or NOTCON	SEL or NOTCON								
	initman				FALSE	FALSE	FALSE								
Behavior	Outline Box Color														
	Selection														
	Ghost_Alarm														
	AUTO														
	AUTO														
	AUTO														
	MAN														
	High_Alarm														
	High_Alarm														
	Low_Alarm														
	JNLPRINT														
	PRINTER														
	JOURNAL														
	Display_BG														

Display Components Reference - Large Control Boxes

Blinking indicates an Unacknowledged Alarm. The blinking will stop when the alarm is acknowledged. The blinking behavior is set according to the real-time value of ACKSTAT as shown in the table below:

		Value	
Parameter	ackstat	UNAKALRM	
Behavior	UnAckAlarm	TRUE	FALSE

PV Value

This is a 6-character field that displays the PV value of the point. The display format is set according to the “digits” display parameter, and defaults to PVFORMAT.

The foreground and background colors are set according to the real-time value of HIGHAL and HIGHALPR as shown in the table below:

		Value							
Parameter	highal	NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	
	highalpr		EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL	
Behavior	Foreground Color	NOALARM	Emergency_FG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	NOALARM
	Background Color	Display_BG	Emergency_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG

Tag and SetPoint

The large control box switches between displaying Tag and SetPoint through the use of the Tag/SP Switch (tag_sp_btn.pct) display component.

Tag

When the switch is set to "Show Tag", the tagname of the point is displayed. There are multiple versions of the large control box that differs in the number of characters used to display tagname. If no more than 8 characters are used for the tagname, it is best to use the 8-character version which takes less screen space.

SetPoint

When the switch is set to "Show SP", the setpoint is displayed. It includes a "SP" label and a 6-character field that displays the SP value of the point. The display format is set according to the "digits" display parameter, and defaults to PVFORMAT.

Display Components Reference - Large Control Boxes

The foreground and background colors of Tag and SetPoint fields are set according to the real-time value of MODE, NMODE, MODATTR, NMODATTR, PTORST, and INITMAN as shown in the table below:

Behavior		Parameter						Value																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
Background Color	Foreground Color	initman	ptorst	nmodattr	modattr	nmode	mode																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										

Alarm State

The Alarm State is a 2-character field to display the status of the point. The text is set according to the real-time value of ALENBST, HIGHAL, PVSOURCE, PVSTS, and the value of the Advance Control Indicator display parameter as shown in the table below:

		Value																	
								INHIBIT	DISABLE	ENABLE									
Parameter	alenbst																		
	highal	BADOC	OPHI	OPLO					NOALARM	NOALARM	BADPV	OPENTHER	UNREASBL	PVLL	PVHH	PVLO	PVHI	PVROCP	PVROCN
	pvsources				SUB	MAN													
							UNCERTN												
	pvssts ac_label																		
Behavior	Point Status Text	BO	OH	OL	S	M	U	IN	DS	ac_label	BP	OT	UN	LL	HH	LP	HP	RC	RC
																		LD	HD
																			??

Display Components Reference - Large Control Boxes

The color of the text is set according to the Alarm State text and the real-time values of the HIGHAL and HIGHALPR parameters as shown in the table below:

		Value											
		Point Status Text							??	IN	DS	S	M
Parameter		highal	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM
		highalpr	EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL					
Behavior	Point Status Color	Emergency_FG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	ALARM	Low_Alarm	Low_Alarm	Low_Alarm	Low_Alarm	Low_Alarm
												AC_MV	Display_BG

Small Control Boxes

Control Boxes are display components used to provide operator with an easy access to all regulatory control loops displayed on a graphic. GSP includes two sets of control boxes: a) Large and b) Small Control Boxes. Users can decide which size to use. The Large Control Boxes provides the Setpoint, Tagname and Mode information, whereas the Small Control Boxes take less space on the screen and induce less load on the TPN system.

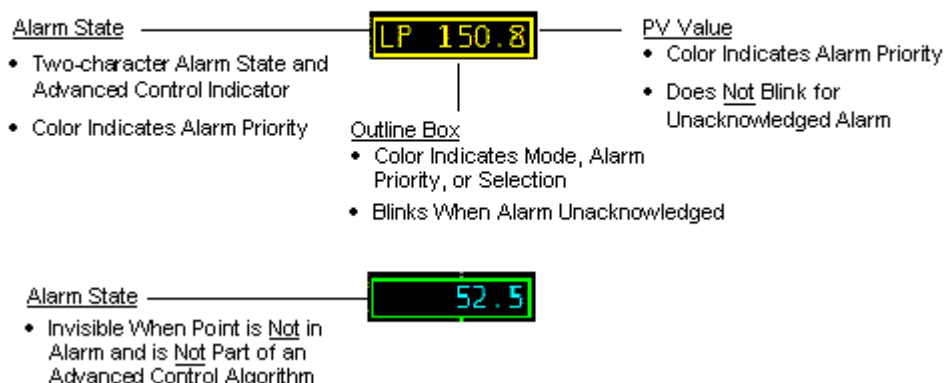
Point Types

Small Control Boxes are designed to be used for Regulatory Control Data Points with PNTFORM=Full, PVSRCOPT=All, and with any of the following algorithms:

- PID
- PID with External Reset Feedback (PidErfb)
- PID with Feed Forward (PidFf)
- Position Proportional (PosProp)
- Ratio Control (RatioCtl, PidRatio)

Basic Description

Small Control Box



Includes a target to invoke faceplate.

Filename

For PM/APM/HPM/AM:

cntl_box_sm_3.pct – Includes 3-character PV value.

cntl_box_sm_4.pct – Includes 4-character PV value.

cntl_box_sm_5.pct – Includes 5-character PV value.

cntl_box_sm_6.pct – Includes 6-character PV value.

cntl_box_sm_7.pct – Includes 7-character PV value.

For HG/PLCG:

cntl_box_sm_hg_3.pct – Includes 3-character PV value.

cntl_box_sm_hg_4.pct – Includes 4-character PV value.

cntl_box_sm_hg_5.pct – Includes 5-character PV value.

cntl_box_sm_hg_6.pct – Includes 6-character PV value.

cntl_box_sm_hg_7.pct – Includes 7-character PV value.

LCN Parameters Referenced

Reference_name	Comment
ackstat collector of point	
point.alenbst	
point.highal	
point.highalpr	
point.initman	
point.mode	
point.ptorst	PM/APM/HPM/AM
point.pv	
point.pvformat	
point.pvsource	PVSRCOPT = All
point.pvsts	

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
digits	Enter Number of Digits After Decimal Point: (0~3, default uses PVFORMAT)	-1	
ac_label	Enter 2-Character Advance Control Indicator: (e.g., AC, CV, MV, etc.)		

The 2-character Advance Control Indicator can be used to remind the operators that the point is part of an advance control algorithm. If characters are entered, they will be displayed in the Alarm State field when the point is not in alarm state.

Behavior

Outline Box

The line color of the Outline Box is set according to the real-time value of HIGHAL, HIGHALPR, ACKSTAT, MODE, PTORST, and INITMAN as shown in the table below:

Behavior	Parameter	Value													
		Selected	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Outline Box Color	highal			NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM
	highalpr								EMERGNCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL	
	ackstat			UNAKALRM											
	mode				AUTO	CAS	BCAS								
	ptorst				SEL or NOTCON	SEL or NOTCON	SEL or NOTCON								
	initman				FALSE	FALSE	FALSE								
	Selection														
	Ghost_Alarm														
	AUTO														
	AUTO														
	AUTO														
	MAN														
	High_Alarm														
	High_Alarm														
	Low_Alarm														
	JNLPRINT														
	PRINTER														
	JOURNAL														
	Display_BG														

Blinking indicates an Unacknowledged Alarm. The blinking will stop when the alarm is acknowledged. The blinking behavior is set according to the real-time value of ACKSTAT as shown in the table below:

		Value	
Parameter	ackstat	UNAKALRM	
Behavior	UnAckAlarm	TRUE	FALSE

PV Value

This is a 6-character field that displays the PV value of the point. The display format is set according to the “digits” display parameter, and defaults to PVFORMAT.

The foreground and background colors are set according to the real-time value of HIGHAL and HIGHALPR as shown in the table below:

		Value							
Parameter	highal	NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	
	highalpr		EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL	
Behavior	Foreground Color	NOALARM	Emergency_FG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	NOALARM
	Background Color	Display_BG	Emergency_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG

Point Status

The Point Status is a 2-character field to display the status of the point. The text is set according to the real-time value of ALENBST, HIGHAL, PVSOURCE, PVSTS and the value of the Advance Control Indicator display parameter as shown in the table below:

		Value																	
								INHIBIT	DISABLE	ENABLE									
Parameter	alenbst																		
	highal	BADOC	OPHI	OPLO					NOALARM	NOALARM	BADPV	OPENTHER	UNREASBL	PVLL	PVHH	PVLO	PVHI	PVROCP	PVROCN
	pvsources				SUB	MAN													
							UNCERTN												
	pvssts ac_label																		
Behavior	Point Status Text	BO	OH	OL	S	M	U	IN	DS	ac_label	BP	OT	UN	LL	HH	LP	HP	RC	RC
																		LD	HD
																			??

The color of the text is set according to the Alarm State text and the real-time values of the HIGHAL and HIGHALPR parameters as shown in the table below:

Behavior	Parameter	Value															
		Point Status Text	highal	highalpr	Emergency_FG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	ALARM	Low_Alarm	Low_Alarm	Low_Alarm	Low_Alarm	Low_Alarm	AC_MV

Output Boxes

There are three types of Output Boxes included in the GSP package:

Controller Output Boxes with Tag

Analog Output Boxes with Tag

Output Value Only

These are display components that allow the display of an output value (OP). At runtime, the display component will invoke the faceplate or update the information in the faceplate when it is selected.

Controller Output Boxes With Tag

Controller Output Boxes are display components provided to display the output value, mode, and alarm information for controller points with an output only (no PV). It should not be used for direct analog output points. Use the Analog Output Boxes (anlg_op_box) for direct analog output points.

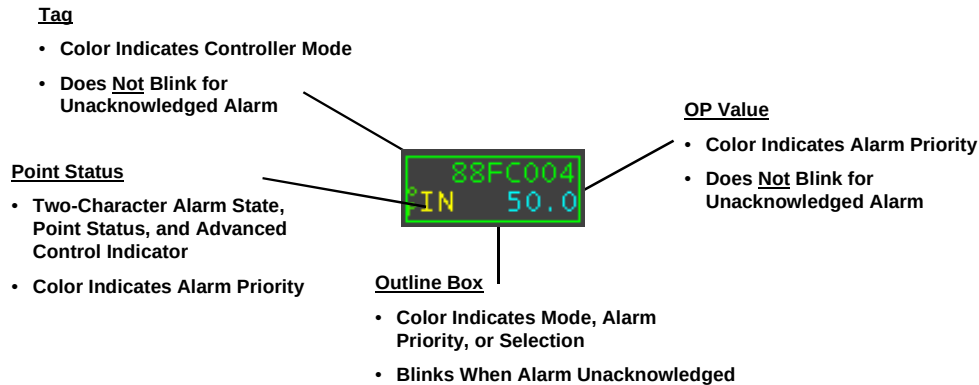
There are four versions of the Controller Output Boxes that differ only in the alignment or the number of characters used to display the tagname. Use the 8-character version if possible. Use the 16-character version if you need to display a tagname longer than 8

characters. Use the 19-character version if the tagname includes a network gateway identifier.

Point Types

Controller Output Boxes are designed to be used for Regulatory Control Data Points with an output only (no PV) and PNTFORM=Full. An example is an Auto/Manual station. The display components can be used for output only controllers on all NIM box types (PM, APM, HPM, etc.).

Controller Output Box



Basic Description

The color of Tag indicates Mode. Non-normal mode is displayed in reverse video. Point Status (2 characters) shows the point status or alarm state. Includes a target to invoke faceplate.

Filename

cntl_op_box_lt.pct – Includes 8-character tagname left aligned.
cntl_op_box_rt.pct – Includes 8-character tagname right aligned.
cntl_op_box_16.pct – Includes 16-character tagname centered.
cntl_op_box_19.pct – Includes 19-character tagname centered.

LCN Parameters Referenced

Reference_name	Comment
ackstat collector of point	
point.alenbst	
point.highal	
point.highalpr	
point.initman	
point.modattr	
point.mode	
point.nmodattr	
point.nmode	
point.op	
point.ophifl	
point.oplofl	

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
digits	Enter Number of Digits After Decimal Point: (0~3)	1	
ac_label	Enter 2-Character Advance Control Indicator: (e.g., AC, CV, MV, etc.)		

The 2-character Advance Control Indicator can be used to remind the operators that the point is part of an advance control algorithm. If characters are entered, they will be displayed in the Alarm State field when the point is not in alarm state.

Behavior

Outline Box

The line color of the Outline Box is set according to the real-time value of HIGHAL, HIGHALPR, ACKSTAT, MODE, and INITMAN as shown in the table below:

		Value													
		Selected	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Parameter	highal		NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM
	highalpr								EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL	
	ackstat		UNAKALRM												
	mode			AUTO	CAS	BCAS									
	initman			FALSE	FALSE	FALSE									
Behavior	Outline Box Color	Selection	Ghost_Alarm	AUTO	AUTO	AUTO	MAN	High_Alarm	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	Display_BG	

Blinking indicates an Unacknowledged Alarm. The blinking will stop when the alarm is acknowledged. The blinking behavior is set according to the real-time value of ACKSTAT as shown in the table below:

		Value	
Parameter	ackstat	UNAKALRM	
Behavior	UnAckAlarm	TRUE	FALSE

Output Value

This is a 6-character field that displays the OP value of the point. The display format is set according to the “digits” display parameter, and defaults to 1.

The foreground and background colors are set according to the real-time value of HIGHAL and HIGHALPR as shown in the table below:

		Value							
Parameter	highal	NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	
	highalpr		EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL	
Behavior	Foreground Color	NOALARM	Emergency_FG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	NOALARM
	Background Color	Display_BG	Emergency_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG

Tag

There are multiple versions of the Controller Output Box that differ in the alignment or the number of characters used to display a tagname. If no more than 8 characters are used for the tagname, it is best to use the 8-character version, which takes less screen space.

Behavior

The foreground and background colors of the Tag field are set according to the real-time value of MODE, NMODE, MODATTR, NMODATTR, and INITMAN as shown in the table below:

Behavior		Parameter					Value												
Background Color	Foreground Color	initman	nmodattr	modattr	nmode	mode													
Display_BG	AUTO	FALSE	=modattr		=mode	AUTO													
Display_BG	AUTO	FALSE	NONE		=mode	AUTO													
Display_BG	AUTO	FALSE	=modattr		=mode	CAS													
Display_BG	AUTO	FALSE	NONE		=mode	CAS													
Display_BG	AUTO	FALSE	=modattr		=mode	BCAS													
Display_BG	AUTO	FALSE	NONE		=mode	BCAS													
Display_BG	MAN		=modattr		=mode														
Display_BG	MAN		NONE		=mode														
AUTO	Display_BG	FALSE			≠mode	AUTO				≠mode	AUTO								
AUTO	Display_BG	FALSE	≠NONE		≠modattr	AUTO				≠modattr	AUTO								
AUTO	Display_BG	FALSE			≠mode	CAS				≠modattr	CAS								
AUTO	Display_BG	FALSE	≠NONE		≠modattr	CAS					CAS								
AUTO	Display_BG	FALSE			≠mode	BCAS				≠modattr	BCAS								
AUTO	Display_BG	FALSE	≠NONE		≠modattr	BCAS					BCAS								
MAN	Display_BG				≠mode					≠modattr									
MAN	Display_BG				≠modattr					≠modattr									
Display_BG	MAN																		

Point Status

The Point Status is a 2-character field to display the status of the point. The text is set according to the real-time value of ALENBST, HIGHAL, and the value of the Advance Control Indicator display parameter as shown in the table below:

		Value						
Parameter					INHIBIT	DISABLE	ENABLE	
	alenbst							
	highal	BADOC				NOALARM	NOALARM	
	ophifl		TRUE					
	oplofl			TRUE				
	ac_label							
Behavior	Point Status Text	BO	OH	OL	IN	DS	ac_label	??

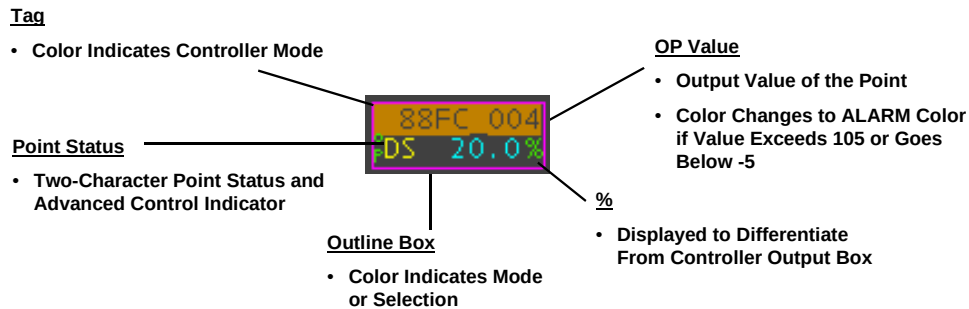
The color of the text is set according to the Alarm State text and the real-time values of the HIGHAL and HIGHALPR parameters as shown in the table below:

Behavior	Parameter	Value											
		Point Status Text											
	highal		NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM
	highalpr		EMERGENCY	HIGH	LOW	UNLPRINT	PRINTER	JOURNAL					
Point Status Color		Emergency_FG	High_Alarm	Low_Alarm	UNLPRINT	PRINTER	JOURNAL	ALARM	ALARM	ALARM	Low_Alarm	Low_Alarm	AC_MV
													Display_BG

Analog Output Boxes With Tag

Analog Output Boxes are display components provided to display the output value and mode for direct analog outputs. These are typically used for HIC's (hand indicating controller) to show the position of valves. A "%" is displayed to the right of the output value to differentiate from the Controller Output Boxes. If a control algorithm is used, such as Auto/Manual station, the Controller Output Boxes (cntl_op_box) should be used instead.

Analog Output Box



Point Types

Analog Output Boxes are designed to be used for AO points with PNTFORM=Full. The display components can be used for all NIM box types (PM, APM, HPM, etc.) for points that have all parameters listed in the LCN Parameters Referenced table below. For points with OP parameter only, use the Output Value Only Box.

Basic Description

The color of Tag indicates Mode. Non-normal mode is displayed in reverse video. Point Status (2 characters) shows the point status or alarm state. Includes a target to invoke faceplate.

Filename

anlg_op_box_lt.pct – Includes 8-character tagname left aligned.
 anlg_op_box_rt.pct – Includes 8-character tagname right aligned.
 anlg_op_box_16.pct – Includes 16-character tagname centered.
 anlg_op_box_19.pct – Includes 19-character tagname centered.

LCN Parameters Referenced

Reference_name	Comment
point.alenbst	
point.modattr	
point.mode	
point.nmodattr	
point.nmode	
point.op	

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
digits	Enter Number of Digits After Decimal Point: (0~2)	1	
ac_label	Enter 2-Character Advance Control Indicator: (e.g., AC, CV, MV, etc.)		

The 2-character Advance Control Indicator can be used to remind the operators that the point is part of an advance control algorithm. If characters are entered, they will be displayed in the Alarm State field when the point is not in alarm state.

Behavior

Outline Box

The line color of the Outline Box is set according to the real-time value of MODE as shown in the table below:

		Value					
Parameter	Selected	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE
	mode		AUTO	CAS	BCAS		
Behavior	Outline Box Color	Selection	AUTO	AUTO	AUTO	MAN	Display_BG

Output Value

This is a 5-character field that displays the OP value of the point. The display format is set according to the “digits” display parameter, and defaults to 1.

The output value foreground color is NOALARM (default to cyan) normally. When output value exceeds 105% or goes below –5%, the color will change to ALARM (default to yellow).

Tag

There are multiple versions of the Analog Output Box that differ in the alignment or the number of characters used to display tagname. If no more than 8 characters are used for the tagname, it is best to use the 8-character version, which takes less screen space.

Display Components Reference - Output Boxes

The foreground and background colors of the Tag field are set according to the real-time value of MODE, NMODE, MODATTR, and NMODATTR as shown in the table below:

		Value											
		Parameter											
	mode	=mode	AUTO										
	nmode	=mode	=mode	AUTO									
	modattr												
	nmodattr	=modattr	NONE										
	Foreground Color	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	MAN	MAN	Display_BG	Display_BG	Display_BG	Display_BG
	Background Color	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	AUTO	AUTO	AUTO	AUTO

Point Status

The Point Status is a 2-character field to display the status of the point. The text is set according to the real-time value of ALENBST, HIGHAL, and the value of the Advance Control Indicator display parameter as shown in the table below:

		Value		
		INHIBIT	DISABLE	ENABLE
	alenbst			
	ac_label			
	Point Status			
	Text	IN	DS	ac_label
				??

The color of the text is set according to the Alarm State as shown in the table below:

		Value				
Parameter	Point Status Text	??	IN	DS	≠	" "
Behavior	Point Status Color	ALARM	Low_Alarm	Low_Alarm	AC_MV	Display_BG

Output Value Only Boxes

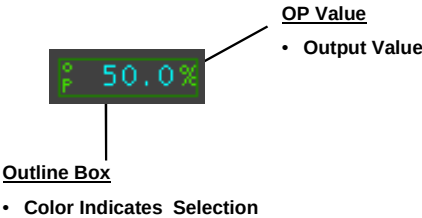
Output Value Only Boxes are display components provided for output of a loop. No tag or indication of mode is provided. It is typically used to display a valve position, not a PV. A “%” is displayed to the right of the output value.

Point Types

The display components can be used for all box types (PM, HG, etc.).

Basic Description

Output Value Only Box



Includes a target to invoke faceplate.

Filename

op_value_box_3.pct – Includes 3-character output value.
op_value_box_4.pct – Includes 4-character output value.
op_value_box_5.pct – Includes 5-character output value.
op_value_box_6.pct – Includes 6-character output value.

LCN Parameters Referenced

Reference_name	Comment
point.op	

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
digits	Enter Number of Digits After Decimal Point: (0~3)	1	

Behavior**Output Value**

This is a field that displays the OP value of the point. The display format is set according to the “digits” display parameter, and defaults to 1.

Control Valves

Control Valves are designed to show the locations of control valves relative to the process lines. The vertical and horizontal orientations are provided to allow process line to go through the valves as needed according to the display. The set of valves that includes an output bar is especially useful when coupled with small control boxes which do not have an output bar.

Point Types

Control Valves can be used for Regulatory Control Data Points that are connected to valves. It should be configured with PNTFORM=Full, and with any of the following algorithms:

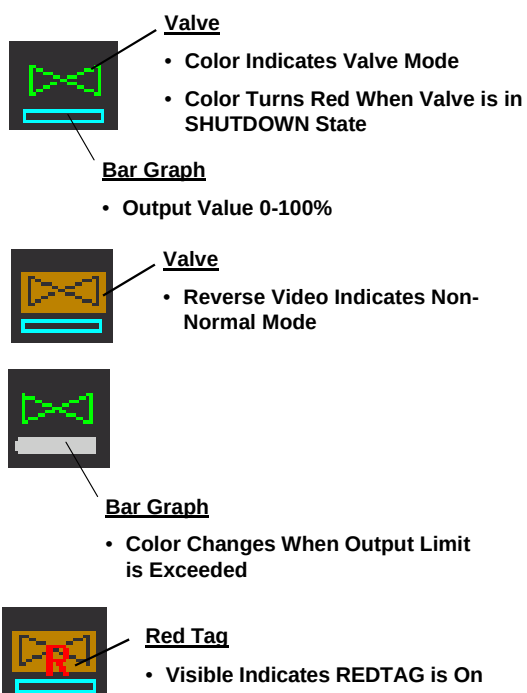
- PID
- PID with External Reset Feedback (PidErfb)
- PID with Feed Forward (PidFf)
- Position Proportional (PosProp)
- Ratio Control (RatioCtl)
- Auto Manual
- Ramp Soak
- Regulatory Data Point on HG/PLCG/AM

Basic Description

There are two sets of Control Valves provided – one with Bar Graph showing the OP value and one without. The bar graph changes color when OP exceeds the high or low output limit.

Each set of Control Valves includes several valve symbols with different orientation and shape. The behaviors of these valves are identical. Users may choose from the set according to the valve type (e.g., 3-way) and piping orientation required on the display. Users should not rotate the component at build-time because it will cause the “R” (Red Tag indicator) to rotate as well.

Control Valves



Filename

For PM/APM/HPM:

	Filename	Description
	cntl_vlv_hor.pct	Horizontal
	cntl_vlv_hor_hat.pct	Horizontal w/ Hat
	cntl_vlv_hor_3w.pct	Horizontal 3-Way
	cntl_vlv_hor_hat_3w.pct	Horizontal w/ Hat, 3-Way
	cntl_vlv_ver.pct	Vertical

Display Components Reference - Control Valves

	Filename	Description
	cntl_vlv_ver_hat.pct	Vertical w/ Hat
	cntl_vlv_ver_3w.pct	Vertical 3-Way
	cntl_vlv_ver_hat_3w.pct	Vertical w/ Hat, 3-Way
	cntl_vlv_op_hor_btm.pct	Horizontal Valve at Bottom, w/ Op Bar
	cntl_vlv_op_hor_btm_hat.pct	Horizontal Valve at Bottom, w/ Op Bar, Hat
	cntl_vlv_op_hor_btm_3w.pct	Horizontal Valve at Bottom, w/ Op Bar, 3-Way
	cntl_vlv_op_hor_btm_hat_3w.pct	Horizontal Valve at Bottom, w/ Op Bar, Hat, 3-Way
	cntl_vlv_op_hor_top.pct	Horizontal Valve on Top, w/ Op Bar
	cntl_vlv_op_hor_top_hat.pct	Horizontal Valve on Top, w/ Op Bar, Hat
	cntl_vlv_op_hor_top_3w.pct	Horizontal Valve on Top, w/ Op Bar, 3-Way
	cntl_vlv_op_hor_top_hat_3w.pct	Horizontal Valve on Top, w/ Op Bar, Hat, 3-Way
	cntl_vlv_op_ver_lt.pct	Vertical Valve at Left, w/ Op Bar
	cntl_vlv_op_ver_lt_hat.pct	Vertical Valve at Left, w/ Op Bar, Hat
	cntl_vlv_op_ver_lt_3w.pct	Vertical Valve at Left, w/ Op Bar, 3-Way
	cntl_vlv_op_ver_lt_hat_3w.pct	Vertical Valve at Left, w/ Op Bar, Hat, 3-Way

Display Components Reference - Control Valves

	Filename	Description
	cntl_vlv_op_ver_rt.pct	Vertical Valve at Right, w/ Op Bar
	cntl_vlv_op_ver_rt_hat.pct	Vertical Valve at Right, w/ Op Bar, Hat
	cntl_vlv_op_ver_rt_3w.pct	Vertical Valve at Right, w/ Op Bar, 3-Way
	cntl_vlv_op_ver_rt_hat_3w.pct	Vertical Valve at Right, w/ Op Bar, Hat, 3-Way

For HG/PLCG/AM:

	Filename	Description
	cntl_vlv_hor_hg.pct	Horizontal
	cntl_vlv_hor_hat_hg.pct	Horizontal w/ Hat
	cntl_vlv_hor_3w_hg.pct	Horizontal 3-Way
	cntl_vlv_hor_hat_3w_hg.pct	Horizontal w/ Hat, 3-Way
	cntl_vlv_ver_hg.pct	Vertical
	cntl_vlv_ver_hat_hg.pct	Vertical w/ Hat
	cntl_vlv_ver_3w_hg.pct	Vertical 3-Way
	cntl_vlv_ver_hat_3w_hg.pct	Vertical w/ Hat, 3-Way
	cntl_vlv_op_hor_btm_hg.pct	Horizontal Valve at Bottom, w/ Op Bar
	cntl_vlv_op_hor_btm_hat_hg.pct	Horizontal Valve at Bottom, w/ Op Bar, Hat

Display Components Reference - Control Valves

	Filename	Description
	cntl_vlv_op_hor_btm_3w_hg.pct	Horizontal Valve at Bottom, w/ Op Bar, 3-Way
	cntl_vlv_op_hor_btm_hat_3w_hg.pct	Horizontal Valve at Bottom, w/ Op Bar, Hat, 3-Way
	cntl_vlv_op_hor_top_hg.pct	Horizontal Valve on Top, w/ Op Bar
	cntl_vlv_op_hor_top_hat_hg.pct	Horizontal Valve on Top, w/ Op Bar, Hat
	cntl_vlv_op_hor_top_3w_hg.pct	Horizontal Valve on Top, w/ Op Bar, 3-Way
	cntl_vlv_op_hor_top_hat_3w_hg.pct	Horizontal Valve on Top, w/ Op Bar, Hat, 3-Way
	cntl_vlv_op_ver_lt_hg.pct	Vertical Valve at Left, w/ Op Bar
	cntl_vlv_op_ver_lt_hat_hg.pct	Vertical Valve at Left, w/ Op Bar, Hat
	cntl_vlv_op_ver_lt_3w_hg.pct	Vertical Valve at Left, w/ Op Bar, 3-Way
	cntl_vlv_op_ver_lt_hat_3w_hg.pct	Vertical Valve at Left, w/ Op Bar, Hat, 3-Way
	cntl_vlv_op_ver_rt_hg.pct	Vertical Valve at Right, w/ Op Bar
	cntl_vlv_op_ver_rt_hat_hg.pct	Vertical Valve at Right, w/ Op Bar, Hat
	cntl_vlv_op_ver_rt_3w_hg.pct	Vertical Valve at Right, w/ Op Bar, 3-Way
	cntl_vlv_op_ver_rt_hat_3w_hg.pct	Vertical Valve at Right, w/ Op Bar, Hat, 3-Way

LCN Parameters Referenced

Reference_name	Comment
point.initman	
point.modattr	
point.mode	
point.nmodattr	
point.nmode	
point.op	With OP Bar
point.ophifl	With OP Bar
point.oplofl	With OP Bar
point.ptorst	PM/APM/HPM
point.redtag	
point.shutdown	PM/APM/HPM

Display Parameters

Parameter	Prompt	Default	Required Entry
Point	Enter Tag: (Enclosed the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes

Behavior

Valve

The color of the valve is set according to the values of INITMAN, MODATTR, MODE, NMODATTR, NMODE, PTORST, and SHUTDOWN parameters as shown in the table below:

Behavior		Parameter								Value																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
Background Color	Foreground Color	shutdown	initman	ptorst	nmodattr	modattr	nmode	mode																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													</

Bar Graph

The color of the Bar Graph is normally NOALARM. When OP value exceeds the high or low output limit, the Bar Graph changes to OP_Limit color. It is set according to the values of OPHIFL and OPLOFL parameters as shown in the table below:

		Value		
parameter	ophifl	TRUE		
	oplofl		TRUE	
Behavior	Bar Graph Color	OP_Limit	OP_Limit	NOALARM

Selector Boxes

Selector Boxes are designed to be used on display to help describe a control strategy. Users can enter a descriptor up to 8 characters, such as "HI SEL" or "LO SEL."

Point Types

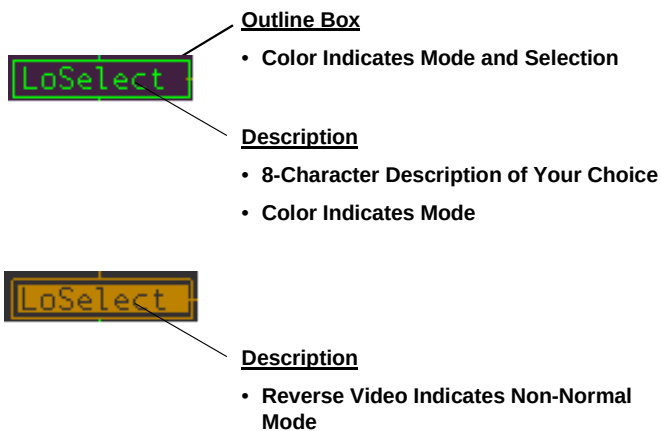
Selector Boxes can be used for Regulatory Control Data Points with PNTFORM=Full and with any of the following algorithms:

- Override Selector (ORSel) on PM/APM/HPM
- Override High Selector (Overhisl) HG
- Override Low Selector (Overlosl) HG
- Control Override Selector (OrSel) on AM

Basic Description

The selector boxes can be used on a display to help describe a control strategy. Users can enter an 8-character descriptor, such as "HI SEL" or "LO SEL."

Selector Boxes



Filename

For PM/APM/HPM/AM:

cntl_selector.pct

For HG:

cntl_selector_hg.pct

LCN Parameters Referenced

Reference_name	Comment
point.initman	
point.modattr	
point.mode	
point.nmodattr	
point.nmode	
point.ptorst	PM/APM/HPM/AM

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
Description	Enter up to 8 characters for description		Yes

The 8-character Advance Control Indicator can be used to remind the operators that the point is part of an advance control algorithm. If characters are entered, they will be displayed in the Alarm State field when the point is not in alarm state.

Behavior

Description

Displays the value entered for Description in the display parameter.

Outline Box

The color of the Outline Box is set according to the real-time value of MODE, NMODE, MODATTR, NMODATTR, PTORST, and INITMAN as shown in the table below:

		Value													
Parameter	Selected	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
	mode		AUTO	AUTO	CAS	CAS	BCAS	BCAS			AUTO	AUTO	CAS	CAS	BCAS
	nmode		=mode	=mode	=mode	=mode	=mode	=mode	=mode	=mode	#mode		#mode		#mode
	modattr										#nmodattr		#nmodattr		#nmodattr
	nmodattr		=modattr	NONE	=modattr	NONE	=modattr	NONE	=modattr	NONE	#NONE		#NONE		#NONE
	ptorst		SEL or NOTCON	SEL or NOTCON	SEL or NOTCON	SEL or NOTCON	SEL or NOTCON	SEL or NOTCON			SEL or NOTCON	SEL or NOTCON	SEL or NOTCON	SEL or NOTCON	SEL or NOTCON
	initman		FALSE	FALSE	FALSE	FALSE	FALSE	FALSE			FALSE	FALSE	FALSE	FALSE	FALSE
Behavior	Description Color														
	Background Color		Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG
	Outline Box Color	Selection	AUTO	AUTO	AUTO	AUTO	AUTO	AUTO	MAN	MAN	AUTO	AUTO	AUTO	AUTO	MAN

Level Bars

Level Bars are designed to provide a vertical or horizontal bar graph of the process variable. Most often, they are added inside or alongside of a process vessel to show the vessel level. These are one of the few GSP display components that are designed to be resizable at build-time.

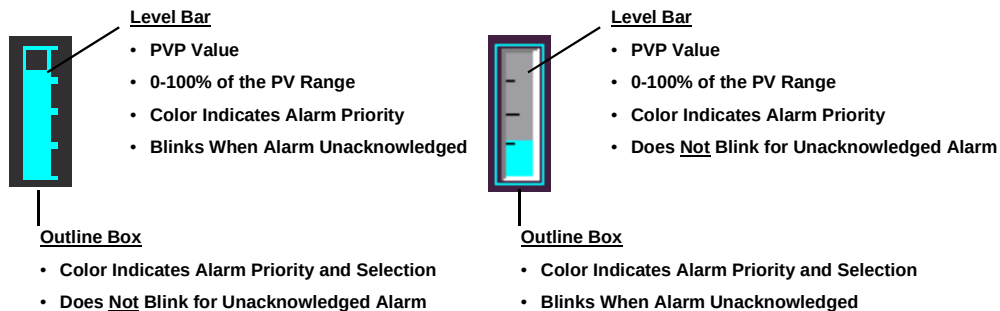
Point Types

Level Bars can be used for the Regulatory PV and Analog Input point types with PNTFORM=Full.

Basic Description

The Level Bars are provided to display dynamic levels. They can be resized at build-time as needed. The Bar Graph shows the PV in percentage of PV range (PVP). The Color indicates alarm priority. Blinking indicates unacknowledged alarm.

Level Bar



Filename

	Filename	Description
	level_bar_hor.pct	Horizontal
	level_bar_ver.pct	Vertical
	level_bar_3D_hor.pct	Horizontal
	level_bar_3D_ver.pct	Vertical

LCN Parameters Referenced

Reference_name
ackstat collector of point
point.highal
point.highalpr
point.pvp

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag for Secondary Loop: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes

Behavior

The color of the Level Bar and Outline Box are set according to the real-time value of HIGHAL, HIGHALPR, and ACKSTAT as shown in the table below:

Parameter	Selected										
		TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
			NOALARM	±NOALARM	±NOALARM	±NOALARM	±NOALARM	±NOALARM	±NOALARM	±NOALARM	
				EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL		
			UNAKALRM								
Behavior	Level Bar Color		NOALARM	Emergency_FG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	NOALARM	
	Outline Color	Selection	Ghost_Alarm	Emergency_BG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	NOALARM	

Cascade Symbols

Cascade Symbols are designed to show the configuration and status of a cascade on the display. Use a line with an arrowhead between controllers in a cascade arrangement. The arrow should point in the direction from the primary controller to the secondary controller in the control loop. Users can choose either the solid line or the dotted line, but keep it consistent. They have the same behavior, and both can be sized and rotated as needed.

Point Types

Cascade symbols are used to display cascade status on the graphic. They are designed to be used for any of the Regulatory Control Data Points that are configured as the secondary controller in a cascade.

Basic Description

	Cascade Arrowhead
	Cascade Continuation Line
	Cascade Dotted Line
	Cascade Line

To use these components, add the cascade line between the primary controller and the secondary controller, scale (resize) or rotate the line as needed. Then add the arrowhead symbol pointing from the primary towards the secondary in the control loop. Enter the secondary controller as the tag.

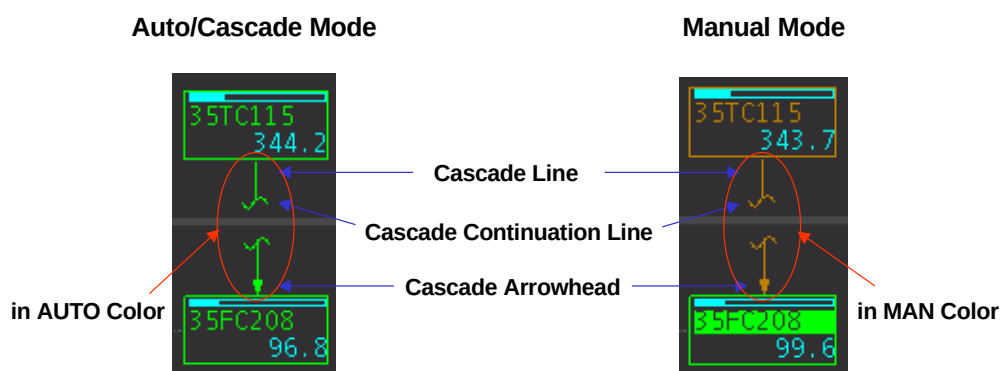
The status of the cascade is shown by changing the color of the line and arrow when the cascade is made and broken. To be consistent with the control box color behavior:

The AUTO color is used to show that the secondary controller is in cascade (primary controller should be in AUTO mode).

The MAN color is used to show that the secondary controller is not in cascade (e.g., auto or manual). In this case the secondary controller is not in cascade, the primary controller should be in MAN color to indicate initialization or manual mode.

Display Components Reference - Cascade Symbols

The following examples show how the cascade status can be displayed using the cascade symbols:



Filename

For PM/APM/HPM:

casc_arrow.pct – Arrowhead.

casc_continue.pct – Squiggle line to indicate continuation.

casc_dline.pct – Dotted line.

casc_line.pct – Line.

For HG/PLCG/AM:

casc_arrow_hg.pct – Arrowhead.

casc_continue_hg.pct – Squiggle line to indicate continuation.

casc_dline_hg.pct – Dotted line.

casc_line_hg.pct – Line.

LCN Parameters Referenced

Reference_name	Comment
point.casreq	Referenced by PM/APM/HPM
point.initman	
point.mode	
point.rcasopt	Referenced by PM/APM/HPM

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag for Secondary Loop: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes

Behavior

The color of the symbols are set according to the real-time value of MODE, INITMAN, RCASOPT, and CASREQ as shown in the table below:

		Value				On Error
Parameter	mode		CAS	BCAS		
	initman		FALSE	FALSE		
	rcasopt	≠NONE				
	casreq	REQUEST				
Behavior	CascColor	MAN	AUTO	AUTO	MAN	MAN

Measurement Boxes

Measurement Boxes are designed to show PV and point status. It is highly recommended that either the engineering units or the tagname be added to the display to allow the operator to know the type of measurement. To reduce clutter on the screen, users may use the show/hide feature to toggle the visibility of text at the operator's request. Refer to "Building Process Displays" on how to accomplish this.

Visually, the major difference between the Measurement Boxes and the Small Control Boxes is the Outline Box. The Outline Box for Measurement Boxes is only visible when the point is in alarm state or selected, whereas the Outline Box for Control Boxes is always visible.

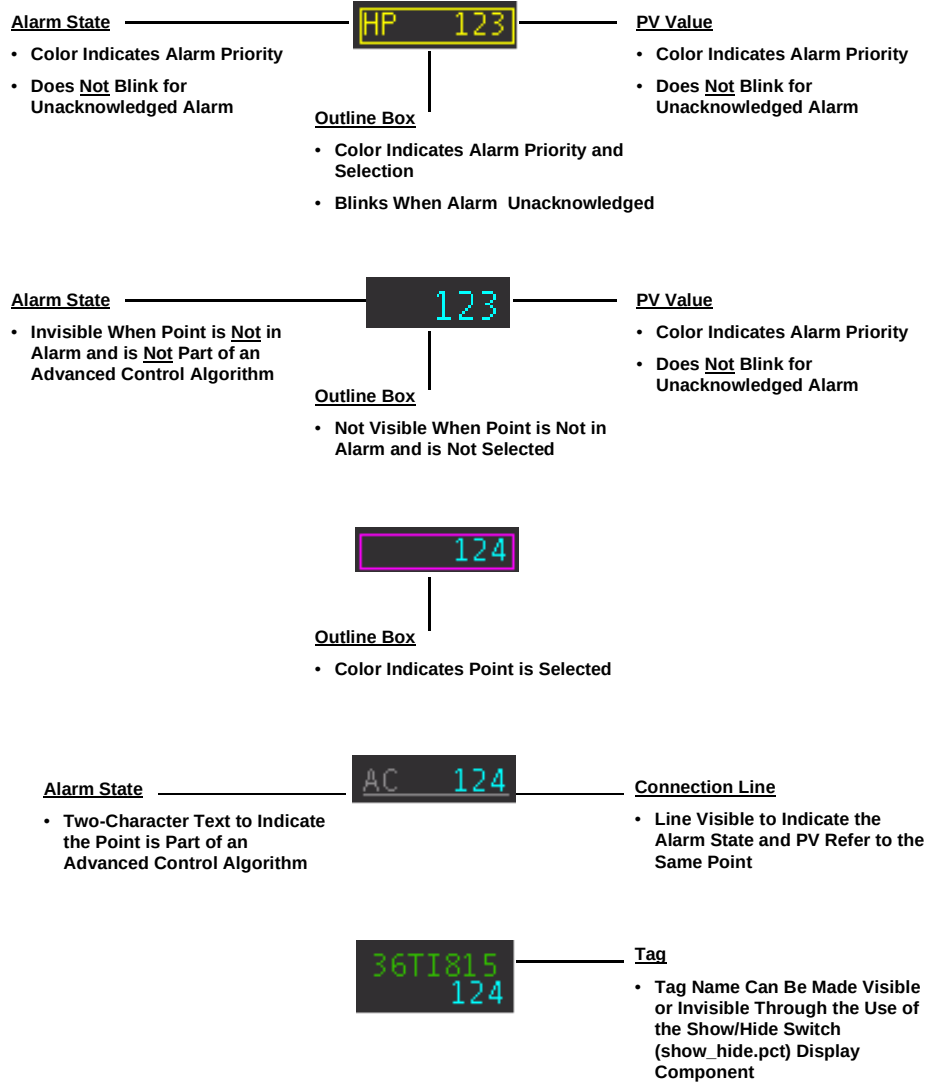
The Measurement Boxes will show a Connection Line between the two-character ac_label (same location as the alarm state) and the PV if the point is part of an Advanced Control Algorithm. This is included to make sure that the operator knows that the label and the PV belong to the same point, since there will be no Outline Box to group these two pieces of information when the point is not in alarm state.

Point Types

Measurement boxes are used to display the PV and point status. They can be used for Regulatory PV Points on PM/APM/HPM and Regulatory Data Points on HG/PLCG/AM with PNTFORM=Full, PVSRCOPT=All.

Basic Description

Measurement Box



Outline Box only visible when the point is selected or in alarm state.

Show or hide tagname through the use of the Show/Hide Switch (show_hide.pct) display component.

Includes a target to invoke faceplate.

Filename

meas_box_3.pct – Includes 3-character PV value.
 meas_box_4.pct – Includes 4-character PV value.
 meas_box_5.pct – Includes 5-character PV value.
 meas_box_6.pct – Includes 6-character PV value.
 meas_box_7.pct – Includes 7-character PV value.
 meas_box_tag_3.pct – Includes 3-character PV value and tag.
 meas_box_tag_4.pct – Includes 4-character PV value and tag.
 meas_box_tag_5.pct – Includes 5-character PV value and tag.
 meas_box_tag_6.pct – Includes 6-character PV value and tag.
 meas_box_tag_7.pct – Includes 7-character PV value and tag.

LCN Parameters Referenced

Reference_name	Comment
ackstat collector of point	
point.alenbst	
point.highal	
point.highalpr	
point.pv	
point.pvformat	
point.pvsource	PVSRCOPT = All
point.pvsts	

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
digits	Enter Number of Digits After Decimal Point: (0~3, default uses PVFORMAT)	-1	
ac_label	Enter 2-Character Advance Control Indicator: (e.g., AC, CV, MV, etc.)		

The 2-character Advance Control Indicator can be used to remind the operators that the point is part of an advance control algorithm. If characters are entered, they will be displayed in the Alarm State field when the point is not in alarm state.

Behavior

Outline Box

The line color of the Outline Box is set according to the real-time value of HIGHAL, HIGHALPR, and ACKSTAT as shown in the table below:

		Value									
Parameter	Selected	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
	highal		NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	
	highalpr			EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL		
	ackstat		UNAKALRM								
Behavior	Outline Box Color	Selection	Ghost_Alarm	Emergency_BG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	Display_BG	

Blinking indicates an Unacknowledged Alarm. The blinking will stop when the alarm is acknowledged. The blinking behavior is set according to the real-time value of ACKSTAT as shown in the table below:

		Value	
Parameter	ackstat	UNAKALRM	
Behavior	UnAckAlarm	TRUE	FALSE

Behavior		Parameter	Value
Background Color	Foreground Color	highalpr	highal
		Display_BG	NOALARM
		Emergency_BG	Emergency_FG
		Display_BG	High_Alarm
		Display_BG	Low_Alarm
		Display_BG	JNLPRINT
		Display_BG	PRINTER
		Display_BG	JOURNAL
Display_BG	NOALARM		

Point Status

The Point Status is a 2-character field to display the status of the point. The text is set according to the real-time value of ALENBST, HIGHAL, PVSOURCE, PVSTS, and the value of the Advance Control Indicator display parameter as shown in the table below:

		Value																	
		alenbst	highal	pvsources	pvs	ac_label	INHIBIT	DISABLE	ENABLE										
Parameter	alenbst																		
	highal	BADOC	OPHI	OPLO					NOALARM	NOALARM	BADPV	OPENTHER	UNREASBL	PVLL	PVHH	PVLO	PVHI	PVROCP	PVROCN
	pvsources				SUB	MAN													
	pvs						UNCERTN												
	ac_label																		
Behavior	Point Status Text	BO	OH	OL	S	M	U	IN	DS	ac_label	BP	OT	UN	LL	HH	LP	HP	RC	RC
																		LD	HD
																			??

When the point is not in alarm but other state text is to be displayed (e.g., Advance Control Indicator), there will be a Connection Line between the state text and the PV. This helps the operators to recognize that the Advance Control Indicator is part of the measurement box display, since the Outline Box will not be visible in this case.

Display Components Reference - Measurement Boxes

The color of the text is set according to the Alarm State text and the real-time values of the HIGHAL and HIGHALPR parameters as shown in the table below:

		Value											
		Point Status Text							??	IN	DS	S	M
Parameter		highal	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	NOALARM
		highalpr	EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL					
Behavior	Point Status Color	Emergency_FG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	ALARM	Low_Alarm	Low_Alarm	Low_Alarm	Low_Alarm	Low_Alarm
												AC_MV	Display_BG

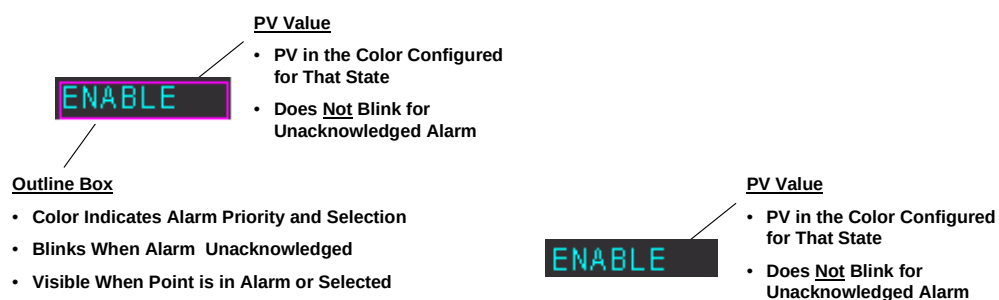
Digital State Boxes

Point Types

Digital State Boxes are used to display the PV State of digital points. They can be used for digital input, digital composite, device control, and flag points with PNTFORM=Full.

Basic Description

Digital State Box



The Digital State Boxes display the state of the PV in the color configured for that state. The outline of the components shows the alarm and acknowledgment status. The component includes a target to invoke faceplate.

Filename

For PM/APM/HPM Digital Input and Flag:

dig_stat_3.pct – Include 3-character PV state.

dig_stat_4.pct – Include 4-character PV state.

dig_stat_5.pct – Include 5-character PV state.

dig_stat_6.pct – Include 6-character PV state.

dig_stat_7.pct – Include 7-character PV state.

dig_stat_8.pct – Include 8-character PV state.

For HG/PLCG Digital and Flag, AM Flag:

dig_stat_hg_3.pct – Include 3-character PV state.

dig_stat_hg_4.pct – Include 4-character PV state.

dig_stat_hg_5.pct – Include 5-character PV state.

dig_stat_hg_6.pct – Include 6-character PV state.

dig_stat_hg_7.pct – Include 7-character PV state.

dig_stat_hg_8.pct – Include 8-character PV state.

For PM/APM/HPM Digital Composite and Device Control:

dc_stat_3.pct – Include 3-character PV state.

dc_stat_4.pct – Include 4-character PV state.

dc_stat_5.pct – Include 5-character PV state.

dc_stat_6.pct – Include 6-character PV state.

dc_stat_7.pct – Include 7-character PV state.

dc_stat_8.pct – Include 8-character PV state.

LCN Parameters Referenced

Reference_name	Module	Comment
ackstat collector of point		
point.highal		
point.highalpr		
point.nostates	PM/APM/HPM	Digital Composite / Device Control
point.pv		
point.s0boxclr	PM/APM/HPM	
point.s1boxclr	PM/APM/HPM	
point.s2boxclr	PM/APM/HPM	NOSTATES = 3
point.statetxt(0)	PM/APM/HPM	
point.statetxt(1)	PM/APM/HPM	
point.statetxt(2)	PM/APM/HPM	NOSTATES = 3
point.lboxclr	HG/PLCG/AM	
point.state1	HG/PLCG/AM	
point.state2	HG/PLCG/AM	
point.uboxclr	HG/PLCG/AM	

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes

Behavior

PV_Value

Displays the PV State of the point. The color is set to the color configured for that state except when the point is in alarm with emergency alarm priority. In this case, the color is set to "Emergency_FG" on "Emergency_BG."

Behavior		Parameter										Value									
		Background Color	Foreground Color	s2boxclr	statetxt2	nostates	s1boxclr	s0boxclr	statetxt1	statetxt0	highalpr	highal	PV Value								
Emergency_BG	Emergency_FG	Emergency_BG	Emergency_FG								EMERGENCY	#NOALARM									
Display_BG	TDC_RED	Display_BG	TDC_RED					RED					statetxt0								
Display_BG	TDC_YELLOW	Display_BG	TDC_YELLOW					YELLOW					statetxt0								
Display_BG	HALF_WHITE	Display_BG	HALF_WHITE					WHITE					statetxt0								
Display_BG	TDC_GREEN	Display_BG	TDC_GREEN					GREEN					statetxt0								
Display_BG	TDC_CYAN	Display_BG	TDC_CYAN					CYAN					statetxt0								
Display_BG	TDC_BLUE	Display_BG	TDC_BLUE					BLUE					statetxt0								
Display_BG	TDC_MAGENTA	Display_BG	TDC_MAGENTA					MAGENTA					statetxt0								
Display_BG	TDC_RED	Display_BG	TDC_RED				RED						statetxt1								
Display_BG	TDC_YELLOW	Display_BG	TDC_YELLOW				YELLOW						statetxt1								
Display_BG	HALF_WHITE	Display_BG	HALF_WHITE				WHITE						statetxt1								
Display_BG	TDC_GREEN	Display_BG	TDC_GREEN				GREEN						statetxt1								
Display_BG	TDC_CYAN	Display_BG	TDC_CYAN				CYAN						statetxt1								
Display_BG	TDC_BLUE	Display_BG	TDC_BLUE				BLUE						statetxt1								
Display_BG	TDC_MAGENTA	Display_BG	TDC_MAGENTA				MAGENTA						statetxt1								
Display_BG	TDC_RED	Display_BG	TDC_RED	RED	3								statetxt2								
Display_BG	TDC_YELLOW	Display_BG	TDC_YELLOW	YELLOW	3								statetxt2								
Display_BG	HALF_WHITE	Display_BG	HALF_WHITE	WHITE	3								statetxt2								
Display_BG	TDC_GREEN	Display_BG	TDC_GREEN	GREEN	3								statetxt2								
Display_BG	TDC_CYAN	Display_BG	TDC_CYAN	CYAN	3								statetxt2								
Display_BG	TDC_BLUE	Display_BG	TDC_BLUE	BLUE	3								statetxt2								
Display_BG	TDC_MAGENTA	Display_BG	TDC_MAGENTA	MAGENTA	3								statetxt2								
Display_BG	ALARM	Display_BG	ALARM																		

The line color of the Outline Box is set according to the real-time value of HIGHAL, HIGHALPR, and ACKSTAT as shown in the table below:

		Value									
		Selected	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
Parameter				NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM	
	highal				EMERGENCY	HIGH	LOW	JNLPRINT	PRINTER	JOURNAL	
	highalpr										
	ackstat			UNAKALRM							
Behavior	Outline Box Color	Selection	Ghost_Alarm	Emergency_BG	High_Alarm	Low_Alarm	JNLPRINT	PRINTER	JOURNAL	Display_BG	

Blinking indicates an Unacknowledged Alarm. The blinking will stop when the alarm is acknowledged. The blinking behavior is set according to the real-time value of ACKSTAT as shown in the table below:

		Value	
Parameter	ackstat	UNAKALRM	
Behavior	UnAckAlarm	TRUE	FALSE

Digital Alarm Graphic Symbols

GSP includes two types of Digital Alarm display components: Digital Alarm Graphic Symbols and Digital Alarm Message Boxes. Users should decide which type of alarm components they prefer and use them consistently throughout the project.

Point Types

Digital Alarm Graphic Symbols can be used for digital input, digital composite, device control, and flag points on all modules with PNTFORM=Full.

Basic Description

The graphic symbol includes a triangle shape and a single character letter. The triangle pointing up indicates a high alarm and triangle pointing down indicates a low alarm.

Symbol color indicates the Alarm Priority. When the point is not in alarm, the symbol can be made visible or invisible through the use of Show/Hide Switch (show_hide.pct) display component.

Some locations use hard-wired panel alarms for emergency alarms that are annunciated through an external alarm system. For these panel alarms, set the value for display parameter “panel” to true. The graphic symbol will be displayed in “Emergency” priority color even if the alarm priority is set to “JNLPRINT”, “PRINTER”, or “JOURNAL.”

Users can enter a single character string to indicate the type of alarm, for example:

- F for Flow
- L for Level
- P for Pressure
- T for Temperature
- V for Vibration

Includes a target to invoke faceplate.

Filename

There are four symbols in the Graphic Symbol set. Users should choose the correct one from the set that corresponds to the hard alarm wired for the instrumentation (e.g., dig_alm_sym_h.pct for high alarm switch).

	Filename	Description
	dig_alm_sym_hh.pct	High-High Alarm Switches
	dig_alm_sym_h.pct	High Alarm Switches
	dig_alm_sym_l.pct	Low Alarm Switches
	dig_alm_sym_ll.pct	Low-Low Alarm Switches

LCN Parameters Referenced

Reference_name
ackstat collector of point
point.alenbst
point.highal
point.highalpr

Display Parameters

Parameter	Prompt	Default	Required Entry
point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
point_type	Enter a single character for type: (e.g., F/L/P/T for Flow, Level, Pressure or Temperature)		
panel	Journaled Panel Alarm? (True/False)	false	

Behavior

Graphic Symbol

The color of the graphic symbol indicates the alarm priority when the point is in alarm state. It is not visible when the point is not in alarm, except the following situations:

When the point is “Disabled” and the point is not in alarm, a “DS” string and the Outline box are displayed.

When the point is “Inhibited”, a “IN” string and the Outline box are displayed (disregarding the alarm state).

When the Show/Hide flag is set to Show, the graphic symbol is displayed in “Symbol” color (dark blue by default).

For panel alarm (display parameter “panel”) set to true.

	Color indicates Alarm Priority.
	If the show/hide toggle is set to “Show”, the symbol will be visible in “Symbol” color when point is “Enabled”, and not in alarm state. If the show/hide toggle is set to “Hide” in the above situation, the Graphic Symbol will be invisible, and <u>not</u> selectable.

Outline Box

The box is visible in “Selection” Color when the point is selected.

The box is visible in “ALARM” Color when the “DS” or “IN” text is displayed.

	Visible in “Selection” color when point is selected.
	When point is “Disabled” and not in alarm state.
	When point is “Inhibited.”

						Value											
						Parameter						Behavior					
						ackstat						Letter & Symbol Color	Inhibit Flag	Disable Flag	Selectable	Background Color	Outline Box Color
															TRUE		Selection
							UNAKALRM					Ghost Alarm	FALSE	FALSE	TRUE		Display_BG
												Display_BG	FALSE	TURE	TRUE	Display_BG	Display_BG
												Display_BG	TRUE	FALSE	TRUE	Display_BG	ALARM
												Symbol	FALSE	FALSE	TRUE	Display_BG	ALARM
												Display_BG	FALSE	FALSE	TRUE	Display_BG	Display_BG
												Display_BG	FALSE	FALSE	FALSE	Display_BG	Display_BG
												Emergency_FG	FALSE	FALSE	TRUE	Emergency_FG	Emergency_BG
												Emergency_FG	FALSE	FALSE	TRUE	Emergency_FG	Emergency_BG
												Emergency_FG	FALSE	FALSE	TRUE	Emergency_FG	Emergency_BG
												Emergency_FG	FALSE	FALSE	TRUE	Emergency_FG	Emergency_BG
												High_Alarm	FALSE	FALSE	TRUE	Display_BG	Display_BG
												Low_Alarm	FALSE	FALSE	TRUE	Display_BG	Display_BG
												JNLPRINT	FALSE	FALSE	TRUE	Display_BG	Display_BG
												PRINTER	FALSE	FALSE	TRUE	Display_BG	Display_BG
												JOURNAL	FALSE	FALSE	TRUE	Display_BG	Display_BG
												Display_BG	FALSE	FALSE	FALSE	Display_BG	Display_BG
												Display_BG	FALSE	FALSE	FALSE	Display_BG	Display_BG

Digital Alarm Message Boxes

GSP includes two types of Digital Alarm display components: Digital Alarm Graphic Symbols and Digital Alarm Message Boxes. Users should decide which type of alarm components they prefer and use them consistently throughout the project.

Point Types

Digital Alarm Message Boxes can be used for digital input, digital composite, device control, and flag points on all modules with PNTFORM=Full.

Basic Description

Users can use these components to display a predefined alarm message when the point is in alarm state. It also includes a target to invoke faceplate.

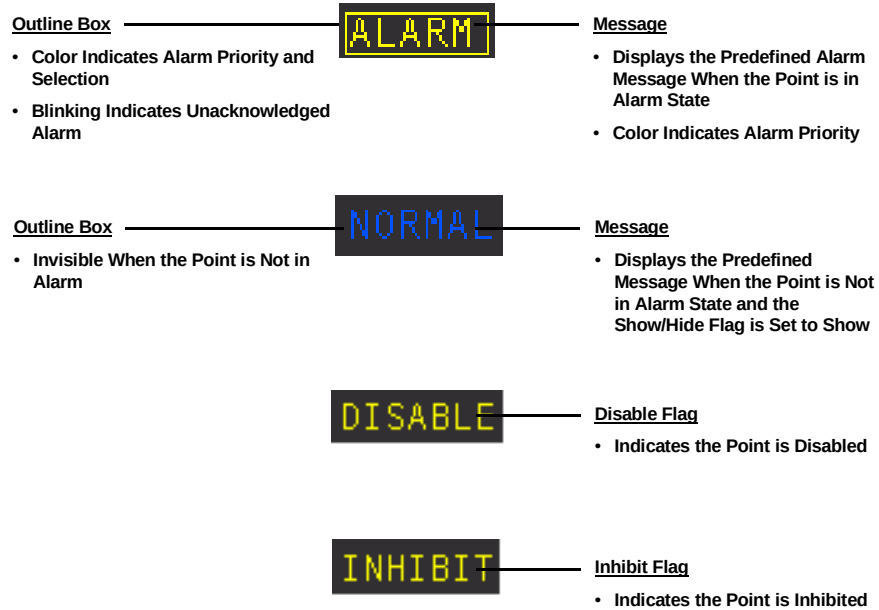
Choose the component with the correct size for the message you want to display. For example:

- To display “PAL” to indicate the point is Pressure Alarm Low, use dig_alm_msg_3.pct.
- To display “MCC TRBL” to indicate MCC Trouble, use dig_alm_msg_8.pct.

Message color indicates the Alarm Priority. When the point is not in alarm, the message can be made visible or invisible through the use of Show/Hide Switch (show_hide.pct) display component.

Some locations use hard-wired panel alarms for emergency alarms that are annunciated through an external alarm system. For these panel alarms, set the value for display parameter “panel” to true. The message will be displayed in “Emergency” priority color even if the alarm priority is set to “JNLPRINT”, “PRINTER”, or “JOURNAL.”

Digital Alarm Message Box



Filename

- dig_alm_msg_3.pct – Include 3-character alarm condition string.
- dig_alm_msg_4.pct – Include 4-character alarm condition string.
- dig_alm_msg_2x4.pct – Include two 4-character alarm condition strings.
- dig_alm_msg_5.pct – Include 5-character alarm condition string.
- dig_alm_msg_6.pct – Include 6-character alarm condition string.
- dig_alm_msg_8.pct – Include 8-character alarm condition string.

LCN Parameters Referenced

Reference_name
ackstat collector of point
point.alenbst
point.highal
point.highalpr

Display Parameters

Parameter	Prompt	Default	Required Entry
Point	Enter Tag: (Enclose the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
nrm_msg	Enter the message to be displayed for normal condition. nrm_top_msg & nrm_btm_msg for the 2x4 components	See Table Below	
alm_msg	Enter the message to be displayed for alarm condition. alm_top_msg & alm_btm_msg for the 2x4 components	See Table Below	Yes
panel	Journalized Panel Alarm? (True/False)	False	

Behavior**Message**

Users can overwrite the default alarm message and normal message which are part of the display parameters of the component.

	Default Alarm Message	Default Normal Message	Disabled	Inhibited
3-Character	ALM	NRM	DIS	INH
4-Character	ALRM	NORM	DIS	INH
2x4 Character	ALRM	NORM	DIS	INH
5-Character	ALARM	NORML	DIS	INH
6-Character	ALARM	NORMAL	DIS	INH
8-Character	ALARM	NORMAL	DISABLE	INHIBIT

If the point is not in alarm state while the point is Disabled or Inhibited, the Alarm Enable Status will be displayed.

If the point is not in alarm state while the point is Enabled, you can use the show/hide flag to either display a predefined normal state message or make the component invisible.

Outline Box

The color of the outline box indicates the alarm priority. The blinking outline box indicates the alarm is unacknowledged. When the point is selected, the outline box changes to the “Selection” color, and the faceplate will be invoked for the point.

Display Components Reference - Digital Alarm Message Boxes

		Value									
		Parameter									
Behavior	Selectable	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
	DS_Visible	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
	IN_Visible	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
	Alarm Flag	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
	Message Color	Ghost_Alarm	Ghost_Alarm	Display_BG	Display_BG	Display_BG	Display_BG	Emergency_FG	Emergency_FG	Emergency_FG	Emergency_FG
	Background Color	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Display_BG	Emergency_BG	Emergency_BG	Emergency_BG	Emergency_BG
	Outline Color	Selection	Ghost_Alarm	ALARM	ALARM	Display_BG	Display_BG	Emergency_BG	Emergency_BG	Emergency_BG	Emergency_BG
	ackstat		JNAKALRM								
	panel							TRUE	TRUE	TRUE	TRUE
	highalpr							EMERGENCY	JNLPRINT	PRINTER	JOURNAL
	highal		NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM
	alenbst			DISABLE	INHIBIT						
	showhide					TRUE	FALSE				
Parameter	Selected	TRUE									
	ackstat		JNAKALRM								
	panel							TRUE	TRUE	TRUE	TRUE
	highalpr							EMERGENCY	JNLPRINT	PRINTER	JOURNAL
	highal		NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM
	alenbst			DISABLE	INHIBIT						
	showhide					TRUE	FALSE				
	Selected	TRUE									
	ackstat		JNAKALRM								
	panel							TRUE	TRUE	TRUE	TRUE
	highalpr							EMERGENCY	JNLPRINT	PRINTER	JOURNAL
	highal		NOALARM	NOALARM	NOALARM	NOALARM	NOALARM	#NOALARM	#NOALARM	#NOALARM	#NOALARM
	alenbst			DISABLE	INHIBIT						
	showhide					TRUE	FALSE				

Digital Valves

GSP includes four types of Digital Valve components:

Valve with No Actuator

Solenoid Valve (SOL)

Valve with Handle (T-Valve)

Motor Operated Valve (MOV)

The behaviors of these components are the same. They are to be used for dynamic valves that have limit switches coming to the DCS to indicate status. To maximize performance and flexibility of these components, the state text values are to be entered during build-time as display parameters.

These Digital Valve components do not display alarm state. Use the Digital State Boxes or the Digital Alarm Graphic Symbols to monitor alarm states.

Point Types

Digital Valves can be used for Digital Composite and Device Control points that are connected to valves. Two sets of display components are provided – one for PM/APM/HPM, one for HG. The only difference between the two sets is the default display parameters.

Basic Description

There are several display parameters the user needs to enter values for when using the digital valve components.

	Description
	Default: Filled indicates valve in State1 for PM/APM/HPM, or State0 for HG (matching the upper box on TDC DETAIL page). Color is set according to GSP Color Constants Vlv_State1 or Vlv_hg_State0.
	Default: Hollow indicates valve in State0 for PM/APM/HPM, or State1 for HG (matching the middle box on TDC DETAIL page). Color is set according to GSP Color Constants Vlv_State0 or Vlv_hg_State1.
	Default: Hollow indicates valve in State2 (matching the lower box on TDC DETAIL page). Color is set according to GSP Color Constants Vlv_State2 or Vlv_hg_State2.
	Default: Hollow indicates valve in Transition State (neither contact made). Color is set according to GSP Color Constants Moving.

	Description
	Default: Hollow indicates valve in BadPV State (both contact made) or fail to retrieve value from LCN. Color is set according to GSP Color Constants BadPV.
	Outline Box: Indicates the point is selected.

Filename

For PM/APM/HPM valve with 2 States:

	Filename	Description
	dig_vlv_hor.pct	Horizontal Valve w/ No Actuator
	dig_vlv_hor_sol.pct	Horizontal Solenoid Valve
	dig_vlv_hor_T.pct	Horizontal T-Valve w/ Handle
	dig_vlv_hor_mov.pct	Horizontal Motor Operated Valve
	dig_vlv_ver.pct	Vertical Valve w/ No Actuator
	dig_vlv_ver_sol.pct	Vertical Solenoid Valve
	dig_vlv_ver_T.pct	Vertical T-Valve w/ Handle
	dig_vlv_ver_mov.pct	Vertical Motor Operated Valve

Display Components Reference - Digital Valves

For PM/APM/HPM valve with 3 States:

	Filename	Description
	dig_vlv_hor_3states.pct	Horizontal Valve w/ No Actuator
	dig_vlv_hor_sol_3states.pct	Horizontal Solenoid Valve
	dig_vlv_hor_T_3states.pct	Horizontal T-Valve w/ Handle
	dig_vlv_hor_mov_3states.pct	Horizontal Motor Operated Valve
	dig_vlv_ver_3states.pct	Vertical Valve w/ No Actuator
	dig_vlv_ver_sol_3states.pct	Vertical Solenoid Valve
	dig_vlv_ver_T_3states.pct	Vertical T-Valve w/ Handle
	dig_vlv_ver_mov_3states.pct	Vertical Motor Operated Valve

For HG valve:

	Filename	Description
	dig_vlv_hor_hg.pct	Horizontal Valve w/ No Actuator
	dig_vlv_hor_sol_hg.pct	Horizontal Solenoid Valve
	dig_vlv_hor_T_hg.pct	Horizontal T-Valve w/ Handle
	dig_vlv_hor_mov_hg.pct	Horizontal Motor Operated Valve
	dig_vlv_ver_hg.pct	Vertical Valve w/ No Actuator
	dig_vlv_ver_sol_hg.pct	Vertical Solenoid Valve
	dig_vlv_ver_T_hg.pct	Vertical T-Valve w/ Handle
	dig_vlv_ver_mov_hg.pct	Vertical Motor Operated Valve

LCN Parameters Referenced

Reference_name	Comment
point.pv	
Point.op	Motor Operated Valve Transition state is displayed when PV $\neq$ OP, which indicates the valve is moving.

Display Parameters

The user needs to enter the State0_Text, State1_Text and State2_Text according to the database configuration on LCN for the point.

We have provided default values to make it easier for the display builders. If the default values do not match the ones configured on your TPN database, you may want to customize these components by modifying the default values to match your database.

The color of the valve is set according to the values of the PV State. The State0_Text, State1_Text, State2_Text, Moving_Text, and BadPV_Text in display parameters are used to determine the state of the valve. The State0_Fill, State1_Fill, and State2_Fill in display parameters are used to determine whether the valve is displayed with filled color or stays hollow. These components also include a target to invoke faceplate.

Parameter	Prompt	Default	Required Entry
Point	Enter Tag: (Enclosed the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
State0_Text	Enter the State0 Text configured: (Default to CLOSED/OPEN)	CLOSED for PM/APM/HPM OPEN for HG	Yes
State1_Text	Enter the State1 Text configured: (Default to OPEN/CLOSED)	OPEN for PM/APM/HPM CLOSED for HG	Yes
State2_Text	Enter the State2 Text configured:		For 3-State Point
State0_Fill	Fill valve when in State0? (Default to False/True)	False for PM/APM/HPM True for HG	Yes
State1_Fill	Fill valve when in State1? (Default to True/False)	True for PM/APM/HPM False for HG	Yes
State2_Fill	Fill valve when in State2?	False	For 3-State Point
Moving_Text	Enter Moving Text: (Default to MOVING/INBETWN)	MOVING for PM/APM/HPM INBETWN for HG	Yes
BadPV_Text	Enter BadPV Text: (Default to BAD/BADPV)	BAD for PM/APM/HPM BADPV for HG	Yes

Behavior

The color of the valve is set according to the values of the PV State. For Motor Operated Valve, the value of the OP is also referenced. The behavior of the valve is determined as shown in the table below:

		Value									On Error
			State0_Text	State0_Text	State1_Text	State1_Text	State2_Text	State2_Text	Moving	BadPV	
Parameter	PV Value										
	OP Value	≠ PV									
	State0_Text										
	State1_Text										
	State2_Text										
	Moving_Text										
	BadPV_Text										
	State0_Fill		TRUE	FALSE							
	State1_Fill				TRUE	FALSE					
	State2_Fill						TRUE	FALSE			
Behavior	Valve Outline Color	Moving	Vlv_State0	Vlv_State0	Vlv_State1	Vlv_State1	Vlv_State2	Vlv_State2	Moving	BadPV	BadPV
	Valve Fill Color	Display_BG	Vlv_State0	Display_BG	Vlv_State1	Display_BG	Vlv_State2	Display_BG	Display_BG	Display_BG	Display_BG
	Moving Text	Moving Text	Blank	Blank	Blank	Blank	Blank	Blank	Moving Text	BadPV Text	BadPV Text

Digital Pumps and Motors

GSP includes two types of Digital Pump and Motor components: two states and three states.

Various shapes and orientations of pumps/motors are provided. The behaviors of these components are the same. They can be used for either pumps or motors to show their status on the display. To maximize performance and flexibility of these components, the state text values are to be entered during build-time as display parameters.

These Digital Pump and Motor components do not display alarm state. Use the Digital State Boxes or the Digital Alarm Graphic Symbols to monitor alarm states.

Point Types

Digital Pump and Motor components can be used for Digital Composite and Device Control points that are connected to pumps and motors. Two sets of display components are provided – one for PM/APM/HPM, one for HG. The only difference between the two sets is the default display parameters.

Basic Description

There are several display parameters the user needs to enter values for when using the digital pump/motor components.

	Description
	Default: Filled indicates pump in State1 for PM/APM/HPM, or State0 for HG (matching the upper box on TDC DETAIL page). Color is set according to GSP Color Constants Pmp_State1 or Pmp_hg_State0.
	Default: Hollow indicates pump in State0 for PM/APM/HPM, or State1 for HG (matching the middle box on TDC DETAIL page). Color is set according to GSP Color Constants Pmp_State0 or Pmp_hg_State1.
	Default: Hollow indicates pump in State2 (matching the lower box on TDC DETAIL page). Color is set according to GSP Color Constants Pmp_State2 or Pmp_hg_State2.
	Default: Hollow indicates pump in Transition State (neither contact made). Color is set according to GSP Color Constants Moving.
	Default: Hollow Indicates pump in BadPV State (both contact made) or fail to retrieve value from LCN. Color is set according to GSP Color Constants BadPV.
	Outline Box: Indicates the point is selected.

Filename

For PM/APM/HPM pump with 2 States:

	Filename
	dig_pump_1_lt.pct
	dig_pump_1_rt.pct
	dig_pump_2.pct
	dig_pump_3.pct

Display Components Reference - Digital Pumps and Motors

For PM/APM/HPM pump with 3 States:

	Filename
	dig_pump_3states_1_lt.pct
	dig_pump_3states_1_rt.pct
	dig_pump_3states_2.pct
	dig_pump_3states_3.pct

For HG pump with 2 States:

	Filename
	dig_pump_1_lt_hg.pct
	dig_pump_1_rt_hg.pct
	dig_pump_2_hg.pct
	dig_pump_3_hg.pct

LCN Parameters Referenced

Reference_name	Comment
point.pv	Can be used for either pump or motor

Display Parameters

The user needs to enter the State0_Text, State1_Text and State2_Text according to the database configuration on LCN for the point.

We have provided default values to make it easier for the display builders. If the default values do not match the ones configured on your LCN database, you may want to customize these components by modifying the default values to match your database.

The color of the pump is set according to the values of the PV State. The State0_Text, State1_Text, State2_Text, Moving_Text, and BadPV_Text in display parameters are used to determine the state of the pump. The State0_Fill, State1_Fill, and State2_Fill in display parameters are used to determine whether the pump is displayed with filled color or stays hollow. These components also include a target to invoke faceplate.

Parameter	Prompt	Default	Required Entry
Point	Enter Tag: (Enclosed the tag in brackets [] if it contains special characters such as \$, ! or starts with numbers.)		Yes
Pump_ID	Enter a single character Pump ID:		
State0_Text	Enter the State0 Text configured: (Default to OFF/ON)	OFF for PM/APM/HPM ON for HG	Yes
State1_Text	Enter the State1 Text configured: (Default to ON/OFF)	ON for PM/APM/HPM OFF for HG	Yes

Display Components Reference - Digital Pumps and Motors

State2_Text	Enter the State2 Text configured:		For 3-State Point
State0_Fill	Fill pump when in State0? (Default to False/True)	False for PM/APM/HPM True for HG	Yes
State1_Fill	Fill pump when in State1? (Default to True/False)	True for PM/APM/HPM False for HG	Yes
State2_Fill	Fill pump when in State2?	False	For 3-State Point
Moving_Text	Enter Moving Text: (Default to MOVING/INBETWN)	MOVING for PM/APM/HPM INBETWN for HG	Yes
BadPV_Text	Enter BadPV Text: (Default to BAD/BADPV)	BAD for PM/APM/HPM BADPV for HG	Yes

Behavior

The color of the pump is set according to the values of the PV State. The behavior of the pump is determined as shown in the table below:

		Value								On Error
Parameter	PV Value	State0_Text	State0_Text	State1_Text	State1_Text	State2_Text	State2_Text	Moving	BadPV	
	State0_Text									
	State1_Text									
	State2_Text									
	Moving_Text									
	BadPV_Text									
	State0_Fill	TRUE	FALSE							
	State1_Fill			TRUE	FALSE					
	State2_Fill					TRUE	FALSE			
Behavior	Pump Outline Color	Vlv_State0	Vlv_State0	Vlv_State1	Vlv_State1	Vlv_State2	Vlv_State2	Moving	BadPV	BadPV
	Pump Fill Color	Vlv_State0	Display_BG	Vlv_State1	Display_BG	Vlv_State2	Display_BG	Display_BG	Display_BG	Display_BG
	Pump ID	Display_BG	State0	Display_BG	State1	Display_BG	State2	Moving	BadPV	BadPV
		Display_BG	State0	Display_BG	State1	Display_BG	State2	Moving	BadPV	BadPV

Primmod Status Indicators

The Primmod Status Indicators change color based on the alarm status of the PRIMMODs rather than an individual point. They are typically used in overview display to provide operators with an overall alarm status at a glance.

By design, they are not selectable, and do not acknowledge alarms.

Point Types

Primmod Status Indicators are provided to be used for PRIMMOD points. Typically, the following point types are used:

Flag on PM/APM/HPM

Batch Assign on HG

Any Point Type on AM

Basic Description

The Primmod Status Indicator shows the alarm priorities and acknowledgment status of the PRIMMOD points, thereby enabling the operator to quickly see patterns in the alarms and take action. They change color based on the alarm status of the PRIMMOD rather than an individual point. They are designed to be included on overview displays. Alarms that are referenced by the PRIMMOD collectors cannot be acknowledged.

There are three versions of the Primmod Status Indicator:

Large Horizontal

Large Vertical

Small

Display Components Reference - Primmod Status Indicators

	Dynamics	Description
Large Horizontal Primmod Status Indicator		Appearance when at least one alarm of Emergency, High and Low priorities are active within the referenced PRIMMODs.
		Appearance when at least one alarm of Low alarm is active within the referenced PRIMMODs.
		Appearance when no Emergency, High, or Low alarm is active within the referenced PRIMMODs.
Large Vertical Primmod Status Indicator		Appearance when at least one alarm of Emergency, High and Low priorities are active within the referenced PRIMMODs.
		Appearance when at least one alarm of Low alarm is active within the referenced PRIMMODs.
		Appearance when no Emergency, High, or Low alarm is active within the referenced PRIMMODs.
Small Primmod Status Indicator		Appearance when at least one alarm of Emergency, High and Low priorities are active within the referenced PRIMMODs.
		Appearance when at least one alarm of Low alarm is active within the referenced PRIMMODs.
		Appearance when no Emergency, High, or Low alarm is active within the referenced PRIMMODs.

Blinking indicates unacknowledged alarm status.

Filename

primmod_stat_lg_hor.pct – Large Horizontal
primmod_stat_lg_ver.pct – Large Vertical
primmod_stat_sm.pct – Small

LCN Parameters Referenced

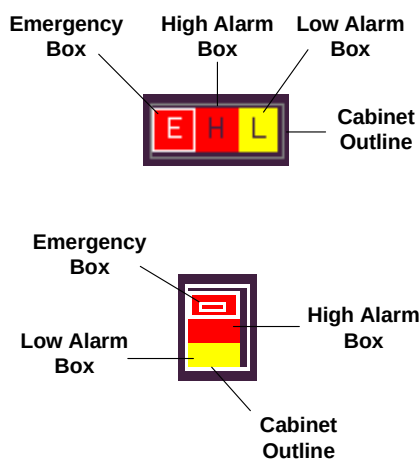
primsts collector of PRIMMOD points.

Display Parameters

Parameter	Prompt	Default	Required Entry
PRIMMOD1~20	Enter PRIMMOD: (e.g., 50FC070) Note: - Do not add [] around tagname. - Up to 20 PRIMMOD points allowed. Do not skip parameter when entering PRIMMOD tags.		

Note: Users must make sure the correct primmod tag is entered. No error message is displayed during validation or run-time when encountering a non-existent primmod tag.

Behavior



Display Components Reference - Primmod Status Indicators

The Box, Line, and Outline colors are set according to the Primmod status of the PRIMMODs as shown in the table below:

		Value					
Parameter	UnAckEm count	>0					0
	AckEm count		>0				0
	UnAckHi count			>0			0
	AckHi count				>0		0
	UnAckLo count					>0	0
	AckLo count						>0 0
Behavior	Emergency Box Color	Emergency_BG	Emergency_BG				Display_BG
	Emergency Line Color	Emergency_FG	Emergency_FG				Display_BG
	Emergency Blink	TRUE	FALSE				FALSE
	Cabinet Outline Color	Emergency_FG	Emergency_FG				Ghost_Alarm
	High Alarm Box Color			High_Alarm	High_Alarm		Display_BG
	High Alarm Blink			TRUE	FALSE		FALSE
	Low Alarm Box Color					Low_Alarm Low_Alarm	Display_BG
	Low Alarm Blink					TRUE FALSE	FALSE

Aggregate Primmod Status Indicators

The Aggregate Primmod Status Indicators change color based on the alarm status of the highest priority alarm of the PRIMMODs rather than an individual point. They are typically used in overview display to provide operators with an overall alarm status at a glance.

By design, they are not selectable, and do not acknowledge alarms.

Point Types

Primmod Status Indicators are provided to be used for PRIMMOD points. Typically, the following point types are used:

Flag on PM/APM/HPM

Batch Assign on HG

Any Point Type on AM

Basic Description

The Aggregate Primmod Status Indicator shows the highest alarm priority and acknowledgment status of the PRIMMOD points, thereby enabling the operator to quickly see patterns in the alarms and take action. They change color based on the alarm status of the PRIMMOD rather than an individual point. They are designed to be included on overview displays. Alarms that are referenced by the PRIMMOD collectors cannot be acknowledged.

Dynamics	Description
	Appearance when the highest active alarm priority referenced by the PRIMMODs is Emergency.
	Appearance when the highest active alarm priority referenced by the PRIMMODs is High.
	Appearance when the highest active alarm priority referenced by the PRIMMODs is Low.
	Appearance when no Emergency, High, or Low alarm is active within the referenced PRIMMODs.

Blinking indicates unacknowledged alarm status.

Filename

primmod_stat_aggregate.pct

LCN Parameters Referenced

primsts collector of PRIMMOD points.

Display Parameters

Parameter	Prompt	Default	Required Entry
PRIMMOD1~20	Enter PRIMMOD: (e.g., 50FC070, don't add brackets [] around tagname.) Note: Up to 20 PRIMMOD points allowed. Do not skip parameter when entering PRIMMOD tags.		

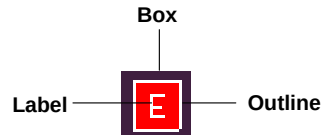
Note: Users must make sure the correct Primmod tag is entered. No error message is displayed during validation or run-time when encountering a non-existent Primmod tag.

Behavior

The Box, Line and Outline colors as well as the Label text are set according to the primmod status of the PRIMMODs as shown in the table below. Note that Unacknowledged alarms have higher priority. For example, if an unacknowledged Low

Display Components Reference - Aggregate Primmod Status Indicators

Priority alarm and an acknowledged High Priority alarm exist, the Status Indicator will show blinking low priority color.



		Value															
Behavior	Parameter																
	primmod1	NOALARM	UNACKEM			UNACKHI		UNACKLO			ACKEM			ACKHI		ACKLO	
	primmod2	NOALARM		UNACKEM			UNACKHI		UNACKLO		ACKEM			ACKHI		ACKLO	
	..primmod20	NOALARM			UNACKEM			UNACKHI		UNACKLO		ACKEM		ACKHI		ACKLO	
		NOALARM			UNACKEM			UNACKHI		UNACKLO		ACKEM		ACKHI		ACKLO	
Behavior	Box FillColor	Display_BG	Emergency_FG	Emergency_BG	Emergency_FG	Emergency_BG	High_Alarm	High_Alarm	High_Alarm	Low_Alarm	Low_Alarm	Low_Alarm	Emergency_FG	Emergency_BG	Emergency_FG	Emergency_BG	Display_BG
	Box Outline Color	Ghost_Alarm	Emergency_FG	Emergency_FG	Emergency_FG	Emergency_FG	High_Alarm	High_Alarm	High_Alarm	Low_Alarm	Low_Alarm	Low_Alarm	Emergency_FG	Emergency_FG	Emergency_FG	Emergency_FG	Ghost_Alarm
	Label Color	Display_BG	Emergency_FG	Emergency_FG	Emergency_FG	Emergency_FG	Emergency_BG	Emergency_BG	Emergency_BG	Emergency_BG	Emergency_BG	Emergency_BG	Emergency_FG	Emergency_FG	Emergency_FG	Emergency_FG	Display_BG
	Label	blank	⌘	⌘	⌘	⌘	⌘	⌘	⌘	⌘	⌘	⌘	⌘	⌘	⌘	⌘	blank
	UnAckAlarm	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE

Custom Parameter Boxes

Point Types

Custom Parameter Boxes are used to display any point.parameter with numeric or text values. They can be used for all point types. For array element of custom AM point, enter the parameter element in the “param” display parameter [e.g., COM_STS(1)].

Basic Description

Custom Parameter Boxes are designed to display the value of a custom point parameter. It provides the flexibility to display values that are not handled by other display components.

The custom parameter name is also displayed to allow the operator to know what is being measured. To reduce clutter on the screen, the custom parameter name is toggled through the show/hide button.

The display does not provide any Alarm State.

The display has a target to call up the Faceplate. The faceplate type is dependent on the point type being displayed by the subpicture.

Filename

Cust_param_4.pct – Displays 4-character custom parameter value.
Cust_param_5.pct – Displays 5-character custom parameter value.
Cust_param_6.pct – Displays 6-character custom parameter value.
Cust_param_7.pct – Displays 7-character custom parameter value.
Cust_param_8.pct – Displays 8-character custom parameter value.

LCN Parameters Referenced

Reference_name	Description
Point.parameter	Point parameter value

Display Parameters

Parameter	Prompt	Example	Default	Required Entry
point	Enter Tag: (enclose tag in [] if it contains special character)	[50FC070]		Yes
IsPointDig	Is this point of type digital? (true/false)		False	No
param	Enter parameter of point to be displayed:	PVAUTO		Yes
param_desc	Enter desc for parameter of point to be displayed:	PVAUTO		Yes
ParamIsBool	Is this parameter of type boolean: (true/false)		False	No
digits	If param is a number, enter number of digits (0-3) after decimal point		2	No

Behavior



← Point Parameter Value



← **Outline Box**
• Visible When Point is Selected



← **Description**
• Displays the Text Entered for param_desc Display Parameter
• Description Can Be Made Visible or Invisible Through the Use of the Show/Hide Switch (show_hide.pct) Display Component

Display Navigation Buttons Reference

General Information

The GUS Display Navigation Buttons are designed to provide a quick way to see the alarm status and navigate to the display that the operators want to bring up.

We recommend including these navigation buttons in the Task Panel. However, they can also be used in GSP process displays.

There are two sets of navigation buttons:

Buttons to be inserted in Task Panel Display directly. They are to be used to invoke High-Level process displays.

Buttons to be inserted in Low-Level Navigation Display which then get inserted in the Task Panel Display. They are to be used to invoke Low-Level process displays.

Both sets of navigation buttons include two sub-sets to show the alarm status:

Buttons with Small Primmod Status Box

Buttons with Aggregate Primmod Status Box

For the buttons in Low-Level Navigation Display, there are two additional choices:

Buttons with normal action to invoke the referenced display.

Buttons that will iconize Native Window if needed before invoking the referenced display. These components are designed to prevent the NativeWindow from showing up between the GUS display call-up under the SafeView environment. They are the components with “-NW” (minimize NW) included in their filename.

Warning:

Be sure to allow Native Window to minimize in the SafeView Configuration if you use the navigation buttons that minimize the Native Window.

There are six display component files for each type of the display navigation button mentioned above. The only difference is the label sizes. The actions are the same within the six component files.

Display Navigation Buttons Reference - General Information

Behavior

	Alarms exist for Emergency, High and Low priorities.	Small Primmod Status Box
	Alarm(s) exist for Emergency priority.	Small Primmod Status Box
	Alarm(s) exist for High priority.	Small Primmod Status Box
	Alarm(s) exist for Low priority.	Small Primmod Status Box
	Highest alarm is Emergency priority.	Aggregate Primmod Status Box
	Highest alarm is High priority.	Aggregate Primmod Status Box
	Highest alarm is Low priority.	Aggregate Primmod Status Box
	No alarm exists connected to the primmods, or no primmod entered for this navigation button at build-time.	Both Type of Boxes
	Selected high-level navigation button and the linked low-level navigation buttons. Note the different background color.	Both Type of Boxes
	Display connected to this button is currently active (on screen). Note the outline box color.	Both Type of Boxes

Filename

Filename	Aggregate/ Small Primmod Status Box	LowLevel/ HighLevel Navigation	Minimize Native Window	Label Size
navbtn_hilvl_aggr_3.pct	Aggregate	HighLevel	No	3
navbtn_hilvl_aggr_4.pct	Aggregate	HighLevel	No	4
navbtn_hilvl_aggr_5.pct	Aggregate	HighLevel	No	5

Display Navigation Buttons Reference - General Information

Filename	Aggregate/ Small Primmod Status Box	LowLevel/ HighLevel Navigation	Minimize Native Window	Label Size
navbtn_hilvl_aggr_8.pct	Aggregate	HighLevel	No	8
navbtn_hilvl_aggr_10.pct	Aggregate	HighLevel	No	10
navbtn_hilvl_aggr_12.pct	Aggregate	HighLevel	No	12
navbtn_hilvl_sm_3.pct	Small	HighLevel	No	3
navbtn_hilvl_sm_4.pct	Small	HighLevel	No	4
navbtn_hilvl_sm_5.pct	Small	HighLevel	No	5
navbtn_hilvl_sm_8.pct	Small	HighLevel	No	8
navbtn_hilvl_sm_10.pct	Small	HighLevel	No	10
navbtn_hilvl_sm_12.pct	Small	HighLevel	No	12
navbtn_lolvl_aggr_3.pct	Aggregate	LowLevel	No	3
navbtn_lolvl_aggr_4.pct	Aggregate	LowLevel	No	4
navbtn_lolvl_aggr_5.pct	Aggregate	LowLevel	No	5
navbtn_lolvl_aggr_8.pct	Aggregate	LowLevel	No	8
navbtn_lolvl_aggr_10.pct	Aggregate	LowLevel	No	10
navbtn_lolvl_aggr_12.pct	Aggregate	LowLevel	No	12
navbtn_lolvl_aggr-NW_3.pct	Aggregate	LowLevel	Yes	3
navbtn_lolvl_aggr-NW_4.pct	Aggregate	LowLevel	Yes	4
navbtn_lolvl_aggr-NW_5.pct	Aggregate	LowLevel	Yes	5
navbtn_lolvl_aggr-NW_8.pct	Aggregate	LowLevel	Yes	8
navbtn_lolvl_aggr-NW_10.pct	Aggregate	LowLevel	Yes	10
navbtn_lolvl_aggr-NW_12.pct	Aggregate	LowLevel	Yes	12
navbtn_lolvl_sm_3.pct	Small	LowLevel	No	3
navbtn_lolvl_sm_4.pct	Small	LowLevel	No	4
navbtn_lolvl_sm_5.pct	Small	LowLevel	No	5

Display Navigation Buttons Reference - General Information

Filename	Aggregate/ Small Primmod Status Box	LowLevel/ HighLevel Navigation	Minimize Native Window	Label Size
navbtn_lolvl_sm_8.pct	Small	LowLevel	No	8
navbtn_lolvl_sm_10.pct	Small	LowLevel	No	10
navbtn_lolvl_sm_12.pct	Small	LowLevel	No	12
navbtn_lolvl_sm-NW_3.pct	Small	LowLevel	Yes	3
navbtn_lolvl_sm-NW_4.pct	Small	LowLevel	Yes	4
navbtn_lolvl_sm-NW_5.pct	Small	LowLevel	Yes	5
navbtn_lolvl_sm-NW_8.pct	Small	LowLevel	Yes	8
navbtn_lolvl_sm-NW_10.pct	Small	LowLevel	Yes	10
navbtn_lolvl_sm-NW_12.pct	Small	LowLevel	Yes	12

LCN Parameters Referenced

Reference_name	Comment
primsts collector of PRIMMOD points	For Small Primmod Status Box
primcnt collector of PRIMMOD points	For Aggregate Primmod Status Box

Display Parameters

Parameter	Prompt	Default	Required Entry
Graphic_Name	Enter graphic name: (e.g., reactor) Note: Only enter the display filename. Do not enter the pathname.		Yes
Button_Text	Enter text for button		Yes
PRIMMOD1~20	Enter PRIMMOD: (e.g., 50FC070). Note: - Do not add [] around tagname. - Up to 20 PRIMMOD points allowed. Do not skip parameter when entering PRIMMOD tags.		

Note: Users must make sure the correct primmod tag is entered. No error message is displayed during validation or run-time when encountering a non-existent primmod tag.

Function Buttons Reference

The Function Buttons are buttons with specific target actions. When selected (using left mouse click or touch screen), they perform a predefined task. Typically, the action occurs on the other GUS displays on the same station.

We recommend inserting these function buttons in the Task Panel if you decide to use them, although they can function in other GSP displays as well.

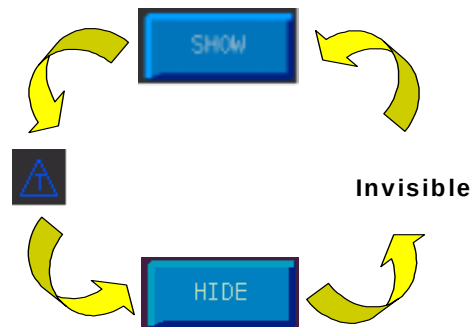
Show/Hide Toggle Button

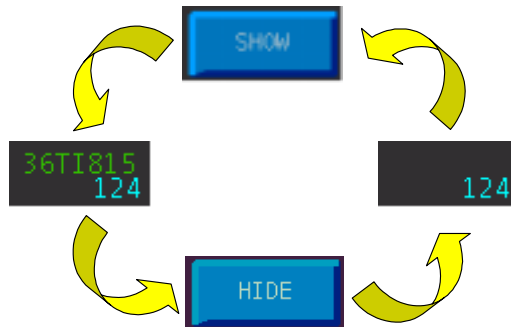
Basic Description

The Show/Hide toggle button is provided to toggle the display components between "Show" and "Hide" state. The effect of the button applies to all the GUS displays on the station. It uses display database variable dispdb.bool255 to toggle between the show/hide state. Refer to "Reserved GUS Display Database Variables" in the Appendix for more information.

The button doesn't reference any LCN or display parameter. The figure below shows the behavior of a digital alarm graphic symbol and a measurement box with tag.

Show/Hide Toggle





Filename

show_hide.pct

Show/Hide Text Toggle Button

Basic Description

The Show/Hide Text toggle button is provided to allow the users to toggle the user-defined display objects between “Show” and “Hide” state. The effect of the button applies to all the GUS displays on the station. An example of its usage is illustrated in the “Build Process Displays” section under “Add Finishing Touches.” It uses display database variable dispdb.bool253 to toggle between the show/hide state. Refer to “Reserved GUS Display Database Variables” in the Appendix for more information.

Filename

show_hide_text.pct

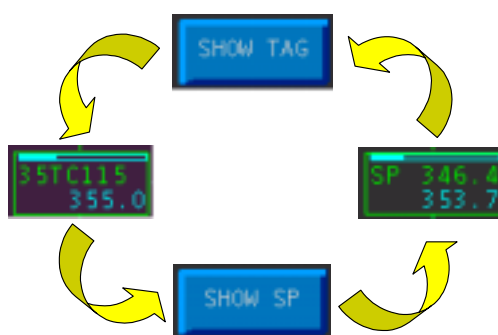
Tag/Setpoint Toggle Button

Basic Description

The Tag/Setpoint toggle button is provided to toggle the display components between "Show Tag" and "Show Setpoint." The effect of the button applies to all the GUS displays on the station.

The button doesn't reference any LCN or display parameter. The figure below shows the behavior of a large control box.

Tag/Set Point Toggle Button



Filename

tag_sp_btn.pct

User Switch

Basic Description

The User Switch button is provided to allow the users to switch the dispdb.int256 among several user-defined states in the GSP process displays. The effect of the button applies to all the process displays running on the station. This is one of the few GSP display components that are designed to be resizable at build-time. Users should test the run-time version to make sure the width is sufficient to display the longest display parameter value entered.



Its usage is similar to what is illustrated in the “Build Process Displays” section under “Add Finishing Touches”.

Filename

User_switch.pct

Display parameters

Parameter	Prompt	Required Entry
Item0	Item for dispdb.int256 = 0	Yes
Item1	Item for dispdb.int256 = 1	No
Item2	Item for dispdb.int256 = 2	No
Item3	Item for dispdb.int256 = 3	No
Item4	Item for dispdb.int256 = 4	No
Item5	Item for dispdb.int256 = 5	No

Note: Do not skip parameter when entering the values for display parameters.

Behavior

At run-time, a pull-down list will appear when the user clicks on the Select Arrow. The user can then click on the item in the pull-down list to select the item. The pull-down list will disappear when the user clicks on the Select Arrow again, or clicks anywhere outside the button.

Example

Goal

Below is an example to illustrate how to implement color standard for different line types. For example, the project decides to show all lines in a neutral color (e.g., light gray), and occasionally show different colors for the following line types:

Process Line

- Instrument Signals
- Steam and Condensate
- Fuel Gas
- Acid and Caustic

Implementation:

Insert a user_switch component in the task panel display.

Enter the following values for the display parameters:

- Item0: Dim Line Color
- Item1: Process Line
- Item2: Instrument Signals
- Item3: Steam & Condensate
- Item4: Fuel Gas
- Item5: Acid & Caustic

Validate and Save the task panel.

- Decide the colors to use for the various line types. Hint: You can use the “GSP Configuration Utility” to define custom colors for this purpose. Refer to the “GSP Configuration Utility” section for details on how to configure user-defined constants. In this example, we configured a constant named “FuelGas” for colored Fuel Gas lines and use TDC_HALF_WHITE for neutral (default) color for all lines.
- Add the following script for all Fuel Gas Lines on the displays:

```
Sub onDataChange ( )  
  If dispdb.int256 = 4 then  
    Me.LineColor = GetSetting("GSP","USER"," FuelGas ")  
  Else  
    Me.LineColor = TDC_HALF_WHITE
```

```
End if  
End sub
```

Runtime User Interface



When user click on the pull-down arrow of the component at runtime, a pull-down list will appear.



After selecting the “Fuel Gas” item, Fuel Gas lines in all process displays will turn to the user-defined “FuelGas” color. All lines will reverse back to TDC_HALF_WHITE when user selects “Dim All Lines” from the pull-down list.

View Point

Basic Description

The View Point component allows the operator to keep track of the process value of a tag. It also allows the operator to call a point into the faceplate that isn't on the current graphic. We recommend inserting the component in taskpanel display.

Functions

View a Tag and Its Current Value in the Taskpanel

This allows the operator to track a particular measurement that the operator needs to keep an eye on. The value will stay displayed in the taskpanel until it is removed. To change the tag, select the view point button, then enter the new tag in the dialog box. To clear the view point, select the view point button, then blank out the tagname in the dialog box.

Send a Tag to the Faceplate

This allows the operator to call up a tag into the faceplate for a point that is not on the current display. If the operator needs to change the SP/OP of a point that is not in the current display, the operator can send it to the faceplate via view point button and not have to change the current display.

Filename

viewpt.pct

Display Parameters

The View Point component requires no display parameter.

Behavior

At display call-up, the component looks like:

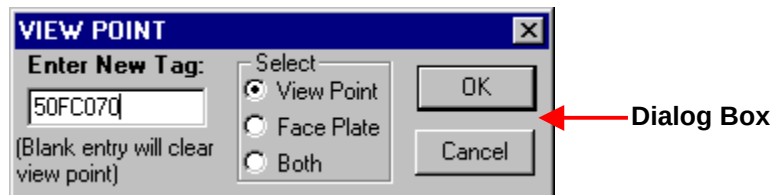


When user clicks on the View Point component, a dialog box will appear on the lower-left corner of the screen. Enter the tagname.

Select Options:

- View Point (Track point.parameter Value in View Point Area)
- Faceplate (Invoke or Update Faceplate)
- Both (Both View Point and Faceplate)

Then click the “OK” button.

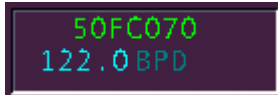


A message “PROCESSING” will appear while data is being collected from TPN.



The Tag, PV, and Engineering Unit will be displayed as below. The PV will be updated tracking the changes on the TPN. Refresh frequency is every 4 seconds.

Function Buttons Reference - View Point



The table below show the value displayed for different point type:

Point Type		Point.parameter Displayed	Format
"ANINNIM", "REGPVNIM", "NUMERNIM", "NUMERCAM", "NUMERCHG", "ANLCMPHG", "ANLINHG"		PV	PVFORMAT
"REGCLNIM"	CTLALGID = "PID", "PIDERFB", "PIDFF", "POSPROP", "RATIOCTL", "PIDPOSPR", "RAMPSOAK"	PV	PVFORMAT
"REGCLNIM"	Others	OP	D1
"ANOUTNIM", "ANLOUTHG"		OP	D1
"CUSTOMAM"		PV	PVFORMAT
"REGHG"	SUPPIO = "INPSUPPR"	OP	D1
"REGHG"	ALGIDDAC="SELOVDDC"	OP	D1
"REGHG"	Others	PV	PVFORMAT
"REGAM"	PVALGID <> "NULL"	PV	PVFORMAT
"REGAM"	CTLALGID <> "NULL"	OP	
"TIMERNIM", "TIMERHG", "TIMERAM"		PV	PVFORMAT
"COUNTHG", "COUNTAM", "CTLCOUNT"		AV	AVFORMAT

Point Type		Point.parameter Displayed	Format
"DIINNIM", "FLAGNIM", "DIGINHG", "FLAGHG", "FLAGAM"		PV	
"DIOUTNIM", "DIGOUTHG"		OP	
"DICMPNIM", "DEVCTL"	NODINPTS= 0	OP	
"DICMPNIM", "DEVCTL"	Others	PV	
"DIGCMPHG"	NMBRINPT=0	OP	
"DIGCMPHG"	Others	PV	

GUS Display Call-up Button

Basic Description

The GUS Display Call-up Button includes a target to call-up GUS Display. Users can add the button to process display or Task Panel display. The button allows the user to specify the pathname of the display file, therefore, it can be used to reference the GUS display files that reside in folder that is not in the Pathname Catalog.

The button is resizable at build-time to allow the user in scaling it to the desired size for the button label.

Filename

pct_btn.pct

Display Parameters

Parameter	Prompt	Required Entry
Picture	Enter the picture filename: (e.g., reactor).	Yes
Label	Enter Label to display on button.	Yes
Path	Enter Path of the GUS display folder: (e.g., c:\GUS_display\).	Not required if the file resides in folder included in Pathname Catalog.

Link Targets

Basic Description

Two new display components were added. They are the same except the Label “L” or “Z”. Our recommendation is to:

Use the Link Target to indicate a display-link to a GUS display at the same level or higher than the current display in the display hierarchy.

Use the Zoom Target to indicate a display-link to a GUS display of greater detail or to the level below the current display in the display hierarchy.

Filename

link_target.pct – Target with “L” label.

zoom_target.pct – Target with “Z” label.

Display Parameters

Parameter	Prompt	Default	Required Entry
Schem	Enter the picture file name: (eg. Reactor)		Yes

Note: Make sure that the picture is in the display search path that you have defined for GUS or, if not, enter the full pathname.

Application Button

Basic Description

The Application Button includes a target to invoke a document, such as the operator manual, the shutdown procedure, etc. The button allows the user to specify both the application filename and the referenced application.

The button is resizable at build-time to allow the user in scaling it to the desired size for the button label.

Filename

app_btn.pct

Display Parameters

Parameter	Prompt	Required Entry
Filename	Enter Filename: (e.g., c:\helpfile\pump.hlp)	Yes
Label	Enter Label to display on button:	Yes
Application	Enter Path for Application: (e.g., c:\msoffice\winword\winword.exe)	Yes

Program Activation Button

Basic Description

The Program Activation Button includes a target to activate a program. If the program is not active when the button is selected (mouse click or touch screen), then the program will be activated and the program window will be in focus. If the program is already active, then the program window will be brought to the front with focus on the window. Note: It will not activate another copy of the same program.

The button is resizable at build-time to allow the user in scaling it to the desired size for the button label.

Filename

activate_btn.pct

Display Parameters

Parameter	Prompt	Required Entry
Filename	Enter Filename: (e.g., explorer.exe)	Yes
Label	Enter Label to display on button	Yes
Path	Enter Path of the file: (e.g., c:\winnt)	Yes
Title	Optional title string to match	

Button Template

Basic Description

This is a button template that the user can use as a starting point and add target action to the Button object. Using this button template provides a consistent look as the other GSP buttons. The customized button should be saved under a different name.

The button is resizable at build-time to allow the user in scaling it to the desired size for the button label.

Filename

button.pct

Display Parameters

Parameter	Prompt	Required Entry
Label	Enter Label to display on button.	Yes

GSP Version

Basic Description

The GSP Version displays the current version number of the GUS Solution Pack that is installed on the GUS. It works with the show/hide toggle button, therefore, can be made visible and invisible on the GSP display. An example is included in the GSPColor.pct.

Filename

version.pct

GSP Components Library

A GSP Components Library is provided to maximize the benefit of the GUS Component Explorer function. The GSP Components Library definition file is installed with the TPS GUS Solution Pack build-time files at the time of installation.

You must register the definition file with, and add it to, the Component Explorer before the GSP Components Library is available in the Display Builder. After registering/adding the component definition file, the GSP Components Library displays in the Component window whenever you open a GUS display.

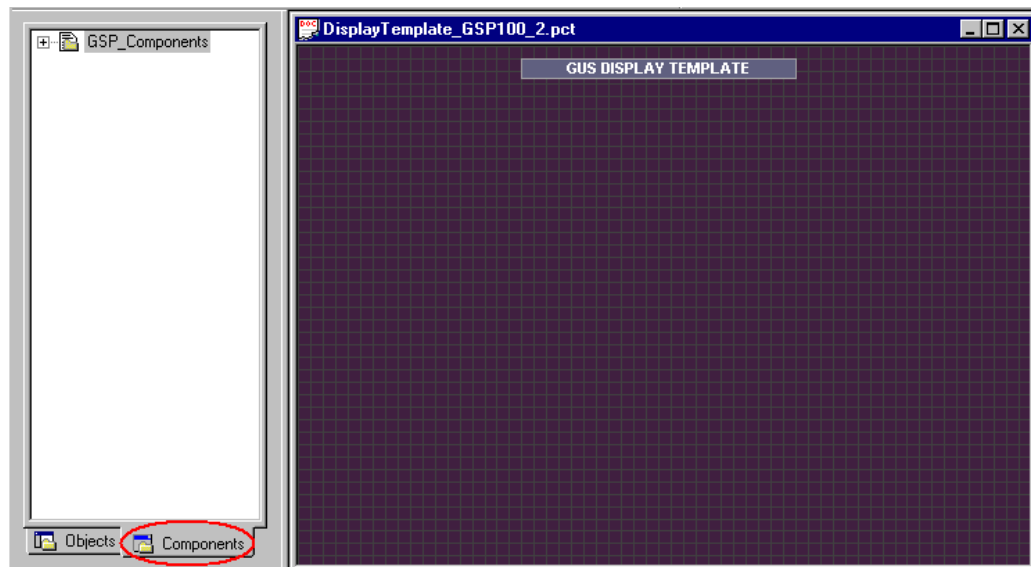
Refer to *Display Builder User's Guide* for instructions on registering/adding the GSP Components Library definition file, for a detailed description of the Component Explorer functionality.

Turn on the Component Explorer

Start the GUS Display Builder.

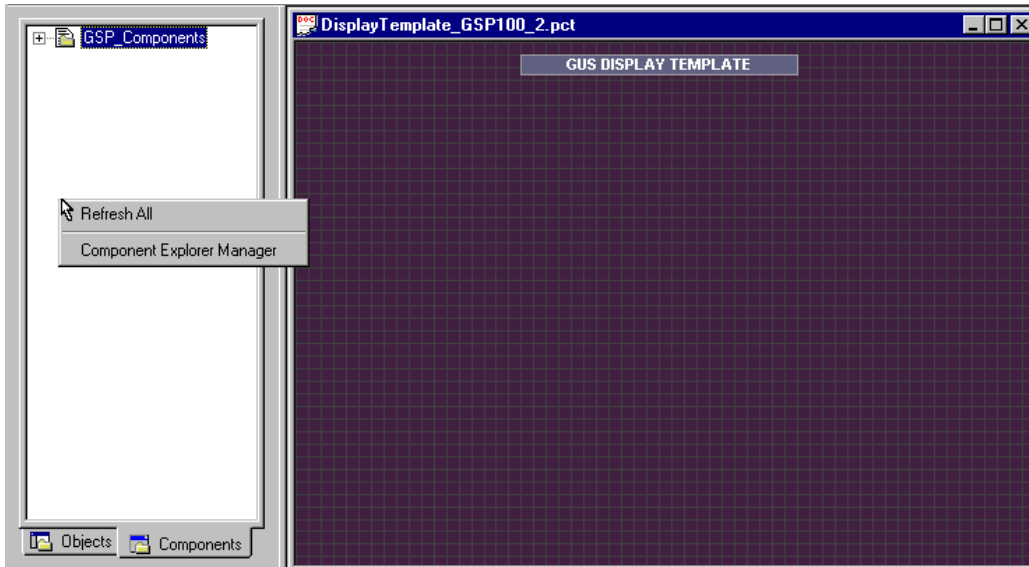
Select GUS Explorer on the View menu (or select the GUS Explorer button on the Standard toolbar). Either the Object Browser or the Component Explorer window is displayed.

If the Object Browser is displayed, select the Components tab. A view of the Components Library is displayed.

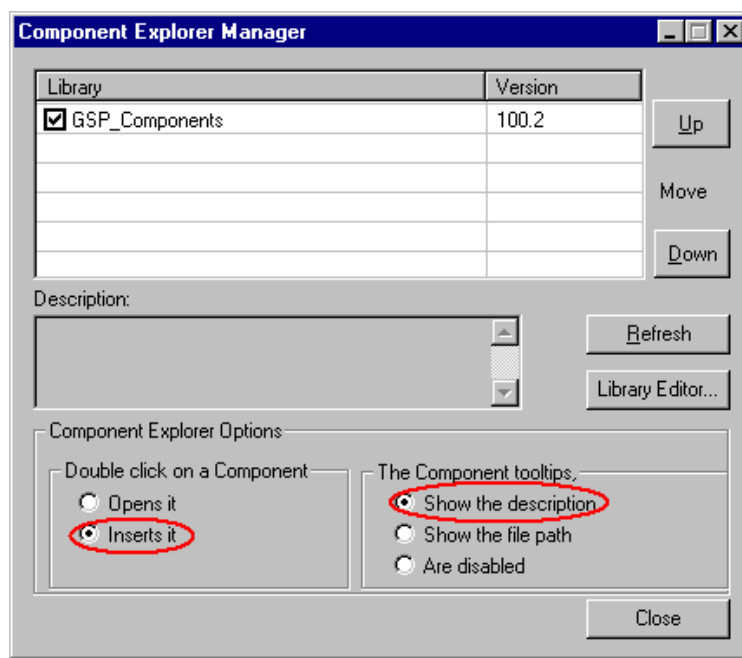


Set Component Explorer Options

Right-click the library or any empty area inside the Component window. The Library Functions pop-up menu is displayed.



Select Component Explorer Manager. The Component Explorer Manager is turned on. GSP recommends selecting the Component Explorer Options as shown below to maximize the usefulness of the GSP Components Library.

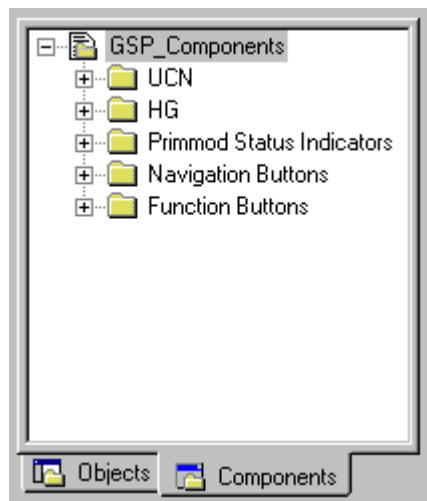


Select “Close” button to turn off the Component Explorer Manager and go back to the GUS Display Builder.

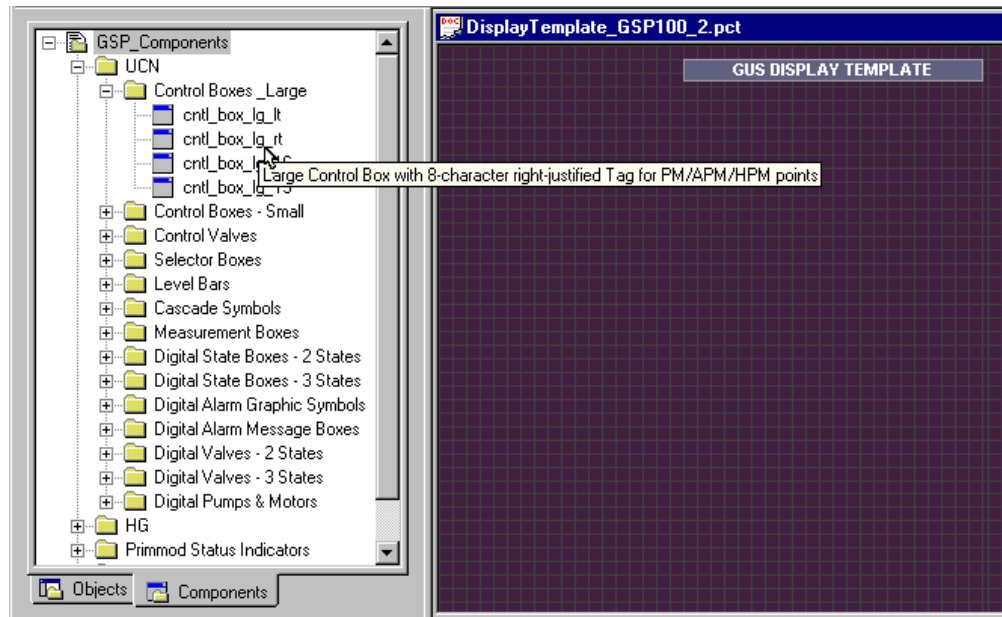
GSP Component Library Hierarchy

Click on the symbol or Double-click on the folder to expand the display to show sub-folders and display components. The Components Library is displayed in the standard Windows hierarchical tree format.

GSP Components are organized into groups as show below. There are sub-folders under each group.



If the option is selected in Component Explorer Manager, a one-line description (a tooltip) is displayed when the cursor is positioned over a display component.



Double-click the component you want to insert.

Refer to “Quick Start”, “Building Process Displays” and “Build Task Panel with Navigation” in this document for more detail on how to build displays.

Refer to Display Components Reference” in this document for more detail on each display component.

GSP Configuration Utility

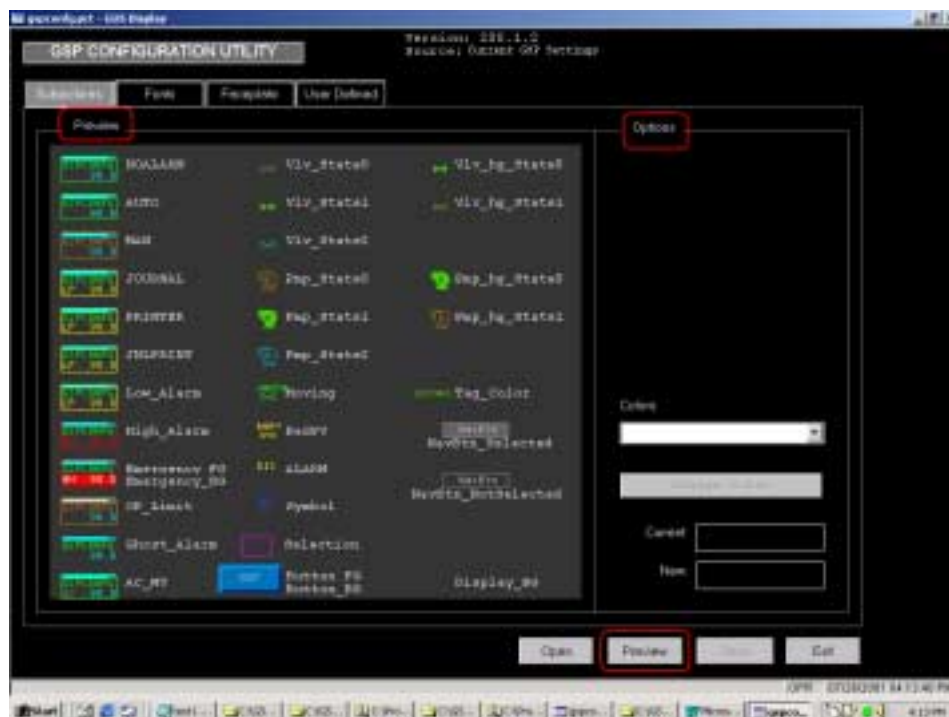
The GSP Configuration Utility provides access to configuration information for the GSP components. It can only run under GSP R200.1 or higher.

This utility was provided at the customers' request. It allows projects to customize the constants used by the GSP display components.

- The configuration utility may only be run by Engineers and Administrators.
- The "Preview" action is required before the customized configuration can be saved. This is to ensure that the customized configuration with the combination of changes will provide a desirable interface to the operators.
- The "Save" action is required before the new configuration can be applied to the current station. This is to ensure that the customized configuration is always saved. After future GSP upgrades, the customized configuration may need to be restored from the saved file.

User can use the utility to modify the alarm colors configured for GUS Displays to match the LCN configured alarm colors. For easy reference, the RGB color constants for TPN alarm colors are provided in the Appendix under GSP Color Constants.

The user can view the various constants by selecting the Subpictures, Fonts, Faceplate, and User Defined tabs in the GSP Utility window. The Preview panel on the left displays the constants as they are configured for the local GUS. The Options panel, on the right contains various buttons to facilitate the modifications to the appropriate constants.



The Preview button is used to preview the modifications made to the constants. Only previewed modifications can be saved, and this is done via the Save button. Once saved, the user has the option of applying the modifications to the local GUS station, or not. The saved configuration file can be opened later via the Open button.

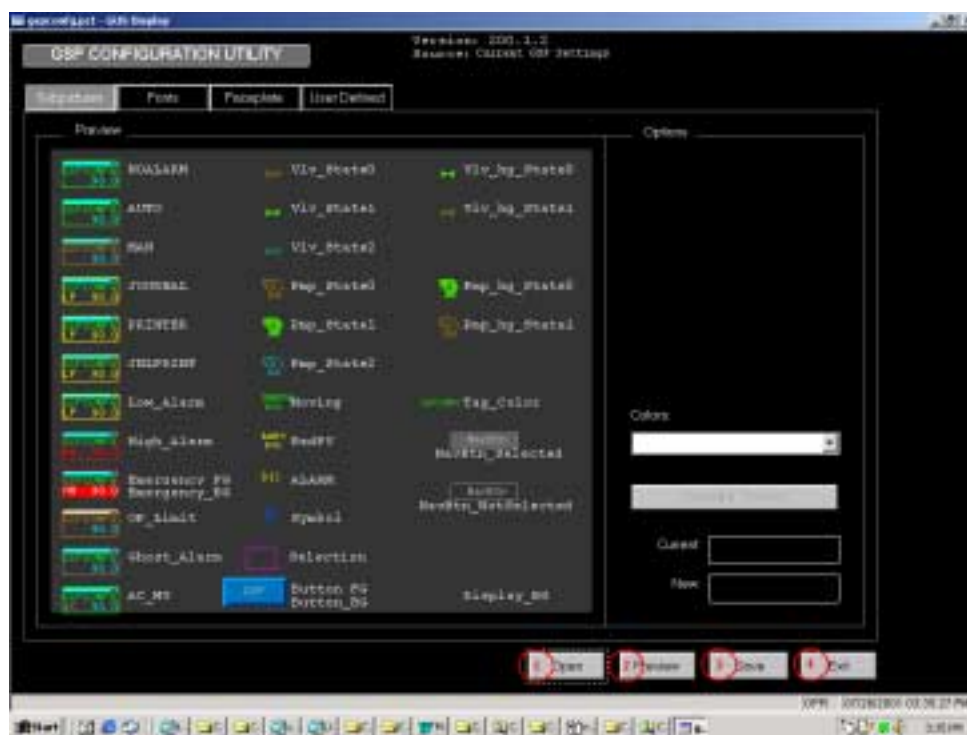
The top right of the GSP Configuration utility contains information important only for the maintenance of the GSP package.

GSP Configuration Utility Common Commands

The utility has four common commands that are described below.

1. Open
2. Preview
3. Save/Apply
4. Exit

In general, the user is required to “Preview” any change before “Save” is allowed. Also, the user is required to “Save” the customized configuration before it can be applied to the station. Only the buttons for the allowed actions are enabled during each step, the other buttons will be disabled (dimmed).

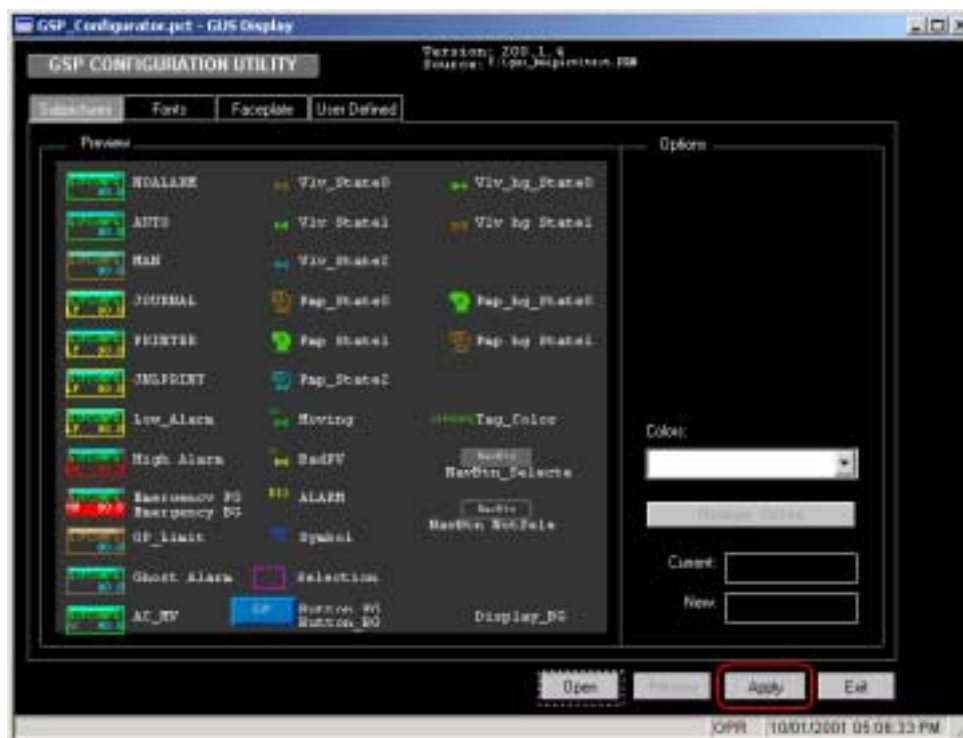


Open an Existing GSP Configuration File

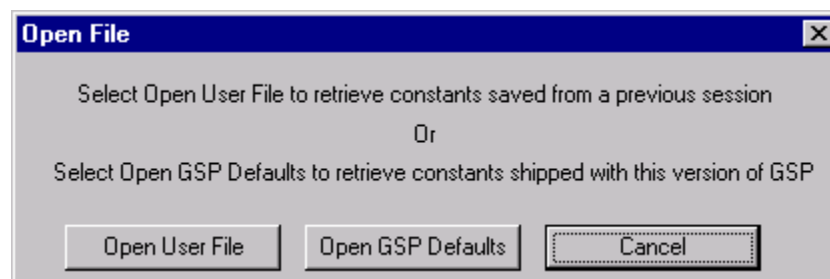
The GSP Configuration Utility automatically starts with the current configuration on the station. The Open button can be used to load a saved configuration file or the GSP default configuration file into the configuration utility.

To open a saved configuration file:

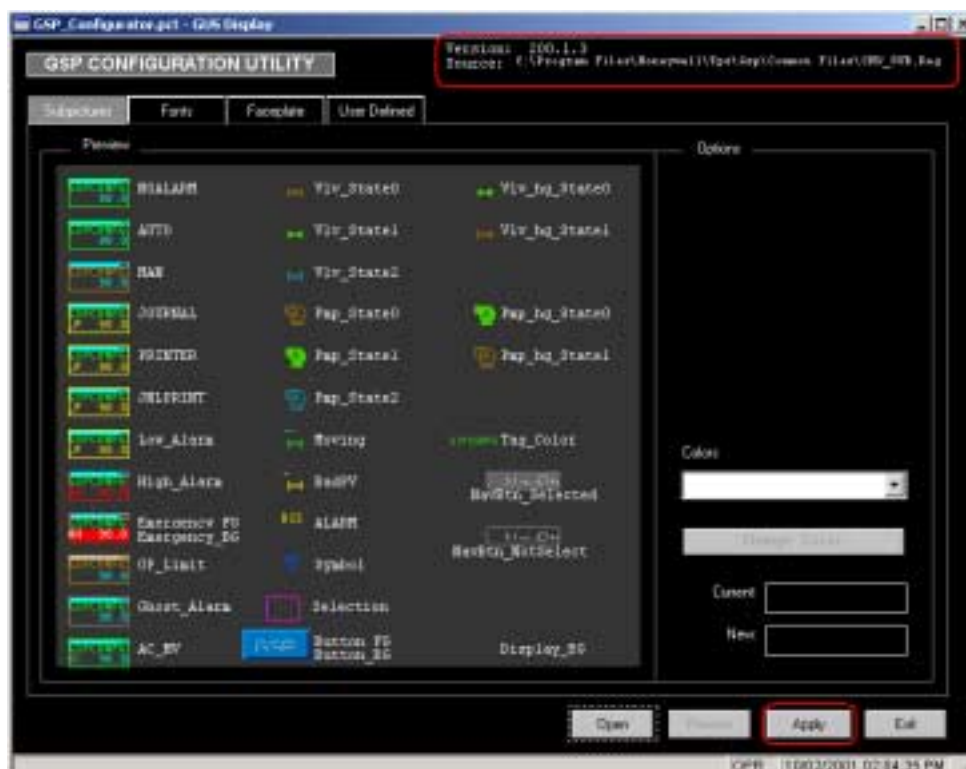
1. Click on the Open button.
2. Click on the Open User File button.
3. Locate the user file (.PRN) and Select it. (Note: The configuration file is automatically previewed once opened.)
4. Click on the Apply button if you want to apply this configuration to the station.
Note: Once the configuration is applied to a station, it will take effect on all GSP displays invoked on this station.



1. To open the GSP Default configuration file:
2. Click on the Open button.
3. Click on the Open GSP Defaults button.



4. Click on the Apply button if you want to apply this configuration to the station.
Note: Once the configuration is applied to a station, it will take effect on all GSP displays invoked on this station.



Note: The configuration file name and version number will be displayed on the upper right of the window as shown below.

Preview Changes to GSP Configurations

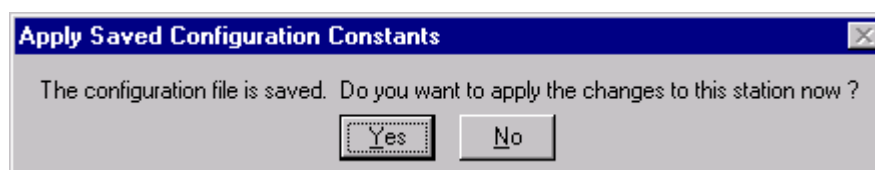
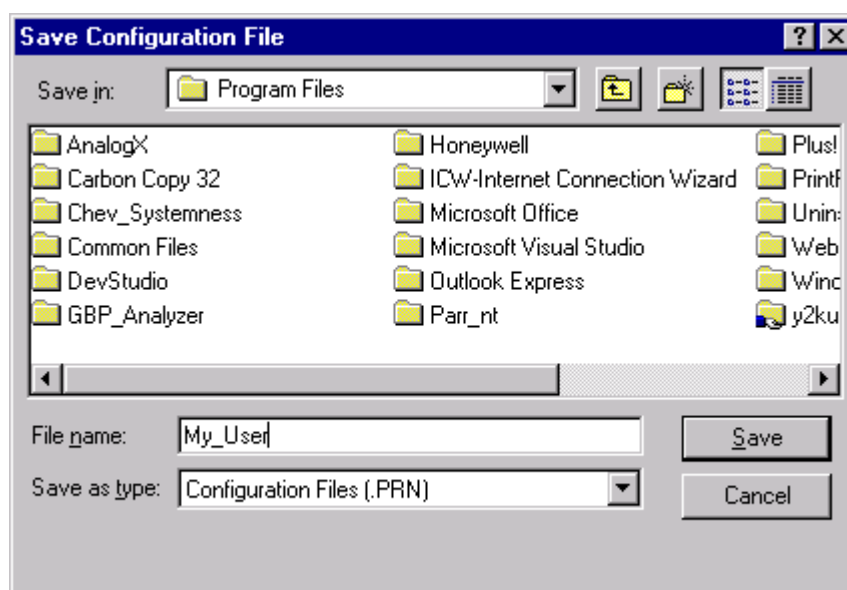
The Preview button is used to view to GSP configuration changes before they are saved to a file and applied to the local machine.

Save the Previewed GSP Configuration File

The Save button is used to save the previewed configuration constants to a .PRN file. The constants in this file can also be applied to the local GUS station.

To save the new configuration constants:

1. Tab through all the pages and preview any change to the GSP configuration constants. (**Note:** Changes to GSP configuration must be previewed before they can be saved and applied.)
2. Once confirmed, click on the Save button.
3. Select the directory and enter the file name. Click on OK. Note: The file will be saved in text format with a default .PRN extension.
4. Select the appropriate option when prompted if the changes are to be applied to the local GUS station. (**Note:** The changes to the constants will not have take effect unless they are applied to the local GUS station. The Preview Panel only displays the changes effective for the current use of the GSP Configuration Utility.)



Exit the GSP Configuration Utility

The Exit button is used to close the GSP Configuration Utility. All unsaved changes will be lost.

GSP Configuration Utility Constants

GSP information is configured in the Configuration Utility via four tab page dialogs:

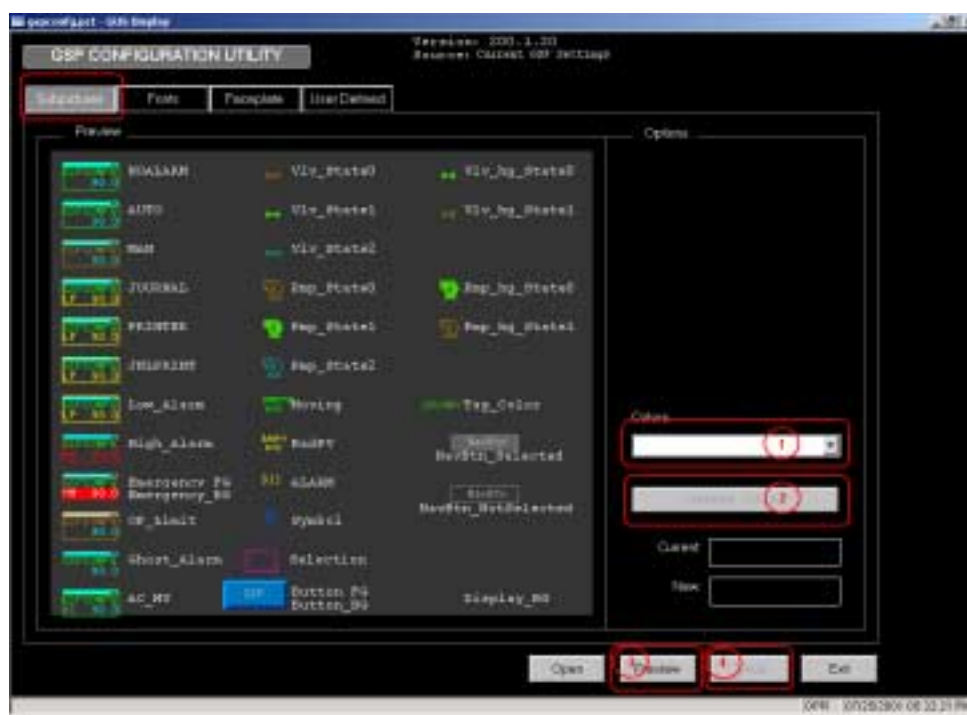
- Subpictures
- Fonts
- Faceplate
- User Defined

Subpictures Page

The Subpictures dialog page contains constants to configure the appearance of GSP display components in response to LCN conditions.

To change a subpicture color constant:

1. Select a color constant from the Colors drop down list. The constant will be highlighted in the Preview Panel.
2. Click on the Change Color button. From the color palette, select a new color.
3. Click on the Preview button to preview the change on the Preview Panel.
4. Click on the Save button when you want to save all the changes you have previewed. Proceed to save the configuration file to the desired location. (Note: changes from the other pages will also be saved.)
5. Select OK/Cancel when asked if the changes are to be applied to the local GUS station.



Fonts Page

The Fonts dialog page contains constants to configure the text fonts and number fonts to be used by subpictures in GSP displays. The default constants from GSP are:

Text Font -- used to display text information (e.g. tag and 2-character alarm state)

- Font Name: monotype.com
- Font Size: 12

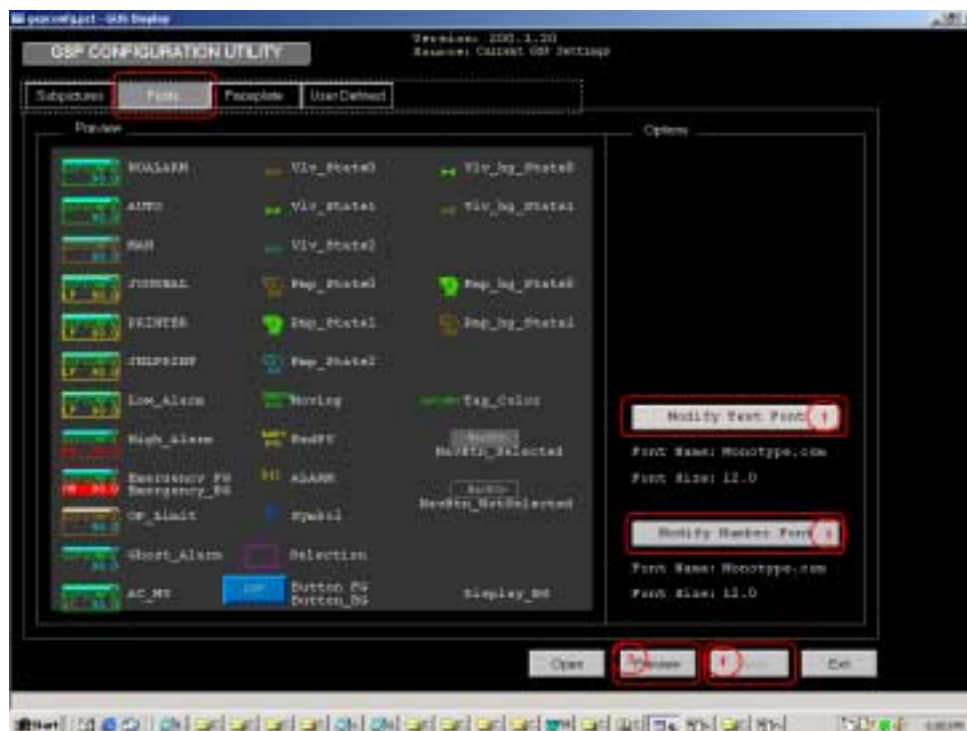
Number Font -- used to display numeric value (e.g. PV values)

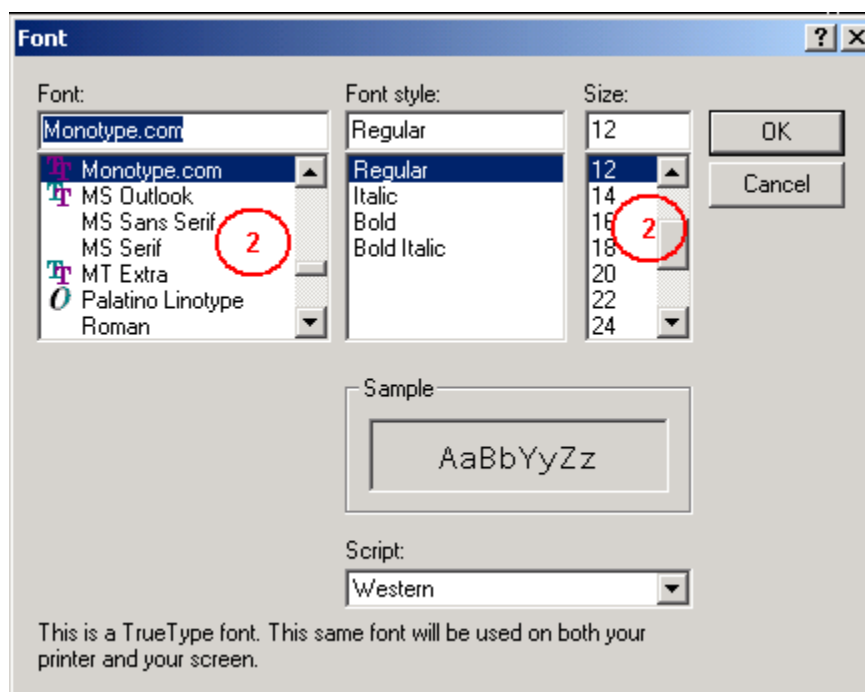
- Font Name: monotype.com
- Font Size: 12

Note: When deciding to use fonts other than the default font provided by GSP, make sure the font you choose is non-proportional and TrueType. Preview the font size in the Fonts Page to make sure that the value to display would not exceed the space allocated by the GSP Display Components. GSP will not be responsible if information gets truncated during runtime when non-default fonts are used.

To change a font constant:

1. Select the appropriate button: Modify Text Font or Modify Number Font.
2. Select a new font name and/or a new font size, then click “OK”. Note: “Font style” and Script settings are not saved by GSP. The default is “Regular”, and any other selection (Italic or Bold) will not take effect.
3. Click on the Preview button to preview the change on the Preview Panel.
4. Click on the Save button. Proceed to save the configuration file to the desired location. (Note: Changes on all pages must be Preview prior to Save which will save all the configuration changes.)
5. Select OK/Cancel when asked if the changes are to be applied to the local GUS station.



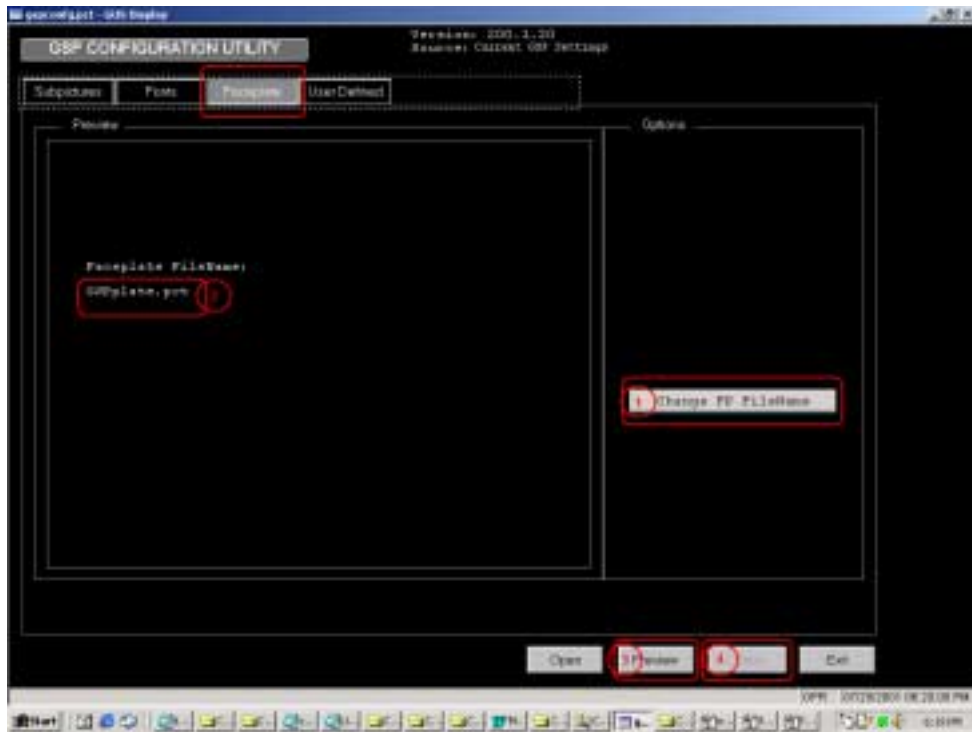


Faceplate Page

The Faceplate dialog page contains the filename of the faceplate that will be invoked when some of the GSP display components are selected at runtime. It should be a GUS display with .pct extension. However if the GUS Windows folder-option is set to "hide file extensions", the extension should be excluded here as well (e.g. GSPplate). No pathname is saved. GSP uses the same GUS displays search path when invoking faceplate.

To change the filename:

1. Click on the Change FP FileName button.
2. Enter the name of the faceplate file you want to use, then click “OK” or “Cancel”. The updated filename will be displayed on the Preview Panel. Note: The faceplate display must reside in one of the folder configured in the display search paths (similar to other GUS displays).
3. Click on the Preview button to test the invocation of the faceplate display.
4. Click on the Save button when you want to save all the changes you have previewed. Proceed to save the configuration file to the desired location. (Note: changes from the other pages will also be saved.)
5. Select OK/Cancel when asked if the changes are to be applied to the local GUS station.



User Defined Page

The User Defined dialog page allows the user to add user constants. These user-defined constants can be used in scripts for customized applications.

Example

Use the utility to defined a constant named "ProjectPhase". Add the script below to a text object on GUS displays:

```
Sub OnDisplayStartup()  
  
    me.text = GetSetting("GSP", "USER", "ProjectPhase")  
  
End Sub
```

When the value for the "ProjectPhase" is set to "Project Phase I", all displays with the above scripted object will display Project Phase I". When you progress to phase II, all you have to do is change the value for the constant "ProjectPhase" to "Project Phase II", and all displays with the above scripted object will be updated accordingly -- **no display change, no re-validation required.**

Refer to example in User Switch under Function Button Reference section for another example of using the user-defined constants.

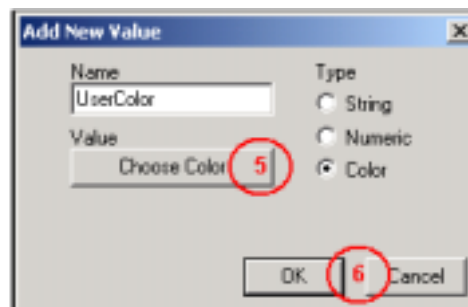
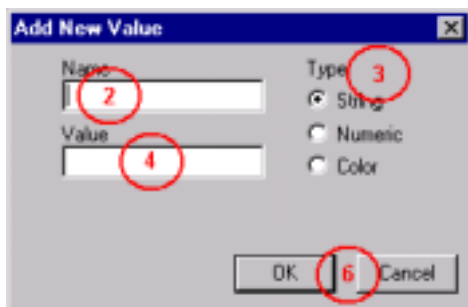
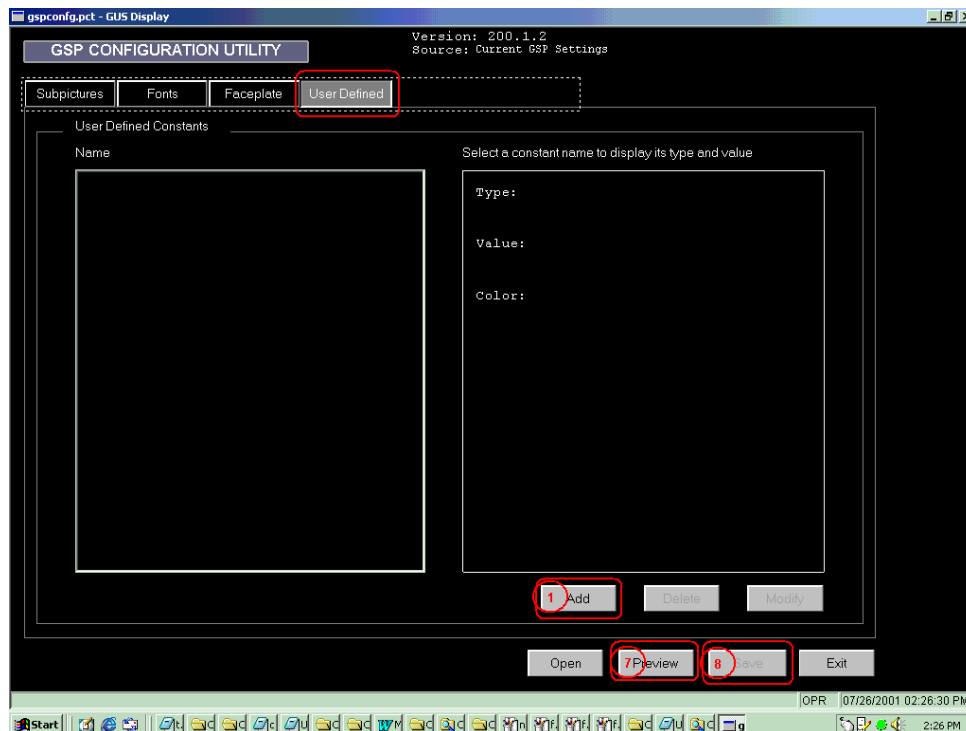
To Add a User Defined Constant

There are two panels in the page. The left panel displays the names of the existing user constants on the local GUS station. The right panel displays the information of the currently selected constant on the left panel.

Click on the Add button

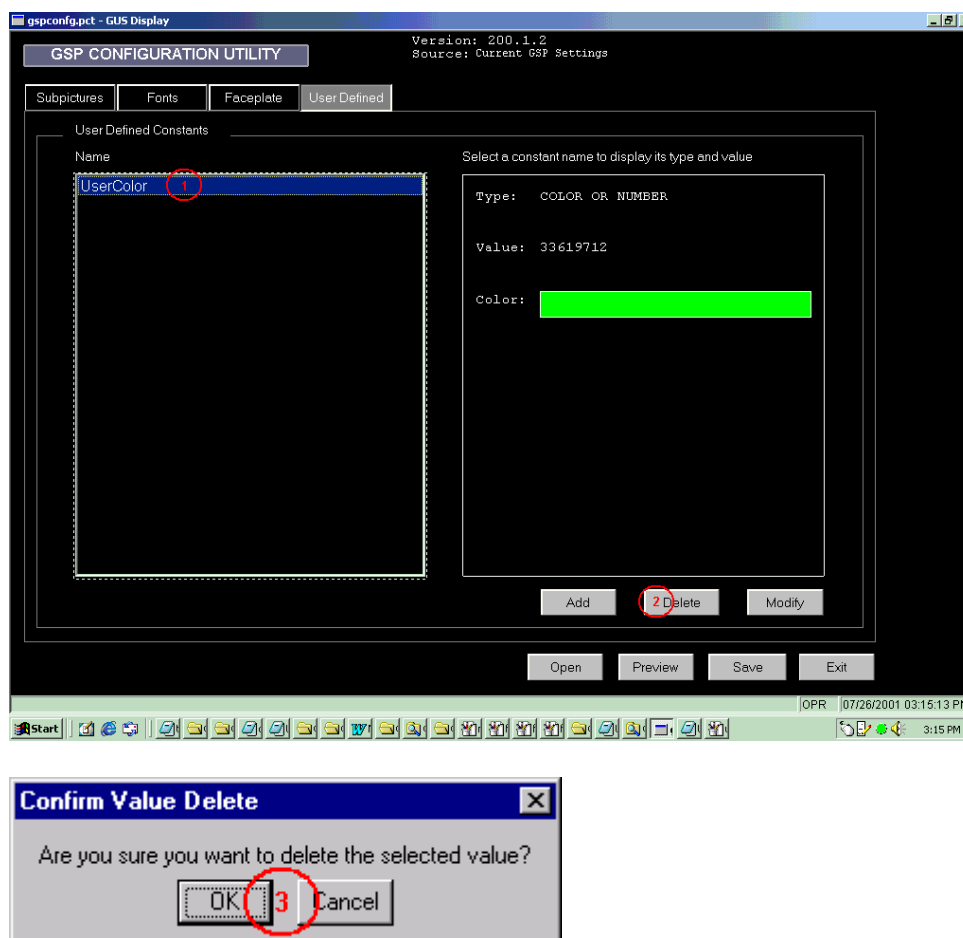
1. Enter the constant name when the "Add New Value" dialog is diaplayed.
2. Select the constant type.
3. Enter an appropriate value for the new constant. Note: no quotes are needed when entering value. Nyumeric value is stored the way it was entered (e.g. 2 or 123.456). Your script in GUS display is responsible for reading the value into a variable of the proper data type.
4. If the new constant is a color, the color palette will appear. Select the color from the basic color palette. Optional: you can click on the "Define Custom Colors>>" button to define colors beyond the basic colors provided.
5. Select OK/Cancel.

- Click on the Save button when you want to save all the changes you have previewed. Proceed to save the configuration file to the desired location. (Note: changes from the other pages will also be saved.)
- Select OK/Cancel when asked if the changes are to be applied to the local GUS station.



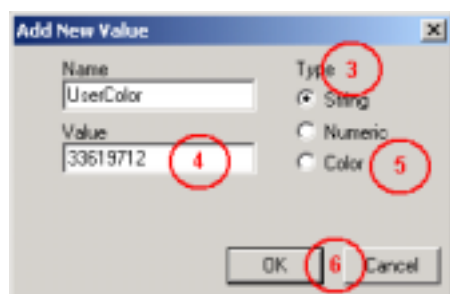
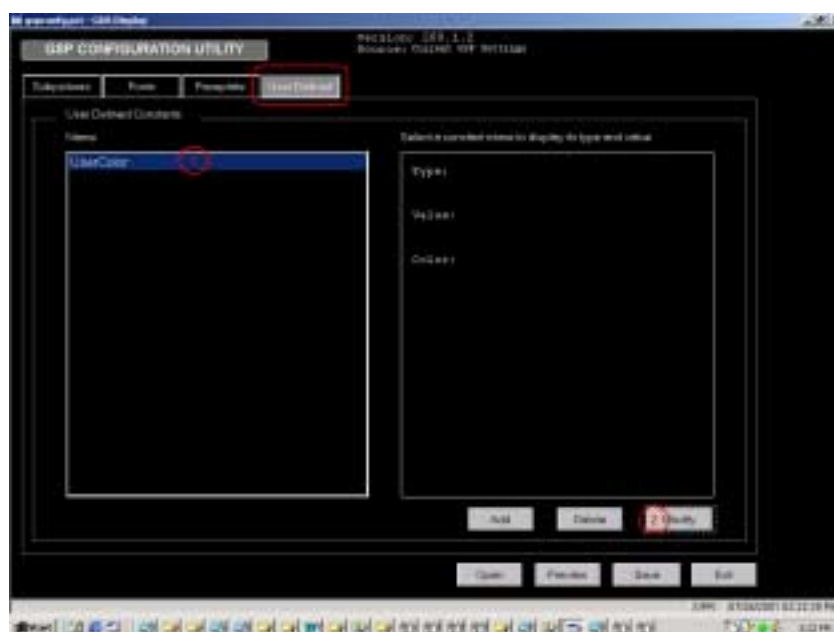
To Delete a User Constant

1. Select a constant name.
2. Click on the Delete button.
3. Confirm the deletion.



To Modify a User Constant

1. Select the constant name.
2. Click on the Modify button.
3. Select constant type.
4. Enter an appropriate value for the new constant.
5. If the new constant is a color, the color palette will appear. Select the new color.
6. Select OK/Cancel.



GSP Upgrade Utility

The GSP Upgrade Utility is provided to help the users to upgrade GSP display components embedded in the GSP displays with minimum effort.

The Upgrade utility may only be run by Engineers and Administrators.

GUS R310 and GSP R100.3 or higher must be installed on the GUS prior to running this utility.

If the displays need to be migrated as part of the GUS upgrade, the displays must be migrated prior to running this utility.

If the off-line (instead of on-line) GUS Display Builder is installed on the GUS, the validation will not be able to validate the displays for TPN data references.

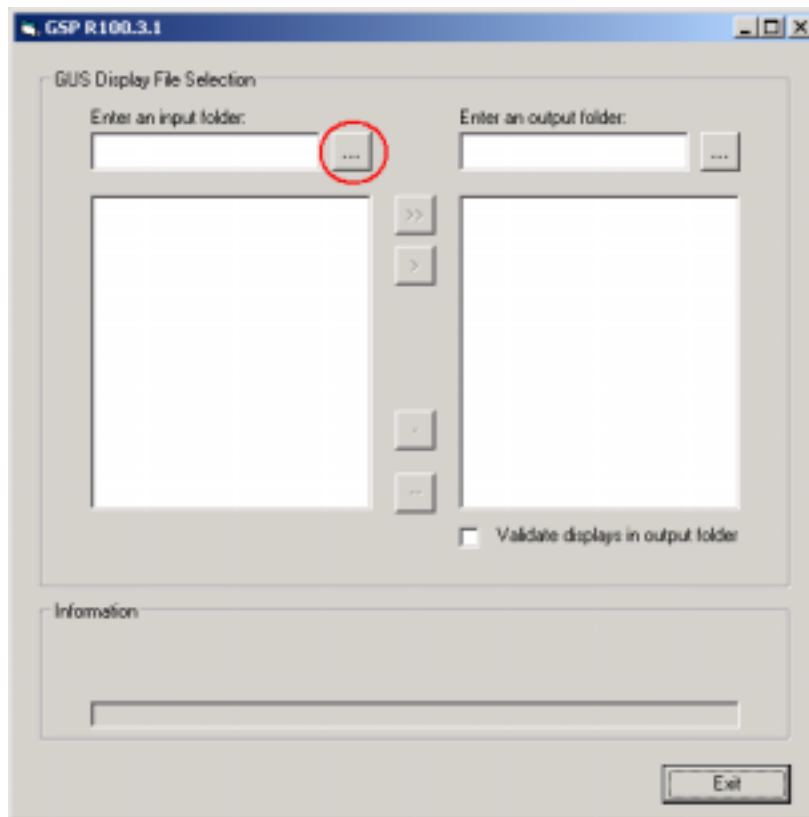
How to Start the Display Upgrade

Locate the folder that consists of the displays you want to upgrade.

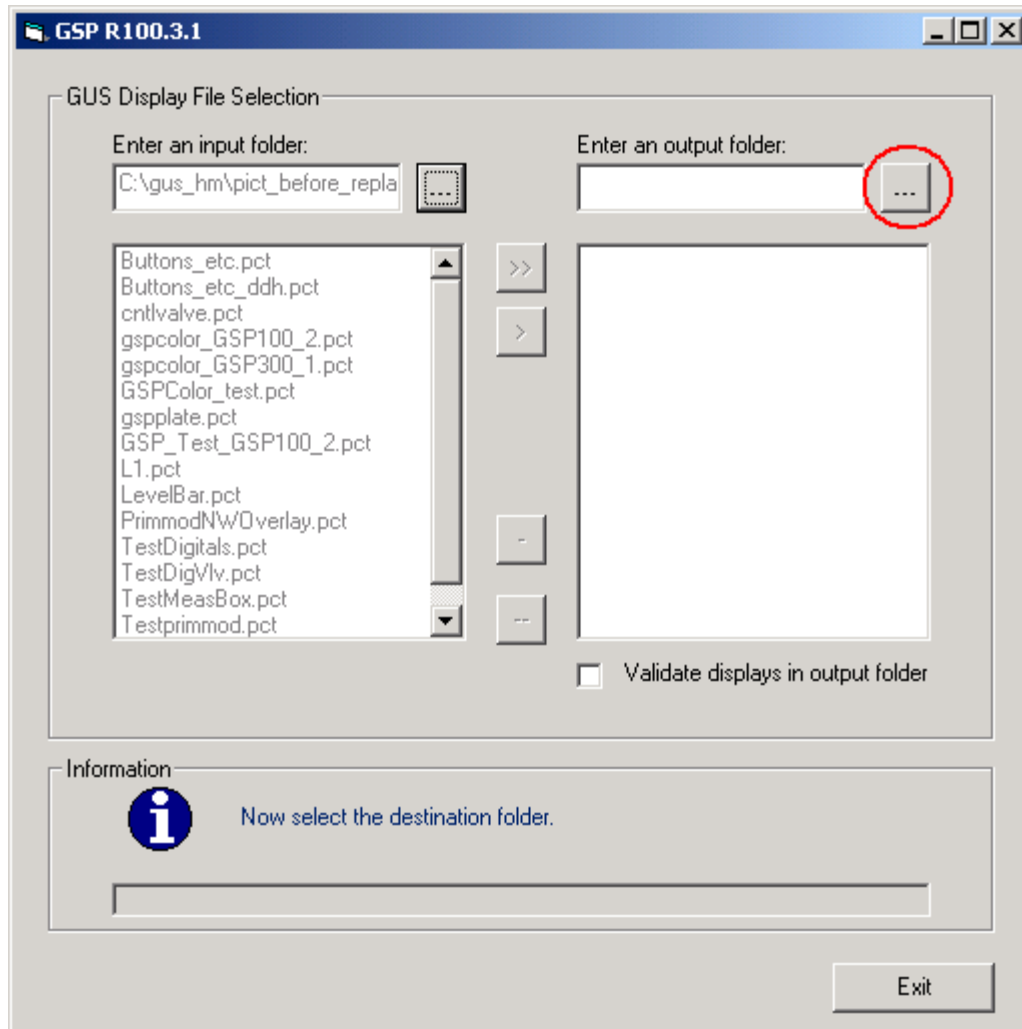
Create a new folder where you want the utility to put the upgraded displays.

To invoke the program, double-click on the executable located in \Program Files\Honeywell\TPS\GSP\Utility\GSP_Upgrade.exe.

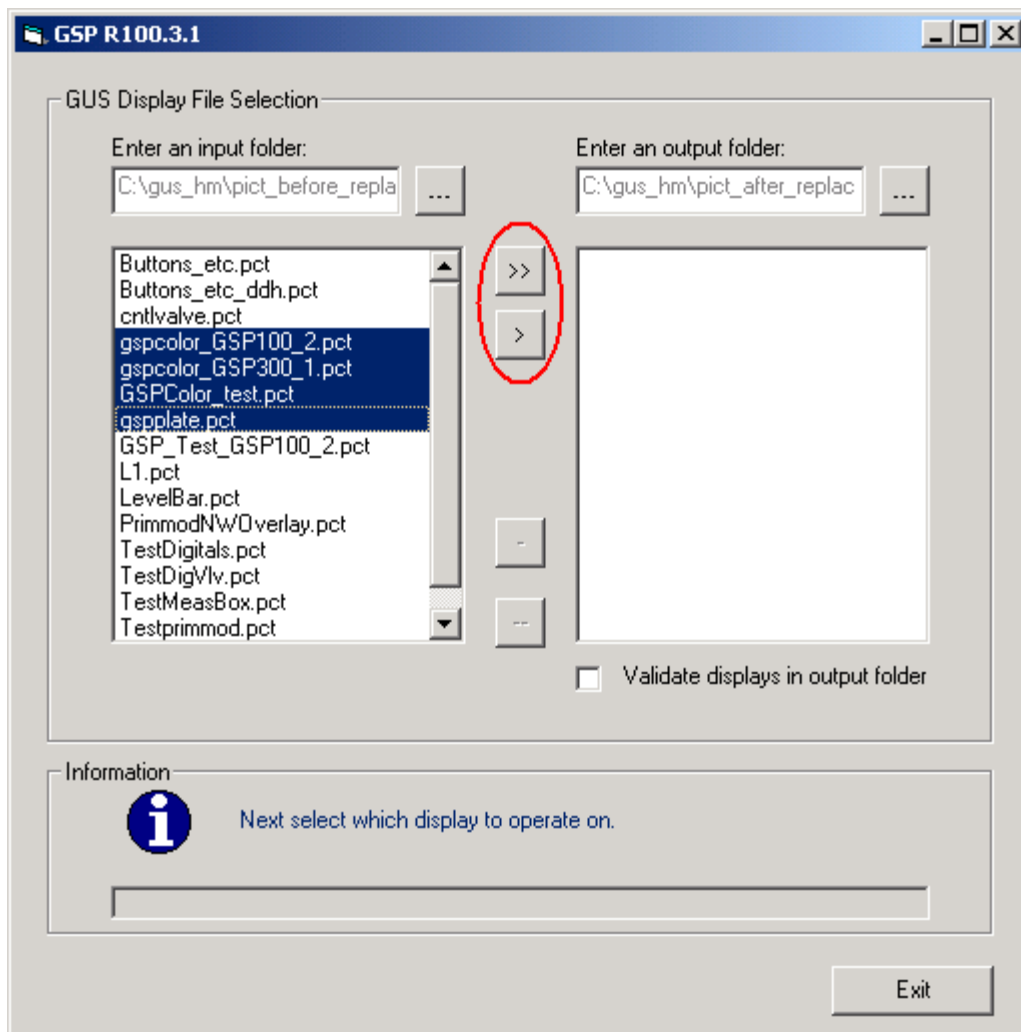
Click on the input folder selection button and select the folder of the existing displays.



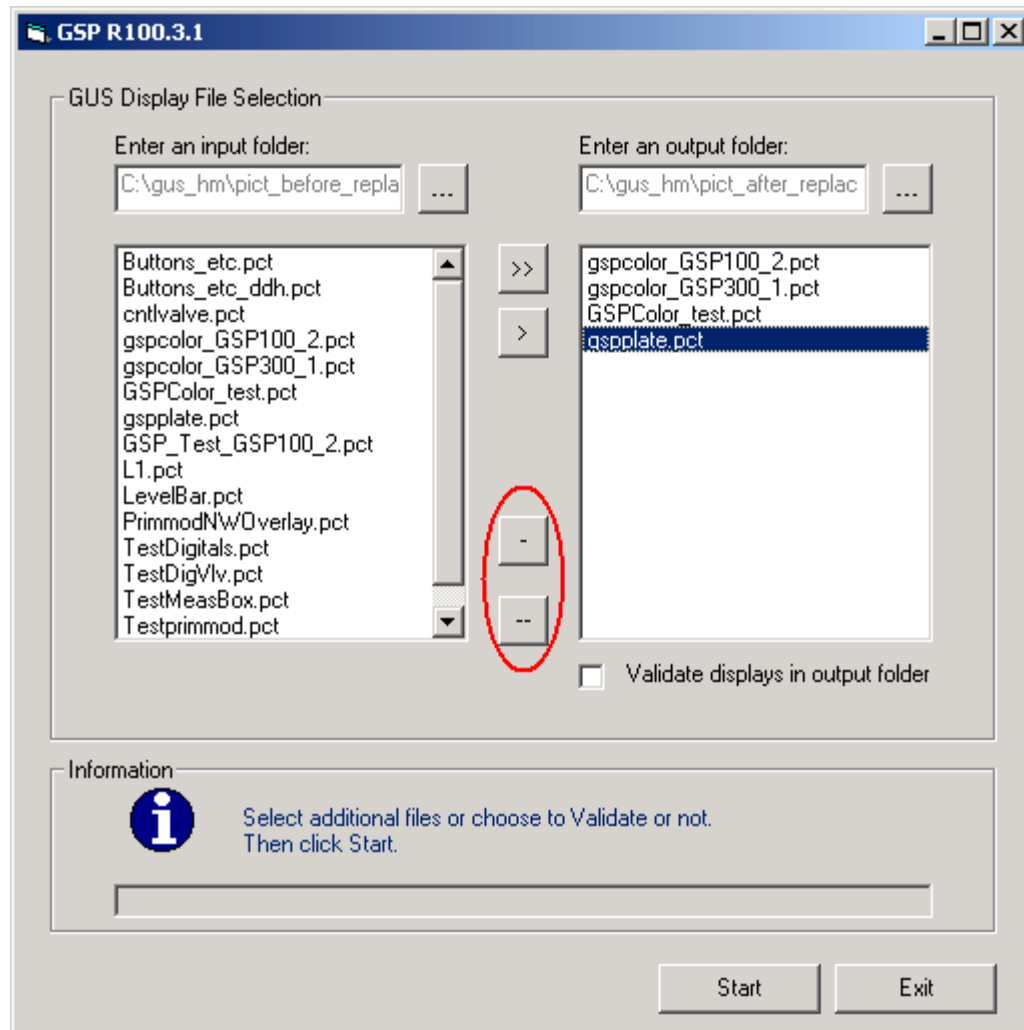
Click on the output folder selection button and select the folder to put the upgraded displays.



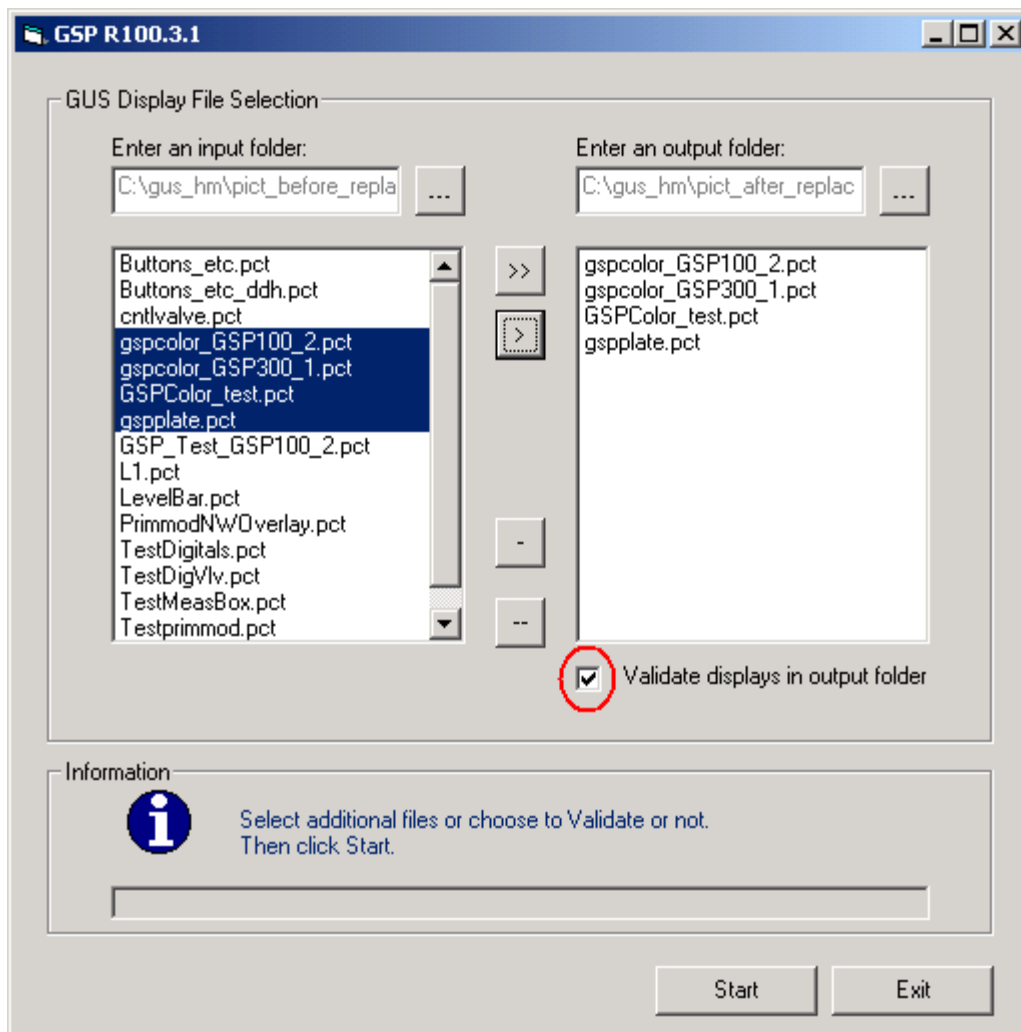
Click on the file(s) in the input folder list to highlight the displays to upgrade. Hold the Shift key or the Ctrl key to select multiple files. Then click on the “>” button to add the highlighted files from the input folder list to the output folder list. By clicking on the “>>” button, all files in the input folder list will be added to the output folder list.



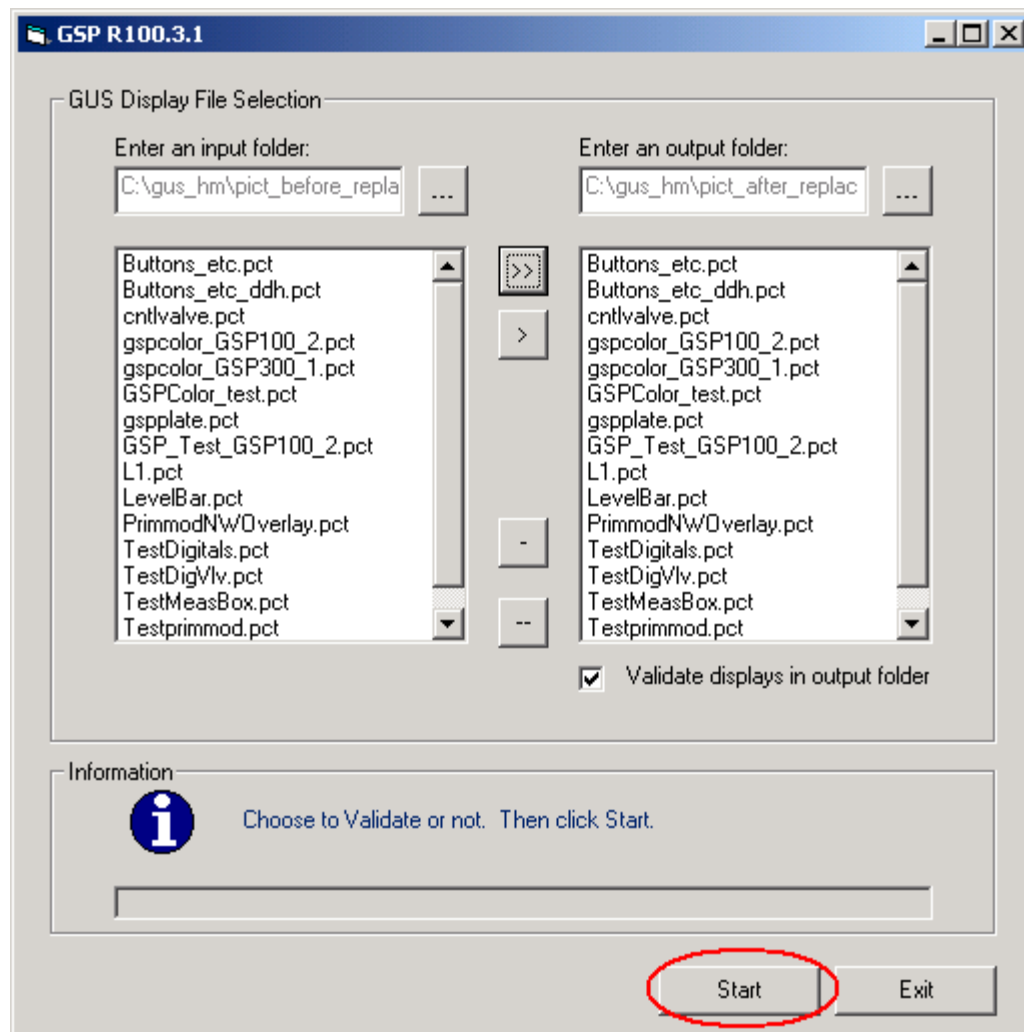
To remove file(s) in the output folder list, click on the files. Hold the Shift key or the Ctrl key to select multiple files. Then click on the “-” button to remove the highlighted file(s) from the output folder list. By clicking on the “- -” button, all files in the output folder list will be removed



Click inside the “Validation” checkbox if you want the program to validate the displays after upgrading the embedded GSP display components. Note: Validation will only be performed for the display if there is no error encountered during the display component upgrade.

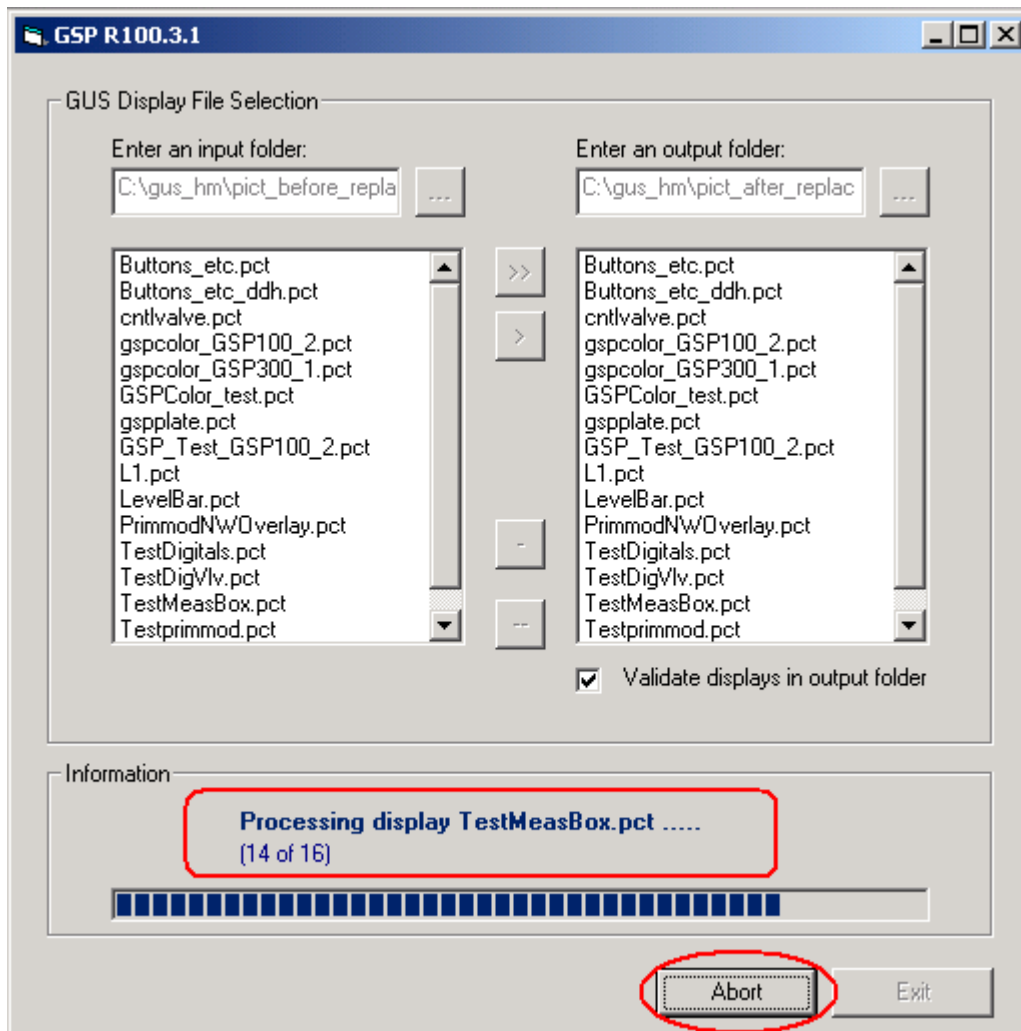


Click the “Start” button to start the GSP display upgrade.



The processing status information will be displayed during the process.

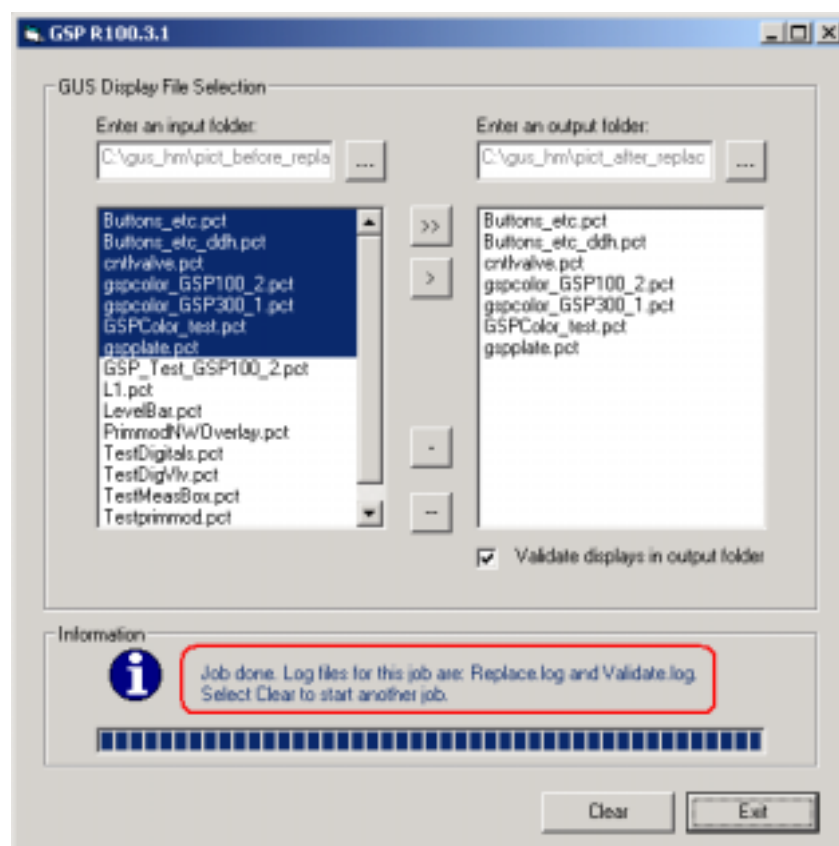
You can click the “Abort” button to abort the upgrade process. Note: For large display with a lot of display components, it may take a few seconds to cancel the upgrade currently in progress. A message “Aborting....” will be displayed. User must wait for the abort process to complete.



A **Replace.log** will be created (or appended if it already exists) in the output folder. It lists the

displays that were upgraded together with any errors encountered during the display component upgrade.

A **Validate.log** will be created (or appended if it already exists) in the output folder if the “Validation” checkbox was checked. It lists the displays that were validated, together with any errors encountered during the validation process.



At this point, you can:

Click “Exit” to exit the program.

Select file(s) in the input folder, then click the “>” button to add them to the output folder list.

Select file(s) in the output folder, then click the “<” button to remove them from the output folder list or click the “< -” button to remove all the displays from the output folder list.

GSP Upgrade Utility - How to Start the Display Upgrade

Click on the input folder selection button or the output folder selection button to work on displays in a different folder.

Appendix

GSP Color Constants -- Default With RGB Settings

Color Constant Name	Red	Green	Blue	Description
AC MV	136	136	136	Medium-light Gray
ALARM	255	255	0	Yellow
AUTO	0	255	0	Green
BadPV	255	255	0	Yellow
Button BG	0	119	191	Blue Gray
Button FG	204	204	204	Light Gray
Display BG	50	50	50	Dark Gray
Emergency BG	255	0	0	Red
Emergency FG	255	255	255	White
Ghost Alarm	136	136	136	Medium-light Gray
High Alarm	255	0	0	Red
JNLPRINT	255	255	0	Yellow
JOURNAL	255	255	0	Yellow
Low Alarm	255	255	0	Yellow
MAN	190	130	0	Brown
Moving	0	255	0	Green
NavBtn NotSelected	50	50	50	Dark Gray
NavBtn Selected	97	97	97	Medium-dark Gray
NOALARM	0	255	255	Cyan
OP Limit	204	204	204	Light Gray
Pmp hg State0	60	210	0	Medium Green
Pmp hg State1	190	130	0	Brown
Pmp State0	190	130	0	Brown
Pmp State1	60	210	0	Medium Green
Pmp State2	0	191	191	Medium Cyan
PRINTER	255	255	0	Yellow
Selection	255	0	255	Magenta
Symbol	0	80	255	Medium Blue
Taq Color	54	191	0	Dark Green
Vlv hg State0	60	210	0	Medium Green
Vlv hg State1	190	130	0	Brown
Vlv State0	190	130	0	Brown
Vlv State1	60	210	0	Medium Green
Vlv State2	0	191	191	Medium Cyan
TDC RED	255	0	0	TDC RED
TDC GREEN	0	255	0	TDC GREEN
TDC YELLOW	255	255	0	TDC YELLOW
TDC BLUE	0	0	255	TDC BLUE
TDC MAGENTA	255	0	255	TDC MAGENTA
TDC CYAN	0	255	255	TDC CYAN
TDC WHITE	255	255	255	TDC WHITE
TDC HALF WHITE	204	204	204	TDC HALF WHITE

Display Component – Point Type Cross-Reference

Display Component	Point Type	Algorithms	Comment
Control Boxes	Regulatory Control	PID PID with External Reset Feedback (PidErfb) PID with Feed Forward (PidFf) Position Proportional (PosProp) Ratio Control (RatioCtl, PidRatio)	PNTFORM=Full PVSRCOPT=All
Output Boxes	Output-Only points		
Control Valves	Regulatory Control	PID PID with External Reset Feedback (PidErfb) PID with Feed Forward (PidFf) Position Proportional (PosProp) Ratio Control (RatioCtl) Auto Manual Ramp Soak Regulatory Data Point on HG/PLCG/AM	PNTFORM=Full
Selector Boxes	Regulatory Control	Override Selector (ORSel) on PM/APM/HPM Override High Selector (Overhisl) HG Override Low Selector (Overlosl) HG Control Override Selector (OrSel) on AM	PNTFORM=Full
Level Bars	Regulatory PV Analog Input		PNTFORM=Full
Cascade Symbols	Regulatory Control	Secondary Controller	
Measurement Boxes	Regulatory PV Regulatory		PNTFORM=Full PVSRCOPT=All
Digital State Boxes	Digital Input Digital Composite		PNTFORM=Full

Appendix - Display Component – Point Type Cross-Reference

Display Component	Point Type	Algorithms	Comment
	Device Control Flag		
Digital Alarm Graphic Symbols	Digital Input Digital Composite Device Control Flag		PNTFORM=Full
Digital Alarm Message Boxes	Digital Input Digital Composite Device Control Flag		PNTFORM=Full
Digital Valves	Digital Composite Device Control		
Digital Pumps & Motors	Digital Composite Device Control		
Digital Alarm Message Boxes	Digital Input Digital Composite Device Control Flag		PNTFORM=Full
Primmod Status Indicators	Flag on PM/APM/HPM Batch Assign on HG Any Point Type on AM		PRIMMOD points
Aggregate Primmod Status Indicators	Flag on PM/APM/HPM Batch Assign on HG Any Point Type on AM		PRIMMOD points

Appendix - Reserved GUS Display Database Variables

Display Component	Point Type	Algorithms	Comment
Custom Parameter Boxes	Any point.parameter with numeric or text values		including array element of custom AM points

Reserved GUS Display Database Variables

GSP reserves Display Data Base variables between 200-256 of all variable types. To avoid conflict with the GSP embedded objects, user must not use the variables reserved. In general, when GSP uses display variables in the standard embedded objects they start with 256 and work down. Users are urged to start with 01 and work up.

String

Variable	Description	Referenced
str256	Tagname selected in faceplate	All GSP display componets with target to callup faceplate: chv_obj.pct
str255	Filename of active GUS displays	Display navigation buttons: - chv_obj.pct - chv_scan.pct
str248	Error message in Task Panel	Display navigation buttons: - TaskPanelTemplate.pct - LowLevelNavTemplate.pct
str247	High-Level display name referenced by the Low-Level displays	High-Level display navigation buttons: LowLevelNavTemplate.pct

Entity

Variable	Description	Referenced
ent256	Point entity selected in faceplate	All GSP Display Componets with target

Appendix - Reserved GUS Display Database Variables

		to callup faceplate
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Boolean

Variable	Description	Referenced
bool256	Toggle display to show Tag/Setpoint (true = show setpoint; false = show tag)	• Large control boxes • tag_sp_btn.pct • chv_obj.pct
bool255	Toggle display to show/hide GSP objects (true = show; false = hide)	• measurement boxes with tag • digital alarm message boxes • cust_param.pct • show_hide.pct • chv_obj.pct
bool254	Triggler OnDataChange script	• DisplayTemplate.pct • chv_obj.pct • app_btn.pct • button.pct • pct_btn.pct
bool253	Toggle display to show/hide user objects (true = show; false = hide)	• show_hide_text.pct • chv_obj.pct

Integer

Variable	Description	Referenced
int256	Switch among several user-defined states	user_switch.pct chv_obj.pct

Appendix - Reserved GUS Display Database Variables

Glossary

Color Constants

GSP-defined color coding to provide consistent component behavior throughout the user interface.

Display Category

One of the GUS display properties. Default GSP display categories are HighLevel, LowLevel, TaskPanel, faceplate. They are used by SafeView to control display locations on the screen.

Display Component

GSP-provided embedded displays such as control valves, control boxes, and alarm indicators used as components for building process displays and task panels.

Faceplate

GSP-provided display that provides an operator interface for viewing and manipulating operating parameters (e.g., setpoint, output, mode, etc.) of a single point. It is based on the GUS faceplate ActiveX control.

Function Buttons

GSP components designed to be inserted on Task Panel to perform station-wide tasks such as toggle the show/hide state of text, or invoke an application.

GSP

GUS Solution Pack

GUS

Global User Station

High-Level Display

Typically, these are overview and summary displays that provide the operator key information to quickly ascertain the operational status of the process area.

IKB

Integrated Keyboard

Low-Level Display

Typically, these are more detailed representations of smaller portions of the process area. If several Low-Level Displays are related to one High-Level Display, build the hierarchical relationship in the task panel to help display navigation.

Master Display Templates

GSP-provided templates to be used as starting point to build GUS displays. They include script and objects required for GSP components to function properly.

Pathname Catalog

Path to search for displays and schematics which is configured on the Area Database of the TPN system.

SafeView

Honeywell window management application used to control the display characteristics of application windows within a Microsoft Windows workspace on a GUS.

Task Panel

A user-built display using GSP components to provide multi-level display navigation, alarm annunciation, and other operator functions.

TPN

TPS Process Network

TPS

TotalPlant Solution

User-Defined Templates

Copy of the master display template with category and size modified to match the user's SafeView configuration.

WDL

Workspace Definition Language used by SafeView.

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